MILITARY SPECIFICATION

TRANSMITTING SET, TELEMETRIC DATA,

AN/AKT-22(V)6

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1.1 SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of AN/AKT-22(V)6, Transmitting Set, Telemetric Data, herein referred to as the equipment, which is an airborne ASW-signal transmitting set that provides a complete data link between acoustic signal acquisition sonobuoys and a shipboard data processing system for both sonic and sonar data. The AN/AKT-22(V)6 provides for the use of either two DIFAR/DICASS and two LOFAR sonobuoys or four LOFAR/RO sonbouoys (See Figure 1).

1.2 Associated equipment. This equipment should operate with the associated equipment listed in 6.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-T-27	Transformers and Inductors (Audio, Power, and High-Power Pulse), General Specification for
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic and Associated Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted (ASG)
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-S-8805/101	Switches, Sensitive, 2 Circuit (5 Amperes) or SPDT (8 Amperes), Unsealed, Low Level Switching
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-22885	Switch, Pushbutton, Illuminated, General Specification for

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MIL-C-39019/2	Circuit Breakers, Magnetic, Low Power, Sealed, Trip-free, One-pole, Auxilliary Contacts
MIL-P-55617	Plastic Sheet, Thin Laminate, Metal Clad, (for printed wiring), Primarily for Multilayer
MIL-R-81877	Radio Sets AN/ARC-159 and AN/ARC-159(V)
MIL-S-81895	Sonobuoy AN/SSQ-62(U)

STANDARDS

Military

MIL-STD-188	Military Communication Systems Technical Standards
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Test Levels and Accept/Reject Criteria for Reliability of Nonexpendable Electronic Equipment
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-2074	Failure Classification For Reliability Testing
MIL-STD-2084	General Requirements For Maintainability of Avionic and Electronic Systems and Equipment
MS 3110	Connectors, Receptacle, Electric, Series 1, Solder Type, Wall Mounting, Flange, Bayonet Coupling

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MS 3338	**Lamp-incandescent, T-1 Bulb, Based, 28-Volt, Integral Lighting**
MS 24655	Switch Toggle, Positive Break, Miniature, Toggle Sealed, Solder Lug, Single Pole, .250 Mounting Bushing

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.1.1 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample(s) shall be subjected to first article inspection (See 4.3 and 6.3).

3.2 Parts and materials. In the selection of parts and materials the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-STD-2084 and MIL-E-5400.

3.2.1 Non-Standard Parts. Approval for the use of nonstandard parts (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained in accordance with MIL-E-5400. Microelectronic devices shall be approved in accordance with MIL-STD-965 (See 6.2.2).

3.2.2 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules. The electronic portions of the equipment shall be functionally modularized in accordance with MIL-STD-2084.

3.3 Design and construction. The equipment shall be in accordance with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total weight. The total weight of the equipment, excluding cables, shall be not greater than 45 pounds.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785 (See 6.2.2).

a. Operational stability. The equipment shall operate with specified performance, continuously or intermittently, for a period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

b. Operating life. The equipment shall have a total operating life of 11,000 hours with minimal servicing and replacement of parts.

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c. Reliability in mean-time-between-failure (MTBF). The equipment shall have 200 hours of mean (operating) time between failures when tested and accepted as specified in 4.5.3.

d. Time totalizing meter. A time totalizing meter conforming to MIL-M-7793 shall be included in the transmitter-multiplexer.

3.3.3 Cabling and connections

a. Cables and connectors. The equipment shall use cables and connectors in accordance with MIL-E-5400.

b. Interconnection cabling. The equipment shall be capable of specified operation using external wiring in accordance with MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and connectors shall not be supplied as part of the equipment.

3.3.4 Control panels. All racks or console mounted control panels shall be in accordance with MIL-C-6781. The configuration of all panels shall be approved by the procuring activity prior to first article testing (See 6.2.2).

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Electromagnetic interference control. The generation of electromagnetic interference (EMI) by the equipment and the vulnerability of the equipment to EMI shall meet the requirements and limits of MIL-STD-461. Existing designs are permitted to have 20 decibels (dB) degradation for radiated emissions from the levels specified in MIL-STD-461.

3.3.7 Provisions for maintainability. The maintainability program, construction and packaging, provisions for test points, and other maintainability parameters shall be in accordance with MIL-STD-2084.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307 (See 6.2.2).

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3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a. Temperature	Room ambient (+25°C ± 10°C)
b. Altitude	Normal ground
c. Vibration	None
d. Humidity	Room ambient up to 90 percent relative humidity
e. Input power voltage	115/208 Vrms, 400 Hz, 3-phase Y and 28.0 VDC as specified in MIL-STD-704

3.3.10 Service conditions. The equipment shall operate as specified under any of the environmental service conditions or reasonable combination of those conditions in accordance with MIL-E-5400 for Class 1A equipment except as modified herein.

a. Vibration. The equipment shall operate as specified when subjected to the vibration requirements of 4.3.2 without isolators.

b. Altitude. The equipment shall operate as specified at any altitude up to 10,000 feet.

3.3.11 Warm-up time. The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall be not greater than five minutes under standard conditions and 15 minutes at extreme service conditions.

3.3.12 Input electrical power

a. Operating power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following sources having characteristics and limits as specified in MIL-STD-704. The power required shall be not greater than the specified amounts.

(1) AC Power (Three Phase Y),
115/208 Vrms, 400 Hz,
500VA

(2) DC Power, 28V,
50 Watts, maximum

b. Lighting power. Input power for panel lighting shall require not more than 0.15 A at 28 VDC.

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c. Degraded performance. Degraded performance will be permitted for voltage transients on the AC bus in accordance with MIL-STD-704. Operation shall return to normal with no resulting damage to the equipment.

d. Standby power. The equipment shall provide for a standby position whereby rated voltage shall be applied to all circuits except the transmitter. The standby power shall be not greater than: 300 VA, 50 Watts D.C. Power and 4.5 Watts Lighting Power.

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerance or values setting forth acceptable variations from the performance under the standard conditions are specified herein.

3.4.1 Operation. When properly interfaced with the AN/ARC-159(V)1 and the AN/ARR-75, the Telemetric Data Transmitting Set AN/AKT-22(V)6 shall provide control for CASS and DICASS sonobuoys and shall relay acoustic signals from selected sonobuoys to a shipboard receiving set and acoustic processor. A block diagram of the AN/AKT-22(V)6 is illustrated on Figure 1. The transmitting set, in the operate mode, shall operate with four LOFAR (AN/SSQ-41 or AN/SSQ-57) sonobuoys or four RO (SSQ-47) sonobuoys or any combination of four LOFAR or RO sonobuoys. When the set is in the DIFAR/DICASS mode it shall be possible to use two of any combination of DIFAR (AN/SSQ-53) or DICASS (AN/SSQ-62) sonobuoys along with two LOFAR (AN/SSQ-41, AN/SSQ-57) sonobuoys. In both operate and DIFAR/DICASS modes, a voice channel shall be provided.

3.5 Detail requirements. The AN/AKT-22(V)6 Transmitting Set shall consist of the items listed below and described in 3.5.1 through 3.5.5.

a. Transmitter-Multiplexer T-1295A/AKT-22(V)

b. Antenna, AS-3033A/AKT-22(V) or AS-3033( )/AKT-22(V)

c. Actuator, Linear Electro-Mechanical TG-229A/AKT-22(V)

d. Control Indicator C-10742/AKT-22(V)

e. Control Generator C-10743/AKT-22(V) or C-10743( )/AKT-22(V)

f. Control, Antenna C-XXXXX/AKT-22(V)

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3.5.1 Transmitter-multiplexer T-1295A/AKT-22(V)

3.5.1.1 Overall unit. The overall transmitter-multiplexer shall meet the following requirements:

a. Function. The T-1295A/AKT-22(V) transmitter-multiplexer shall condition and combine four LOFAR/RO data channels, or two LOFAR channels and two DIFAR/DICASS channels, one voice channel, and up to four trigger signals (locally generated in DICASS mode), into a frequency-division multiplex signal, FM modulate a UHF carrier with the combined multiplex signal and amplify the carrier. The T-1295A/AKT-22(V) shall be convertible to perform completely or partially the functions of the T-1295/AKT-22(V) or T-1220B/AKT-22(V) by the addition, replacement, or transfer of modules and adding minor wiring changes. Provision for modulation by external video or PCM signals shall be provided. The transmitter- multiplexer shall be Intermediate Maintenance Activity (IMA) repairable.

b. Form factor. The transmitter-multiplexer shall be enclosed in a case with the dimensions as illustrated on Figure 2.

c. Weight. The weight of the transmitter-multiplexer shall be not greater than 20 pounds.

d. Contents. The transmitter-multiplexer shall contain the following major assemblies:

(1) Four acoustic voltage controlled oscillators	(see 3.5.1.2)
(2) One 70 kHz DIFAR/DICASS voltage controlled oscillator	(see 3.5.1.2.1)
(3) Four acoustic low-pass filters	(see 3.5.1.3)
(4) One voice low-pass filter	(see 3.5.1.4)
(5) One mixer-amplifier	(see 3.5.1.5)
(6) One trigger filter	(see 3.5.1.6)
(7) One 20-channel FM transmitter	(see 3.5.1.7)
(8) One interstage isolator	(see 3.5.1.8)
(9) One RF power amplifier	(see 3.5.1.9)
(10) One isolator/power monitor	(see 3.5.1.10)

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(11) One telemetry mount (see 3.5.1.11)

(12) One DIFAR mode select module (see 3.5.1.12)

(13) One DIFAR amplifier adapter module (see 3.5.1.13)

(14) One power supply and necessary control circuitry

e. Controls. The T-1295A/AKT-22(V) shall have the following controls on the front panel:

(1) Circuit Breaker	Applies AC power to unit. Protects unit in the event of overload.
(2) Channel Selector Switch	Provides five digit code (ground or open circuit) to transmitter to control transmitter frequency. In RMT position allows control of channel selection from a remote switch.

f. Electrical connections. Connections to external circuits shall be provided in accordance with Table I.

g. Primary power input. 115/208 VAC 400 Hz three phase Y, neutral grounded in accordance with MIL-STD-704.

h. Control voltage input. 28 VDC in accordance with MIL-STD-704.

j. Internal regulated voltage. 28 ± 1.0 VDC and 24 ± .5 VDC.

k. Operating modes

(1) OFF mode no power required.

(2) STANDBY mode shall turn on and apply power to multiplexer circuits. No RF signals shall be generated at any level.

(3) OPERATE mode shall apply operational power to transmitter. One voice channel and four acoustic channels of data from AN/SSQ-41, AN/SSQ-47, or AN/SSQ-57 sonobuoys or the LOFAR portion of data from AN/SSQ-53 sonobuoys and trigger tones shall be multiplexed and transmitted in this mode.

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(4) DIFAR/DICASS mode shall apply operational power to the transmitter. One voice channel and any two channels of data from AN/SSQ-41, or AN/SSQ-57 sonobuoys or the LOFAR portion of data from AN/SSQ-53 or AN/SSQ-62 sonobuoys and the complete data from either two AN/SSQ-53 sonobuoys or two AN/SSQ-62 sonobuoys or one each AN/SSQ-53 and AN/SSQ-62 sonobuoys and trigger tones from the C-10743/AKT-22(V) control generator shall be multiplexed and transmitted in this mode.

m. Operating frequency. The T-1295A/AKT-22(V) shall be capable of transmitting on twenty preset channels between 2200 and 2290 MHz on 0.5 MHz spacing. Any one of the RF channels may be selected by the front panel selector switch, or remotely if so configured. When RF channel selection is made from the front panel selector switch, a means shall be employed to clearly depict the selected channel number.

n. RF carrier deviation. The nominal deviations to the main RF carrier for the various subcarrier and baseband signals are:

(1) DIFAR/DICASS Mode

Subcarrier Frequency (kHz)	Main Carrier deviation (kHz)
12.5	± 13.8
20.8	± 22.5
70.0	± 34.6
165.0	± 81.6
Voice (baseband)	± 9.4
Trigger (each)	± 6.2

(2) Operate Mode

Subcarrier Frequency (kHz)	Main Carrier deviation (kHz)
12.5	± 13.8
20.8	± 22.5
45.8	± 48.8
54.2	± 58.8
Voice (baseband)	± 9.4
Trigger (each)	± 6.2

p. Frequency accuracy. The output frequency shall be within ± 0.003 percent of specified unmodulated carrier frequency in temperature range from 0°C to +45°C and within ± 0.005 percent of specified unmodulated carrier frequency in temperature ranges from -54° C to 0°C and +45°C to +55°C.

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q. RF power output. The RF power output shall be 22 Watts minimum into a 50 ohm resistive load over the frequency range of 2200 to 2290 MHz from -54°C to +55°C.

r. Signal inputs

(1) Four sonic data channels with 10 to 2000 Hz bandwidth.

(2) Six trigger channels within the band of 26 to 38 kHz.

(3) One voice channel with 300 to 3000 Hz bandwidth.

(4) Two DIFAR/DICASS channels with 10 Hz to 17.4 kHz bandwidth.

(5) One of four trigger tone inputs (See 3.5.5.1.b) from the C-10743/AKT-22(V) control generator.

s. Sonic data channel output performance. The sonic data channels shall meet the following performance requirements using the telemetric data receiving set.

(1) Signal to noise ratio. The noise contributed by the AN/AKT-22, when in the operate mode, to each of the four sonic data channels shall be greater than 54 dB below the 100% modulation (+ 2 kHz deviation) of that channel when measured in a bandwidth of 10 to 2000 Hz, and greater than 40 dB below the 100% modulation (+ 28 kHz deviation) of the DIFAR/DICASS channels when measured in a bandwidth of 10 to 20000 Hz.

(2) Phase correlation. Each narrowband sonic data channel shall have equal delay within + 5 degrees tracked over the band from 10 to 500 Hz in the 2 kHz bandwidth channels.

(3) Frequency response. Each narrowband sonic data channel shall have a frequency response within + 1.0 dB from 10 Hz to 500 Hz; the response shall be -3 ± 2 dB at 2 kHz.

t. Voice channel output performance. The voice channel shall meet the following performance requirements using the telemetric data receiving set:

(1) Signal to noise ratio

Operate mode. The noise contributed by the AN/AKT-22 to the voice channel shall be greater than 14 dB below the 100 percent modulation (± 10,000 Hz deviation) of that channel when measured in a bandwidth of 300 Hz to 3 kHz.

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DIFAR/DICASS mode. The noise contributed by the AN/AKT-22 to the voice channel shall be greater than 14 dB below the 100% modulation (+ 10,000 Hz deviation) of that channel when measured in a bandwidth of 300 Hz to 3 kHz.

(2) Distortion. The baseband output shall contain no single harmonic of any of the four subcarrier signals or of the deviated LOFAR subcarriers that is less than 40 dB below any zero signal subcarrier amplitude.

u. Interchangeability with similar units in other systems. The modules used in the T-1295A/AKT-22(V) shall be interchangeable with similar units in the T-1295/AKT-22(V) and the T-1220B/AKT-22(V).

3.5.1.2 Acoustic voltage-controlled oscillators. The acoustic voltage controlled oscillators shall meet the following requirements:

a. Function. The four LOFAR acoustic voltage-controlled oscillators shall be nonrepairable solid-state plug-in modules that convert filtered acoustic signals, received from a corresponding low-pass filter module, into frequency-modulated sine wave restored subcarrier signals for frequency division multiplexing.

b. Input Power. 28 ± 4 VDC at 14mA. maximum.

c. Input signal voltage (Modulating Voltage). ±2.5 volts peak-to-peak at 0 to 2000 Hz for full-scale deviation.

d. Operating frequency assignments and deviation bandwidth.

Design Center Frequency	Design Deviation Bandwidth
12.5 kHz	± 2 kHz
20.8 kHz	± 2 kHz
45.8 kHz	± 2 kHz
54.2 kHz	± 2 kHz

e. Deviation polarity. A positive going input voltage shall result in an increase in output frequency.

f. Harmonic distortion. Less than one percent of any frequency in pass-band.

g. Linearity. Deviation from best straight line shall be less than ± 0.125% of design bandwidth.

h. Amplitude modulation. Less than 10 percent.

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j. Temperature stability. Center frequency shall vary less than ± 2.0 percent of design bandwidth over the temperature range of -20°C to +55°C.

k. Input impedance. 400 kilohms minimum shunted by 5.0 pF maximum for all channels.

m. Controls. The voltage-controlled oscillators shall have the following controls:

(1)	AMPL control	Used to adjust output amplitude 0.5 Vrms to 3.1 Vrms.
(2)	SENS control (gain)	Used to adjust deviation bandwidth.
(3)	FREQ control (balance)	Used to set (zero) operating center frequency.

n. Test point. The voltage controlled oscillator shall have an output test point to monitor its FM subcarrier output signal.

p. Form factor. The dimensions of the voltage controlled oscillators shall be as illustrated on Figure 3.

3.5.1.2.1 70 kHz DIFAR/DICASS voltage controlled oscillator. The 70 kHz voltage controlled oscillator shall meet the requirements of 3.5.1.2 with the following exceptions:

a. Function. The 70 kHz voltage controlled oscillator shall be a nonrepairable solid-state plug-in module that converts DIFAR/DICASS signals, received from the DIFAR amplifier module, into frequency modulated subcarrier signals. The output signal voltage shall be a squarewave.

b. Input signal voltage (Modulating Voltage). ± 10 volts peak-to-peak at 10 Hz to 17.4 kHz for full-scale deviation.

c. Operating frequency assignment and deviation bandwidth.

Design center frequency	70.0 kHz
Design deviation bandwidth	± 28 kHz

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3.5.1.3 Acoustic low-pass filter (sonic data channels). The low-pass filters shall meet the following requirements:

a. Function. The four acoustic low-pass filters shall be nonrepairable solid-state plug-in modules that provide signal conditioning for the four LOFAR sonic data channel inputs to the multiplexer. Each shall also provide DC power, isolated from ground, to power the corresponding acoustic voltage controlled oscillator modules.

b. Input power. 28 VDC at 35mA. maximum supplied from the internal regulated power of T-1295A/AKT-22(V).

c. Input signal voltage. 16 Vrms, maximum.

d. Input impedance. The filters shall have a differential input impedance of one megohm minimum shunted by a maximum of 5.0 pF.

e. Common mode rejection. The common mode rejection of the acoustic low-pass filter differential input shall be 50 dB minimum.

f. Frequency range (bandpass). The acoustic input signal shall be filtered to a bandpass of 10 to 2000 Hz. The passband shall be flat within ± 0.5 dB between 10 Hz and 500 Hz, rolling off to -3 ± 1 dB at 2 kHz. The signal shall be down 12 dB minimum at 8 kHz.

g. Output level. 1.77 Vrms at 100 Hz with 16 Vrms input.

h. Output power. 28 + 3 VDC at 10 mA, isolated from power input ground.

j. Controls. The LOFAR low-pass filter shall have an ADJUST control to adjust the signal level of the amplifier output circuit in the low-pass filter to within ± 0.1 Vrms of specified output level.

k. Test points. The low-pass filter shall have output and output return test points.

m. Form factor. The dimensions of the LOFAR low-pass filter shall be as illustrated on Figure 4.

3.5.1.4 Voice-channel low-pass filter. The voice-channel low-pass filter shall meet the requirements specified in 3.5.1.3 with the following exceptions:

a. Function. The voice-channel low-pass filter shall be a nonrepairable solid-state plug-in module that provides signal conditioning for the voice-frequency channel input to the multiplexer. It shall also provide DC power, isolated from ground, to power the mixer-amplifier.

b. Input signal voltage. 250 millivolts rms, maximum.

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c. Frequency range (bandpass). The input shall be filtered to a bandpass of 300 Hz to 3 kHz. The passband shall be flat within + 1 dB from 300 Hz to 2 kHz rolling off to -3 + 1 dB at 3 kHz. The roll off shall be 6 dB per octave, minimum, beginning at 3 kHz.

d. Output level. The output level shall be adjustable to 80 millivolts rms for a 250 millivolts rms input signal at 1000 Hz.

e. Form factor. The dimensions of the voice low pass filter shall be as illustrated on Figure 5.

f. Output power. 28 ± 3 VDC at 10 mA, isolated from power input ground.

3.5.1.5 Mixer-amplifier. The mixer-amplifier shall meet the following requirements:

a. Function. The mixer-amplifier shall be a nonrepairable solid-state plug-in module that amplifies the summed outputs of the four voltage-controlled oscillators, the voice-frequency low-pass filter and the trigger-filter to provide an FM-multiplexed output signal to the transmitter. The mixer-amplifier shall provide sufficient amplification and gain control to drive the transmitter through the isolation transformer assembly.

b. Input power. Isolated 28 ± 3 VDC at 10mA maximum from the voice low-pass filter.

c. Input signal voltage. 2 Vrms, maximum.

d. Frequency range (bandpass). The frequency response shall be flat from 300 Hz to 250 kHz within ± 0.25 dB.

e. Voltage gain. 20 minimum, adjustable to 2.5.

f. Input impedance. 100 kilohms, minimum, shunted by 20 pF, maximum.

g. Output impedance. 50 Ohms, maximum.

h. Amplitude linearity. ± 0.5 percent from best straight line.

j. Distortion. The total distortion contributed by the mixer-amplifier shall be not greater than 0.5 percent.

k. Controls. The mixer-amplifier shall have a multiturn GAIN control for adjusting the amplitude of the output signal.

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m. Test points. The mixer-amplifier shall have input and output test points.

IN - Used to monitor the composite input signal.
OUT - Used to monitor the composite output signal.

n. Form factor. The dimensions of the mixer-amplifier shall be as illustrated on Figure 6.

3.5.1.6 Trigger-filter. The trigger-filter shall meet the following requirements:

a. Function. The trigger-filter shall be an IMA repairable plug-in module that shall provide passive filtering for up to four sonobuoy trigger signals.

b. Passband. 26 to 38 kHz ± 1.5 dB.

c. Stopband. 20 dB down, below 22.8 kHz and above 43.8 kHz.

d. Input impedance. Each of the four identical inputs shall have an input impedance of 10 kilohm, nominal.

e. Input signal voltage. The input signal voltage to each of the four inputs will be 1 ± 0.1 Vrms.

f. Output impedance. 10 kilohm, nominal.

g. Output signal voltage. Adjustable up to 50 mVrms for an input signal of one Vrms.

h. Controls. The trigger-filter shall have a gain control to provide adjustment from zero to 50 mV with one Volt input.

j. Form factor. The dimensions of the trigger-filter shall be as illustrated on Figure 7.

3.5.1.7 FM transmitter, 20 channel. The 20-channel FM transmitter shall meet the following requirements:

3.5.1.7.1 Overall. The 20-channel FM transmitter shall be a factory repairable encapsulated, solid-state, crystal controlled unit consisting of two separable units; an RF modulator (see 3.5.1.7.2) and a frequency synthesizer (see 3.5.1.7.3). The transmitter design shall be based on a phase-locked loop with the RF modulator oscillator operating at the output frequency. The frequency synthesizer shall produce a reference signal to which the RF modulator oscillator can be locked. The oscillator output shall be amplified to the required output level. A direct 70 MHz input interface shall be provided.

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a. Function. The transmitter shall be a solid-state, 20-channel selectable transmitter operating at S-band between 2200 and 2290 MHz. The transmitter shall be capable of synthesizing any of the ninety assigned IRIG channels at S-band. Selection of twenty of these IRIG channels shall be made by the programming of a Read Only Memory (ROM) in the transmitter. Channel selection shall be accomplished by a remote control panel.

b. Input power. 28 ± 4 VDC at 2.7 A maximum. Internal DC polarity protection shall be provided.

c. Modulation. True FM.

d. Modulation response. DC to 1 MHz ± 1.0 dB.

e. Modulation input impedance. 10,000 Ohm resistive minimum, shunted by 30 pF maximum.

f. Deviation sensitivity. ± 125 kHz ± 10 percent for 2.8 volts peak-to-peak input.

g. Carrier deviation ± 1 MHz maximum.

h. Linearity. ± 1 percent of best straight line for deviation up to one MHz.

j. Overshoot and droop. Two percent maximum, measured at ± 300 kHz deviation.

k. Intermodulation distortion. Any two tones up to one MHz in frequency with a combined deviation of one MHz shall not cause intermodulation exceeding -28 dB.

m. Total harmonic distortion. Two percent maximum for any tone frequency up to one MHz at peak deviation of one MHz.

n. Incidental FM. Less than 10 kHz peak-to-peak.

p. Incidental AM. Less than five percent.

q. Modulation return grounding. Connected to case internally.

r. Output power. 6.3 Watts ± 1 dB.

s. Output impedance. 50 Ohms nominal.

t. Load VSWR. 2.0:1 maximum (any angle).

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u. Operating frequency. Any 20 of the 90 assigned IRIG channels from 2200.5 MHz to 2289.5 MHz in one MHz steps may be utilized; however, this unit shall be programmed for the frequencies listed in Table II. Preset channel selection shall be performed by a factory programmed Read Only Memory in the frequency synthesizer.

v. Remote control channel selection. The transmitter shall be prewired at the factory to tune to the desired channels with the binary, five wire code input specified in Table II.

w. Frequency stability. ± 0.003 percent of assigned frequency at ambient; ± 0.005 percent over environments.

x. Form factor. The dimensions of the overall transmitter shall be as illustrated on Figure 8.

3.5.1.7.2 RF Modulator. The RF modulator shall be a solid-state unit designed for use with the frequency synthesizer to generate 20 factory preset IRIG S-band channels. The unit shall be capable of being driven from any VHF source in the 106.525 to 110.975 MHz range that has an output of 20 mW ± 1 dB with 50 Ohms output impedance.

a. Power requirements. +28 + 4 VDC at 2.2 A maximum. Internal polarity protection shall be provided.

b. Modulation. True FM.

c. RF output power. 6.3 Watts ± 1 dB.

d. Output impedance. 50 Ohms nominal.

e. Load VSWR. 1.2:1 maximum, (any angle).

f. Operating frequency. Any frequency from 2200.5 to 2289.5 MHz as determined by the reference frequency generated by the frequency synthesizer.

g. Frequency stability. ± 0.005 percent of the assigned frequency plus or minus the frequency stability of the Vhf reference frequency.

h. Direct 70 MHz input. Input level per MIL-STD-188 (+1 dbm into 75 Ohm impedance).

j. Form factor. The dimensions of the RF modulator shall be as illustrated on Figure 9.

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3.5.1.7.3 Frequency synthesizer. The UHF frequency synthesizer shall be a solid-state unit designed for use with the RF modulator to generate 20 factory preset IRIG S-band channels. Channel selection shall be accomplished by an external control which produces a five bit command code.

a. Power requirements. 28 ± 4 VDC at 600 ± 60 milliamperes.

b. Output frequencies. 20 preselected channels in 200 kHz increments within the 106.525 to 110.975 MHz range. Preset channel selection shall be performed by a factory programmed ROM. The ROM shall be programmed such that the five wire binary input code specified in Table II produces the output signals which command the synthesizer to generate the associated reference frequency.

c. Output power. 15.8 milliwatts minimum into 50 Ohms.

d. Spurious signal level. Harmonics -15 dB; nonharmonics -80 dB.

e. Frequency stability. ± 0.002 percent of selected frequency.

f. Form factor. The dimensions of the frequency synthesizer shall be as illustrated on Figure 10.

3.5.1.8 Interstage isolator. The interstage isolator shall meet the following requirements:

a. Function. The interstage isolator shall be a nonrepairable unit that consists of a terminated coaxial circulator, permitting signals to pass in the forward direction, but absorbing them in the reverse direction.

b. RF power handling capabilities. 10 Watts minimum forward and reverse.

c. Reverse isolation. 20 dB minimum over 0°C to +70°C; 17 dB minimum over -20°C to 0°C and +70°C to +90°C.

d. Frequency range (passband). 2200 to 2300 MHz.

e. Input and output impedance. 50 Ohms.

f. Voltage standing wave ratio. 1.25:1 maximum over 0°C to 70°C; 1.35:1 maximum over -20°C to 0°C and +70°C to +90°C.

g. Forward insertion loss. 0.3 dB maximum.

h. Form factor. The dimensions of the isolator shall be as illustrated on Figure 11.

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3.5.1.9 RF power amplifier. The RF power amplifier shall meet the following requirements:

a. Function. The RF power amplifier shall be a factory-repairable unit that provides additional RF amplification of the transmitter output signal. The power amplifier shall contain an integral output low-pass filter.

b. Operating voltage. 24 ± 1.0 VDC at 4.5 A maximum.

c. Input/output impedance. 50 Ohms nominal.

d. Input/output frequency range. 2200 to 2290 MHz.

e. Input VSWR. 1.7:1 maximum.

f. Input RF power. 6.0 Watts, nominal.

g. Output RF power. 22 Watts minimum at transmitter-multiplexer output, at an operating baseplate temperature of -20°C to +75°C.

h. Output VSWR. 1.5:1 at any phase angle.

j. Phase shift and delay. No discontinuities over the operating frequency range.

k. Form factor. The dimensions of the power amplifier shall be as illustrated on Figure 12.

3.5.1.9.1 Filter, low-pass S-band. The low pass filter shall meet the following requirements:

a. Operating frequency range. 2200 to 2300 MHz.

b. Harmonic frequency rejection. Second and third harmonic rejection shall be 40 dB minimum.

c. Insertion loss. 0.3 dB, maximum, over the operating frequency range.

d. Input impedance. 50 Ohms nominal.

e. Output impedance. 50 Ohms nominal.

f. VSWR. 1.25:1 maximum.

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3.5.1.10 Isolator/power monitor. The isolator/power monitor shall meet the following requirements:

a. Function. The isolator/power monitor shall be a nonrepairable encapsulated unit that provides transmitter output and external load isolation and shall have integral forward and reverse power monitoring capabilities.

b. RF power handling capability. 40 Watts, minimum, forward and reflected power.

c. Reverse isolation. 20 dB, minimum.

d. Frequency range (passband). 2200 to 2300 MHz.

e. Termination impedance. 50 Ohms.

f. Voltage standing wave ratio. 1.25:1 maximum.

g. Forward insertion loss. 0.3 dB maximum.

h. Power monitoring outputs. 25 ± 2.5 microamperes at 20 Watts forward or reverse, into a 50 microampere meter having an impedance of 20 kilohms. Monitor output terminals shall be isolated from case ground.

j. Form factor. The dimensions of the isolator shall be as illustrated on Figure 13.

3.5.1.11 Telemetry mount. The telemetry mount shall meet the following requirements:

a. Function. The telemetry mount shall be an IMA repairable assembly that provides mounting facilities and interconnecting wiring for the plug-in modules and the transformer assembly. It shall be wired and have provisions for nine low-pass filters, eight voltage controlled oscillators, a 70 kHz VCO, DIFAR mode select module and a mixer-amplifier. The isolation output transformer assembly shall be mounted and wired on the chassis. Signal, power and control connections shall be made through connectors J21 and J22.

b. Form factor. The dimensions of the telemetry mount shall be as illustrated on Figure 14.

c. Electrical connections. The Telemetry mount shall provide electrical connections in accordance with Table III.

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d. Isolation transformer assembly. The isolation transformer assembly shall be a nonrepairable, encapsulated, chassis-mounted assembly that contains the wideband isolation transformer and attenuation network used to couple multiplexed output to the FM transmitter. The isolation transformer assembly shall meet the following requirements:

(1) Transformer ratio. 1:2 voltage step-up.

(2) Output attenuation. 2:1 resistive to produce nominal unity gain overall circuit.

(3) Input voltage. 1 Vrms, maximum.

(4) Primary impedance. 2.5 kilohm.

(5) Secondary impedance. 10 kilohm.

(6) Common mode rejection. 10 megohm, minimum at DC and 40 dB, minimum at 400 Hz.

(7) Response. ± 3 dB from 250 Hz to 110 kHz.

(8) Form factor. The dimensions of the isolation transformer assembly shall be as illustrated on Figure 15.

3.5.1.12 DIFAR mode select module. The DIFAR Mode Select Module shall meet the requirements specified in 3.5.1.12.1 through 3.5.1.12.5.

3.5.1.12.1 Overall. The DIFAR Mode Select Module shall contain a 165 KHz voltage controlled oscillator and bandpass filter, a 25 kHz low-pass filter, a 70 kHz bandpass filter, a summing network/output amplifier and mode select relay.

a. Function. As installed in the T-1295A, the unit shall have two modes, OPERATE and DIFAR/DICASS. In the OPERATE Mode, the control line from the C-10742/AKT-22(V) Control Indicator shall be open and the isolation transformer output shall be connected to the transmitter modulation input. In the DIFAR/DICASS mode, the control line from the C-10742/AKT-22(V) Control Indicator shall be at +28 VDC to command the mode select to connect the DIFAR/DICASS multiplex to the transmitter modulation input.

b. Input power. 28 ± 4 VDC at 35mA. maximum.

c. Controls. The DIFAR mode select module shall have an OUTPUT LEVEL adjustment, for adjusting the output amplitude of the module, and 165 kHz AMPTD adjustment for setting the output amplitude of the 165 kHz VCO.

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d. Test points. The DIFAR mode select module shall have an OUTPUT test point for monitoring the subcarrier output signal. Applying a negative voltage (-28 ± 4 VDC) to pin one (input of the mode select module) shall disable the 165 kHz VCO.

e. Output level adjustment. The overall output shall be adjustable to provide a gain from 0.5 to 2.0 for the composite signal comprized of either of the following:

(1) 4 LOFAR and Voice

(2) 2 LOFAR (1C, 2C), 2 DIFAR and Voice

f. Output impedance. 2.5 kilohm ± five percent.

g. Form factor. The dimensions of the DIFAR mode select module shall be as illustrated on Figure 16.

3.5.1.12.2 Low pass filter section. The low pass filter section shall meet the following requirements:

a. Function. The low pass filter section shall accept the four subcarrier LOFAR multiplex plus voice from the isolation transformer output and pass only the 300 Hz to 22.8 kHz components. The trigger tones and the upper (45.8 and 54.2 kHz) subcarriers shall be removed from the filter output.

b. Input signal voltage. 3V peak-to-peak AC coupled sinewave, maximum.

c. Passband. 100 Hz to 25 kHz ± 2dB.

d. Stopband rejection. 20 dB minimum at 42 kHz; 36 dB minimum at 70 kHz

e. Driving source impedance. 3.9 kilohms ± 10 percent.

f. Insertion loss. 0 dB ± 1 dB from 100 Hz to 23 kHz.

3.5.1.12.3 Bandpass filter section. The bandpass filter section shall meet the following requirements:

a. Function. The bandpass filter section shall act as a sine wave restorer of a VCO (70 kHz ± 28 kHz) square wave. It shall attenuate the modulating signal feed through components from 5 Hz to 18 kHz. This section shall also prevent interference and cross-talk with the other multiplex sources.

b. Input signal voltage. 5.1V peak-to-peak, square wave, maximum.

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c. Passband. 42 kHz to 98 kHz ± 1 dB.

d. Stopband rejection characteristics. -20 dB minimum at 18 kHz and -30 dB minimum at 150 kHz.

e. Driving source impedance. 15 kilohm ± 10 percent.

3.5.1.12.4 165 kHz VCO and bandpass filter section. The VCO and bandpass filter section of the DIFAR Mode Select Module shall meet the following requirements:

a. Function. The voltage controlled oscillator and bandpass filter section shall provide a filtered sine wave subcarrier at 165 kHz ± 28 kHz for modulating signal of 5 Hz to 18 kHz.

b. VCO deviation sensitivity. 165 kHz ± 28 kHz peak deviation for a ± 5.0 VDC input signal.

c. VCO driving source impedance. 10 Ohms, maximum.

d. Deviation polarity. A positive going input voltage shall result in an increase in output frequency.

e. VCO disabling. The 165 kHz VCO shall not oscillate when a negative -28 ± 4 VDC voltage is applied to pin 1 (input) of the DIFAR MODE Select Module.

f. Bandpass filter passband. 137 kHz to 193 kHz ± 1 dB.

g. Bandpass filter stopband characteristics. -20 dB minimum below 98 kHz and -30 dB minimum above 490 kHz.

3.5.1.12.5 Summing network. The summing network shall provide proper impedance terminations for the three filters and sum the individual signals in their proper ratios. The summing network comprising resistors and amplifier, shall have an output impedance of 2.5 kilohms ± 5 percent.

3.5.1.12.6 Summing network amplifier gain. The summing network amplifier shall have a gain range of 0.3 to 2.5. Nominal 1.6 gain.

3.5.1.13 DIFAR amplifier adapter module. The DIFAR amplifier adapter module shall meet the following requirements:

a. Function. The DIFAR amplifier adapter module shall be an IMA repairable unit with dual, isolated input, non-inverting amplifiers to accept DIFAR signals from two types of sonobuoy receivers. It shall select the proper gain configuration to accept the different signal levels from the two receivers.

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b. Gain. Unity in ARR-52 mode; 0.325 in ARR-75 mode.

c. Input signal. The amplifier shall accept the following:

(1) ARR-75 mode. 45 Volts peak-to-peak maximum, differential.

(2) ARR-52 mode. 15 Volts peak-to-peak maximum (from ARR-52A through AM-6695/AKT-22(V), single-ended.

d. Input characteristics. Differential (ground isolated) with 200 kilohms minimum input impedance.

e. Input power. +28 ± 0.5 VDC from the transmitter-multiplexer at 15 milliamperes maximum.

f. Output. 15 Volts peak-to-peak, maximum, with specified input.

g. Frequency passband. 10 Hz to 17.4 kHz.

h. Form factor. The dimensions of the DIFAR amplifier-adapter shall be as illustrated on Figure 17.

3.5.2 Antenna, AS-3033A/AKT-22(V). The antenna shall meet the following requirements:

a. Function. The antenna shall be a nonrepairable, frangible, fuselage-mounted, retractable boom assembly containing a VHF and an S-band antenna. The antenna shall receive RF energy in the band from 162 to 174 MHz and radiate RF energy in the band from 2200 to 2290 MHz (S-band).

b. Transmitting frequency band. 2200 to 2290 MHz.

c. Receiving frequency band. 162 to 174 MHz.

d. Polarization. Vertical polarization in both transmit and receive modes when extended 90 degrees from fuselage and the antenna centerline is in the vertical axis.

e. Voltage standing wave ratio (VSWR). 1.50:1 for transmit band, 2.00:1 for receive band.

f. Gain. Five dB minimum for transmit band; zero dB minimum for receive band.

g. Coupling. Signal coupling from the transmitting antenna to the receiving antenna shall have 70 dB rejection, minimum.

h. Transmitting power capacity. 100 Watts maximum.

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j. Transmit elevation beam width. The transmit antenna element shall have a minimum three dB elevation beam width of 20 degrees centered at the horizon.

k. Transmit azimuth radiation pattern. The transmit antenna element shall have an omnidirectional radiation pattern (within ± 1.25 dB) in the azimuth plane.

m. Phase center. The phase center of the transmitting antenna element shall be at a minimum of 36 inches from the aircraft surface when in operate position.

n. Receive antenna pattern. The receive antenna element shall have an omnidirectional pattern in the azimuth plane.

p. Aerodynamic loading. The antenna shall be capable of sustaining the following steady-state air loads:

(1) Vertical, negligible

(2) Lateral, fore and aft - 64 pounds at the tip, 112 pounds at the center of pressure.

q. Frangibility. The antenna shall fracture in the extended position at a point no greater than 14 inches from the center of the pivot bearing when subjected to the following loads:

(1) Vertical, 1500 pounds

(2) Lateral, fore and aft - 256 pounds at the tip.

r. Weight. The weight of the antenna shall be not greater than 10 pounds.

s. Electrical connections. The antenna shall have the following electrical connectors:

Reference Designation	Receptacle Type	Function
J1	M39012/ 02-0004 Type N	Input to transmit element
J3	M39012/ 29-0001 Type TNC	Output from receive element

t. Form factor. The dimensions of the antenna shall be as illustrated on Figure 18.

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3.5.2.1 Antenna, AS-3033( )/AKT-22(V). The antenna shall meet the following requirements when installed in SH-2F aircraft:

a. Function. The antenna shall be a repairable, frangible, fuselage-mounted retractable boom assembly containing a VHF and an S-band antenna. The antenna shall receive RF energy in the band from 136 to 174 MHz with an instantaneous bandwidth of ± 1 dB and radiate RF energy in the band from 2200 to 2290 MHz (S-band) with an instantaneous bandwidth of ± 1dB.

b. Operating modes. The antenna shall be capable of simultaneous transmit and receive modes of operation.

c. Transmit mode selection. Selection of either the omnidirectional or steered mode of transmission shall be possible. Method of selection shall be via a 28 VDC discrete input. Presence of +28V shall enable directional mode and absence of +28V shall enable omnidirectional.

d. Operating frequency band. The operating frequency bands shall be as follows:

(1) Transmit - 2200 to 2290 MHz instantaneous bandwidth ±1dB

(2) Receive - 136 to 174 MHz instantaneous bandwidth ±1dB

e. Polarization. Vertical polarization in both transmit and receive modes when extended shall be 90 degrees from aircraft fuselage bottom.

f. Voltage standing wave ratio (VSWR). The VSWR shall be 1.50:1 for overall transmit band and 2.00:1 for overall receive band.

g. Coupling. The signal coupling from the transmitting antenna port to the receiving antenna port shall be less than 70 dB.

h. Transmitting power capacity. The transmitting power capacity shall be 100 watts.

j. Transmit elevation beam width. The transmit antenna element shall have a minimum 3 dB elevation beam width of 20 degrees centered at the horizon (perpendicular to the antenna plane) in both the omni-directional and steered modes.

k. Transmit aximuth radiation pattern. The transmit azimuth radiation pattern shall be as follows:

(1) Omnidirectional mode - The transmit antenna shall have an omni-directional radiation pattern (within ± 1.25 dB) in the azimuth plane.

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(2) Steered mode - The main beam shall be not less than 30° or more than 60°. The sidelobes shall be a minimum of 23 dB below main lobe. The center of main beam shall be within 3° of perpendicular of directional antenna.

m. Gain. The antenna gain shall be as follows:

(1) Transmit - Omnidirectional mode: +5 dBi minimum
Steered mode: +13 dBi minimum

(2) Receive - 0 dBi minimum

n. Phase center. The phase center of the directional transmitting antenna elements shall be at a minimum of 36 inches from the aircraft surface when in the operate position. The phase center of the omnidirectional transmitting elements shall be a minimum of 24 inches from the aircraft surface when in the operate position.

p. Antenna steering. When operating in the directional (steered) mode, the antenna shall accept 3 wire synchro data and a 400 Hz reference input for control signals. Input signals for the antenna will indicate the desired pointing direction relative to the aircraft heading. As measured perpendicular to the directional antenna, it shall point within 3° of synchro signal input from a calibrated source.

q. Fault discrete output. When +28 VDC is applied to the antenna to select the directional mode, the fault discrete output shall be 0V when the antenna is pointing correctly in response to input synchro point signals. When the pointing synchro malfunctions and is not pointing correctly in response to the input synchro pointing signals, the fault discrete output shall be +28 VDC capable of driving a #313 external fault lamp.. The lamp shall not illuminate when the servo is slewing within the aircraft turning rates but may light when the unit initially selects directionall until the antenna is pointing correctly.

r. Receiving antenna pattern. The receive antenna element shall have an omnidirectional pattern in the azimuth plane.

s. Aerodynamic loading. The antenna shall be capable of sustaining the following steady-state air loads:

(1) Vertical - negligible

(2) Lateral fore and aft - 64 pounds at tip, 112 pounds at the mid point.

t. Frangibility. It is the intent that the antenna structure fail without structual damage to the aircraft should a landing be required with the antenna partially or fully extended. The antenna shall

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fracture in the extended position at a point no greater than 14 inches from the center of the pivot bearing when subjected to the following loads:

(1) Vertical - 1500 pounds maximum, (TBD) pounds minimum

(2) Lateral, fore and aft - 256 pounds at the tip maximum, (TBD) pounds minimum

u. Weight. The antenna weight shall be not greater than 14 pounds.

v. Drag. The drag of the installed antenna shall be not greater than 40 pounds at speeds of 150 MPH.

w. Electrical connections. The antenna shall have the following electrical connectors:

Ref Des.	Receptable Type	Function
J1	M39012/02-0004 Type N	Input to transmit element
J2	TBD	Pin Connections 1 - ground and 28 VDC return 2 - +28 VDC power & directional antenna select 3 - 115V 400 Hz phase A (R1) 4 - 115V 400 Hz neutral (R2) 5 - 400 Hz synchro direction (S1) 6 - 400 Hz synchro direction (S2) 7 - 400 Hz synchro direction (S3) 8 - fault discrete 9, 10 - spares
J3	M39012/29-0001 Type TNC	Output from receive element

x. Form Factor.

(1) Overall - The overall size of the antenna shall be a minimum required to meet the mechanical and electrical requirements contained herein.

(2) Clevis mount - The form factor of the clevis mount shall be compatible with the AS-3033/AKT-22(V).

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y. Maintainability. The antenna shall be designed to be readily repairable for all moving parts or electronic circuitry. It shall be constructed in a modular manner to enable limited intermediate maintenance activities (IMA) subassembly replacement and depot repair of the antenna sub-assemblies and most antenna failures. Factory repair shall be for major repair and damage repair after being franged (see 3.5.2.1t). Major assemblies deemed replaceable shall be the entire directional sphere, the servo drive mechanical assembly, the servo electronic assembly, the coaxial relay, coaxial slip-ring, the planar directional radiator, and the spherical radome parts.

z. Electromagnetic interference control. The generation of electromagnetic interference by the equipment and the vulnerability of the equipment to EMI shall meet the requirements and limits of MIL-STD-461. The tests shall be performed in accordance with MIL-STD-462. The following tests shall be performed:

a. CE01	e. CS02
b. CE03	f. CS06
c. CE06	g. RS01

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3.5.3 Actuator, electromechanical linear, TG-229A/AKT-22(V). The actuator shall meet the following requirements:

a. Function. The actuator shall be an IMA repairable electromechanical assembly utilized to remotely extend and retract the antenna. It shall contain a retractor mechanism, a reversible motor with a magnetically operated brake, two limit switches and position indicating circuitry.

b. Operating loads. Normal operating load 270 pounds, tension or compression; maximum operating load 540 pounds, tension or compression.

c. Operating stroke. The operating stroke shall be as illustrated on Figure 19.

d. Operating speed. 0.20 inches per second ± 20 percent at normal operating load with 26 VDC applied at +70°F.

e. Static load. Ultimate static load 3780 pounds, tension or compression. Limit static load 1800 pounds, tension or compression.

f. Operating life. 10,000 full stroke cycles at rated load and duty cycle.

g. Duty cycle. One minute ON; five minutes OFF at rated load.

h. Voltage requirement. Normal rated voltage 26 VDC. Unit shall operate over the voltage range of 24 to 29 VDC.

j. Current requirement. Normal operating current 2.5 Amp. Maximum operating current 4.5 Amp.

k. Limit switches. Actuator shall contain internally adjustable limit switches set to limit stroke in accordance with Figure 19.

m. Mechanical stops. Actuator shall contain internal non-jamming mechanical stops to limit overtravel in the event of a limit switch malfunction.

n. Overtravel. Actuator overtravel shall be held to a minimum by means of a separate motor brake.

p. Direction of operation. Actuator shall be electrically reversible.

q. Reversibility. Actuator shall not reverse as a result of application of external loads up to the maximum static load combined with vibration and shock.

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r. Endplay. Actuator endplay shall be not greater than .010 inch TIR when measured under a 10 pound reversing load.

s. Anti-rotation device. Actuator shall contain an anti-rotation device to prevent rotation of extending RAM.

t. Lubrication. Actuator shall be lubricated for rated life at time of manufacture.

u. Weight. The weight of the actuator shall be not greater than four pounds.

v. Form factor. The dimension of the actuator shall be as illustrated on Figure 19.

3.5.4 Control-indicator C-10742/AKT-22(V). The control-indicator shall meet the following requirements:

a. Function. The control-indicator shall be an IMA repairable, self-contained, separately housed unit that serves as the central operating and monitoring unit in the transmitting system. The control-indicator shall perform the following functions: panel lighting, operating mode control, voice control, antenna control, RF power monitoring and sonar trigger channel selection.

b. Form factor. The dimensions of the control-indicator shall be as illustrated on Figure 20.

c. Weight. The weight of the control-indicator shall be not greater than three pounds.

d. Contents. The control-indicator shall contain the following circuits:

(1) Panel lighting. Panel illumination shall be edge lit by three panel indicators which shall be powered by 28 Volts nominal. The edge lit panel shall be in accordance with MIL-P-7788, Class 1-R, Type III.

(2) Operating mode control. A power off mode, standby mode, operate mode, and DIFAR/DICASS mode shall be provided. In the standby and operate modes 28 VDC shall be applied to transmitter-multiplexer to energize the power supply circuits. In the operate mode 28 VDC shall be applied to the transmitter-multiplexer to operate a relay and energize the transmitter. In the DIFAR/DICASS mode, 28 VDC shall be applied to the transmitter-multiplexer to energize the power supply circuits, operate a relay and energize the transmitter, and provide DIFAR/DICASS and/or LOFAR information to the transmitter input.

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(3) Sonar trigger channel selection. The sonar trigger channel selection switches shall control the routing of the receiver channel outputs to the multiplexer.

(4) Transmitter power monitoring. The transmitter power monitoring circuits shall allow both reflected and forward RF power outputs of the transmitter/multiplexer to be monitored on a power meter when the transmitter/multiplexer is in the operate mode or the DIFAR/DICASS mode. A minimum acceptable power indication shall be indicated at the center of the meter.

(5) Voice control. The voice control circuit shall interconnect the aircraft Interphone Communications System (ICS) to enable the operator to send voice transmissions to the shipboard receiving system.

(6) Antenna actuation control. The antenna actuation control circuit shall enable the antenna boom to be extended or retracted and indicate its position.

e. Controls. The control-indicator shall have the following controls:

(1) ANTENNA UP-DN Switch	Controls actuation of Antenna AS-3033A/AKT-22(V) AKT-22(V). In the UP position, the switch routes +28 VDC to the corresponding control elements in antenna actuator circuit to retract boom antenna. In the DN position, the switch routes +28 VDC to the corresponding control elements in antenna actuator circuit to extend antenna. (Note that a wheel interlock circuit in the aircraft can be used to disable the operation of this switch when the aircraft wheels are down.)
(2) MIKE Pushbutton Switch	When pressed, activates aircraft ICS key and routes ICS voice inputs to transmitter-multiplexer for transmission to shipboard telemetry receiving system. (ICS voice transmissions are routed only to aircraft intercom system when pushbutton is in normal position.)
(3) REFL PWR Pushbutton Switch	When pressed and held, applies reflected RF power output indication from isolator AT1 to RF power monitor meter M1. In normal position forward RF power is indicated on the meter located on the Control Indicator.

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(4) TRIGGER A, B, C and D switches — "UP" position, routes sonar trigger inputs from receiver (A, B, C, and D) standard output to the transmitter-multiplexer. "DOWN" position effectively short circuit the corresponding multiplexer inputs to ground.

(5) OFF-STBY-OPR - DIFAR/DICASS — A ganged, four-position selector switch that controls the turn off, turn on, and operation of the transmitter-multiplexer. In the OFF position, the transmitter-multiplexer shall be de-energized. In the STBY position, a +28 VDC standby command shall be applied to the transmitter-multiplexer power control circuit. In the OPR position, a +28 VDC operate command shall be applied to the transmitter-multiplexer power control circuit to energize the transmitter-multiplexer. In the DIFAR/DICASS position, a +28 VDC operate command shall be applied to the transmitter multiplexer power control circuit and the DIFAR mode control line, allowing DIFAR/DICASS mode to be activated.

f. Electrical connection. Connections to external circuits shall be in accordance with Table IV.

3.5.5 Control Generator (C-10743/AKT-22(V)). The control generator (CG) shall meet the requirements of 3.5.5.1 through 3.5.5.4.

3.5.5.1 Overall unit. The control generator shall contain a 12 bit Microprocessor, Intersil type IM-6100 or equal, Random Access Memory (RAM), Programmable Read Only Memory (PROM), a 12 bit bidirectional data bus, and input/output hardware, including Peripheral Interface Elements (PIE) Intersil type IM-6101 or equal, as necessary to meet the system requirements specified herein. The overall control generator shall meet the following requirements:

a. Function. The function of the control generator shall be to provide the appropriate DICASS commands, under operator control, to one of two deployed AN/SSQ-62 DICASS sonobuoys through the transmitter section of either of two AN/ARC-159(V)1 transceivers. Operation shall be accomplished through selective addressing and individual commands or signals to the AN/SSQ-62 sonobuoy in accordance with MIL-S-81895. The unit shall allow the operator to choose either manual or automatic pinging, to control the duration of the ping tone, and in the case of automatic pinging, to select the ping repetition rate. Command capabilities of the control generator shall also allow the airborne operator to lower the depth of the hydrophone on either or both sonobuoys to medium or deep deployment or to scuttle either or both sonobuoys. Control Processor Unit (CPU) flow charts are included as Figure 37.

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b. Timing and control. The control generator shall generate all timing and control signals required to fully operate one or two DICASS sonobuoys. The unit shall exercise remote control of two AN/ARC-159(V)1 transceivers designated as UHF1 and UHF2 and acquire all necessary channel data from two AN/ARR-75 receivers designated as receiver C and receiver D. The timing of the control generator output shall be in accordance with MIL-R-81877 and MIL-S-81895. Every time a ping tone is transmitted, a trigger tone shall be sent to the T-1295A/AKT-22(V) for the same length of time. Trigger tone frequencies shall be in accordance with the following:

Sonar Channel (DICASS Sonobuoy Ping Frequency)	Trigger Tone Frequency
A	26.0 kHz
B	28.4 kHz
C	30.8 kHz
D	33.2 kHz
E (FM↑)	35.6 kHz
F (FM↓)	38.0 kHz

c. Manual Operation. A DICASS buoy shall be pinged every time the corresponding spring-loaded switch is momentarily held in the PING position and released while the MNL indicator is illuminated. The length of the ping tone shall be as indicated by the illuminated section of the 0.1/0.5/1.0 pushbutton (Figure 21).

d. Automatic Operation.

(1) CW Mode. In the CW mode a DICASS buoy shall be pinged automatically when the corresponding DICASS buoy switch is placed to the ON position and the 8 or 16 section of the MNL/8/16 switch is illuminated. When only one buoy is ON, the interval between ping tones shall be 8 or 16 seconds, as indicated by the setting of the MNL/8/16 pushbutton. When both buoys are ON, both buoys shall ping alternately at the rate indicated. The interval between commands shall be evenly spaced; i.e., half of the indicated value (4 or 8 seconds as opposed to 8 or 16 seconds). The interval between ping tones is from the trailing edge of one command tone to the leading edge of the next.

(2) FM Mode. In the FM mode, the command signal generator shall produce a sliding (Sweep) ping tone signal for each of the sonobuoy ping tones. This shall allow for sweeping up or down. When sweeping up the sweep shall start 200 Hz ± 1 Hz below the ping tone center frequency and end one second later 200 Hz ± 1Hz above the center frequency. When sweeping down the sweep shall start 200 Hz ± 1 Hz above the center frequency and end one second later 200 Hz ± 1 Hz below the ping tone center frequency.

e. Depth control. The depth function shall permit either medium or deep deployment of the DICASS sonobuoy hydrophone. The CSG shall generate tones as specified in MIL-S-81895A.

f. Self-test. The Control Generator shall have a Built-In Test (BIT) routine to verify system operational parameters and to generate operator confidence that the CG is able to perform within its design specifications. The BIT shall be automatically initiated whenever the ON/OFF switch is set to on. The BIT shall also be able to be manually initiated by the operator any time the BIT indicator is lit green by pressing the BIT/BUSY switch. The BIT shall include the following tests:

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(1) A sum check of the Read Only Memory (ROM),

(2) A write, read and compare test of the Random Access Memory (RAM).

(3) A check of the 40 Hz real time interrupt hardware to ensure that an interrupt occurs every 25 msec.

(4) Lamp and lamp driver tests.

If the control generator passes all tests, the BIT green light shall be illuminated. If the equipment fails any test, except a lamp test, the BIT amber light shall flash. The BIT circuitry need not have a means of detecting a faulty lamp or lamp driver. The BIT lamp test routine shall be aborted by pressing any switch while the routine is in progress. A switch activation during the routine shall perform no other function at that time.

g. Maintenance Test Mode. The CG shall possess the capability of being placed in a test mode wherein, the front panel keys may be used to generate steady state operating conditions. The maintenance test mode shall be entered by shorting together two test points on the A2 module designated E1 and E2. When operating in the test mode, the front panel key functions shall be in accordance with the following:

(1) MNL/8/16 Switch/Indicator. Used to preset a code to load synthesizers. Pressing the indicator switch shall select one of three nominal frequencies or the null frequency in accordance with the following:

MNL = Frequency 1 (6050 Hz)
8 = Frequency 2 (12100 Hz)
16 = Frequency 3 (27600 Hz)
No Indication = Null Frequency (0 Hz)

(2) UHF 1/UHF 2 Switch/Indicator. Pressing this switch shall select Synthesizer 1, Synthesizer 2, or both synthesizers in accordance with the following:

UHF 1 (Amber) = Synthesizer 1
UHF 2 (Amber) = Synthesizer 2
UHF 1 and UHF 2 = Synthesizer 1 and 2

Whenever UHF 2 is amber only transmitter number two can be energized.

(3) DEPTH Switch/Indicator. Pressing this switch shall load the selected synthesizer(s) with the selected frequency. The DEPTH switch shall light momentarily to confirm the loading.

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(4) ARM C Switch/Indicator. This switch shall select Transmitter 1 or Transmitter 2 in accordance with the following:

ARM C (Amber) = Transmitter 1
ARM C (Green) = Transmitter 2
ARM C (Off) = Not applicable

(5) ARM D Switch/Indicator. This switch shall control the state of the selected transmitter in accordance with the following schedule:

ARM D (Off) = REMOTE and ON/OFF lines deactivated
ARM D (Green) = REMOTE and ON/OFF lines activated
ARM D (Amber) = Transmitter keyed

(6) BUSY Indicator. This indicator shall display the state of the key sense line of the selected transmitter in accordance with the following:

BUSY (On) = Keyed
BUSY (Off) = Not keyed

(7) 0.1/0.5/1.0 Switch/Indicator. This switch shall select the steady state tones which are available at the CG output for observation as appropriate test points in accordance with the following:

1.0 = Address
0.5 = Command
0.1 = Ping and Trigger
No Indication = Null (Inactive state)

(8) SCU Switch/Indicator. Pressing this switch shall cancel previously commanded tests and reset all test conditions. The switch shall light momentarily to confirm the key selection.

(9) BIT Switch/Indicator. Pressing this switch shall execute a routine of tests on the CPU (A3) circuit board. The state of the BIT indicator is interpreted according to the following schedule.

BIT (Green) = Tests in Progress
BIT (Amber) = Detailed Test Report
BIT (Off) = CG Ready for next Command

When the BIT indicator is amber, the results shall be available according to the following (green = pass, amber = fail):

ARM C = ROM Test
DEPTH = RAM Test
ARM D = INTERRUPT Test

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(10) DICASS Buoy Switch "C".

(a). Operating this switch to the "ON" position will cause exit from switch test mode. Note that switch test mode cannot be reinitiated by setting this switch back to the "OFF" position. The CSG must be powered down and then powered up with the DICASS Buoy switch in the "OFF" position to re-enter the switch test mode.

(b). Operating this switch to the PING position initiates the following action:

1 During the first five seconds, the command generator synthesizers will generate the address tones associated with the sonobuoy channel coding which is being provided to the command generator from either receiver C or D depending on which ping switch is activated (in the normal maintenance situation, channel coding is to be provided by the sonobuoy tuning switches on the SM-819/UKM-2B test set). The 1.0 lamp will be lighted amber.

2 During the second five seconds, the ping and trigger tones will be generated and the 0.1 lamp will be amber.

(11) DICASS Buoy Switch "D". Same as above except for paragraph (10)(a).

(12) LOCAL/REMOTE Switch. Pressing this switch enables the Remote I/O port. The remote function and this I/O port are reserved for future use.

h. Form factor. The dimensions of the control generator shall be as illustrated on Figure 21.

j. Weight. The weight of the control generator shall be not greater than eight pounds.

k. Input power. 115 VAC, 400 Hz, single phase, 1.2 A maximum.

m. Contents. The control generator shall consist of the following major assemblies; organized as illustrated on Figure 22.

(1) Module A1 - Front Panel Interface Assembly A1 (See 3.5.5.3.1)

(2) Module A2 - Dual Synthesizer Assembly A2 (See 3.5.5.3.2)

(3) Module A3 - Microprocessor Assembly A3 (See 3.5.5.3.3)

(4) Module A4 - I/O Interface Assembly A4 (See 3.5.5.3.4)

(5) Module A5 - Power Supply Assembly A5 (See 3.5.5.3.5)

(6) Module A6 - FM slide assembly A6 (See 3.5.5.3.6)

(7) Module A7 - Flexible Cable and Rear Mounted Interface Connectors A7 (See 3.5.5.2.3)

(8) Board A8 - Module Interconnect Board A8 (See 3.5.5.2.2)

(9) Front Panel A9 - Front Panel Assembly A9 (See 3.5.5.2.1)

(10) Heat Sink A10 - Power Supply Heat Sink A10 (See 3.5.5.2.4)

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3.5.5.2 Chassis assembly. The control generator chassis assembly shall contain the power supply heatsink and transformer, rear-mounted interface connectors J1 through J5, wiring harness, the front panel assembly, and a module interconnect printed wiring board with five connector socket assemblies to accept printed circuit modules A1 through A5. A sixth connector socket assembly shall be provided on the module interconnect printed wiring board for future expansion.

3.5.5.2.1 Front panel assembly A9. The front panel assembly shall comprise the integrally lighted panel, switches, indicator lamps, illumination circuitry, circuitry to delay activation of 28 VDC keylines during the CG power-up and circuitry to maintain the 5V supply for 50 msec should input power drop out.

a. Form factor. The dimensions of the front panel shall be as illustrated on Figure 21. The front panel shall be in accordance with MIL-P-7788, Class I-R, Type V.

b. Controls. The front panel assembly shall have the following controls, arranged as illustrated on Figure 21 and wired as illustrated on Figure 24 and Table X. All lamps shall operate with amber indicating selected, green indicating available, and blank indicating not available, unless otherwise specified.

(1) ON/OFF switch. Combination toggle switch and circuit breaker which shall apply 115 volt, 400 Hz primary power to the equipment and +28 VDC to the transceiver keyline relays through the delay module.

(2) LOCAL/REMOTE switch/indicator. The remote function (switch/indicator) shall be available for future use. When the CG is powered on the local light shall be lit.

(3) MNL/8/16 switch/indicator. Selects the sonobuoy rate of ping. When the equipment is first turned on, the system shall be in the mode for manual ping operation by means of DICASS BUOY C or DICASS BUOY D switch, and the MNL amber light shall be lit only. Pressing the switch one time shall change the mode to one ping every 16 seconds and light the 16 amber light only. Pressing the switch a second time shall change the mode to one ping every 8 seconds and light the 8 amber light only. Pressing the switch a third time shall return the system to manual ping operation and relight the MNL amber light only.

(4) 0.1/0.5/1.0 switch/indicator. Selects duration of ping. When the equipment is first turned on, the duration of the ping shall be 1.0 second, and the 1.0 amber light shall be lit only. Pressing the switch one time shall change the ping duration to 0.5 second and light the 0.5 amber light only. Pressing the

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switch a second time shall change the ping duration to 0.1 second and light the 0.1 amber light only. Pressing the switch a third time shall return the ping duration to 1.0 second and relight the 1.0 amber light only.

(5) UHF 1/UHF 2 switch/indicator. This switch shall select one of the two AN/ARC-159(V)1 transceivers for transmission of command and ping signals to sonobuoys. When the equipment is first turned on, AN/ARC-159(V)1 No. 2 shall be selected and the UHF 2 amber light shall be lit. The UHF 1 green light shall also be lit if AN/ARC-159(V)1 No. 1 is available for transmission. If AN/ARC-159(V)1 No. 1 is busy, no UHF 1 lights shall be lit. Pressing the switch one time shall select AN/ARC-159(V)1 No. 1, extinguish the UHF 2 amber light, and light the UHF 1 amber light. The UHF 2 green light shall also be lit if AN/ARC-159(V)1 No. 2 is available for transmission. If AN/ARC-159(V)1 No. 2 is busy, no UHF 2 lights shall be lit. Pressing the switch the second time shall reselect AN/ARC-159(V)1 No. 2.

(6) SCU Depth M/D Switch/Indicator. This switch shall activate the hydrophone depth commands or the sonobuoy scuttle command. All segments of the switch shall light green upon application of power.

(a) Depth M/D Switch/Indicator. This switch shall activate the sonobuoy hydrophone depth command. When the equipment is first turned on, the Depth M/D shall light green indicating that the depth command can be activated. Pressing the switch shall activate the M depth command, light the Depth M amber light, light the Arm C and Arm D green lights, and extinguish the SCU and Depth D green lights. When the M Depth amber light is lit the M depth command shall be able to be transmitted to one sonobuoy or the other by pressing either the Arm C or Arm D switch. If the Arm C or Arm D switches are not pressed, pressing the switch a second time shall deactivate the M Depth command, activate the D Depth command, light the D Depth amber light and extinguish the M Depth amber light. The Arm C and Arm D switches shall light green. When the D Depth amber light is lit, the D Depth command shall be able to be transmitted to one sonobuoy or the other by pressing either the Arm C or Arm D switch. If the Arm C or Arm D switches are not pressed, pressing the switch a third time shall deactivate the D Depth command, activate the sonobuoy scuttle command, light the SCU amber light, light the Arm C and Arm D green light and extinguish the M/D Depth lights.

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(b) SCU Switch/Indicator. This switch shall activate the sonobuoy scuttle command. When the equipment is first turned on, the SCU green light shall be lit indicating that the scuttle command can be activated. Pressing the SCU switch three times shall activate the scuttle command, light the SCU amber light, light the Arm C and Arm D green lights and extinguish the Depth M/D green lights. When the SCU amber light is lit, the scuttle command shall be able to be transmitted to one sonobuoy or the other by pressing either the Arm C or Arm D switch. If the Arm C or Arm D switches are not pressed, pressing the SCU switch a fourth time shall deactivate the scuttle command, relight the SCU and Depth M/D green lights and extinguish both arm lights.

(7) ARM C switch/indicator. This switch shall transmit the selected hydrophone depth or sonobuoy scuttle command to sonobuoy C. Pressing either the DEPTH or SCU switch selects the command and lights the ARM C green light. Pressing the ARM C switch shall transmit the command to sonobuoy C and light the ARM C amber light while simultaneously extinguishing the ARM D light. The ARM C light shall be extinguished after the command is sent. If the ARM C switch is pressed while the selected transceiver is busy the indicator shall remain green. When the transceiver becomes available the indicator must again be pressed to activate the command.

(8) ARM D switch/indicator. This switch shall transmit the selected hydrophone depth or sonobuoy scuttle command to sonobuoy D. The ARM D command shall operate in the same manner as ARM C switch and indicator.

(9) BIT/BUSY switch/indicator. This switch shall initiate the self-test, indicate test failures, and indicate when a selected AN/ARC-159(V)1 is being used. When the equipment is first turned on a self-test shall be executed. If the self-test fails the BIT amber light shall blink. If the self-test passes the BIT green light shall be lit indicating that the self-test can again be executed. Pressing the BIT/BUSY switch shall initiate the self-test and light the BIT amber light. The busy indicator will light when an external transceiver is keyed.

(10) DICASS C&D Buoy switches. Shall be toggle switches with three positions, one momentary, which control the pinging of DICASS sonobuoys. In the ON position the selected sonobuoy (C or D) shall be automatically pinged when the MNL/8/16 switch has been used to select either 8-second or 16-second ping intervals between the end of one ping to the beginning of the next ping. In the OFF position the sonobuoy shall not be pinged. When the MNL/8/16 switch has been used to select manual pinging mode the momentary function of the toggle shall be used to manually ping the sonobuoy. When automatic pinging is selected for both

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sonobuoys, (both DICASS BUOY switches are in the ON position) and the MNL/8/16 switch has been used to select either 8-second or 16-second ping intervals, each sonobuoy shall be pinged alternately. If, for example, a 16-second ping interval is selected, sonobuoy C shall be pinged every 16 seconds, and sonobuoy D shall be pinged every 16 seconds. Since each sonobuoy shall be pinged alternately, the interval between the ping command of sonobuoy C and the ping command of sonobuoy D shall be 8 seconds. The 8 and 16 second ping intervals are from the trailing edge of one ping to the leading edge of the next ping.

(11) TRANSMIT indicator. This indicator shall be a legend imbedded in the edge-lit front panel which shall light amber whenever one of the AN/ARC-159(V)1 transceivers is keyed by the control generator. The light shall remain on for the duration of transmission. The TRANSMIT indicator shall be illuminated separately from the remainder of panel and shall be aviation yellow color. Brightness shall be 60 foot lamberts minimum.

(12) CW/FM↑FM↑C and D switches. The CW segment of these switches shall light amber upon application of power. This shall indicate that a ping tone of constant frequency will be transmitted whenever a ping tone is selected (either manual or automatic ping tone operation). When the switch is pressed (once) the FM segment shall light amber and shall produce a positive sloping frequency (200 Hz below to 200 Hz above center frequency) ping tone output. When the switch is pressed again (2nd time) the FM segment shall light amber and shall produce a negative sloping frequency (200 Hz above to 200 Hz below) center frequency ping tone output. When the switch is pressed again (3rd time) the CW segment shall light amber.

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c. Illuminated switches. The front panel assembly shall contain nine illuminated pushbutton switches which shall meet the following requirements:

(1) Illuminating characters shall be hidden before lighted. Black background characters shall be visible in direct sunlight when lighted. All legends shall be non-legible when the system is off. The solid lines shall be white translucent non-lighted.

(2) Illumination color and legend for the switches shall be as illustrated on Figure 25.

(3) The mechanical dimensions of switches shall be as illustrated on Figure 25.

(4) Electrical

Switch Contacts: Gold

Switch Rating:

	Sea Level	50,000 Feet	Maximum Contact Resistance Ohms
Max. Voltage	28 VDC	28 VDC	.025
Max. Current	1 Amp Res. 5 Amp Ind.	1 Amp Res. 5 Amp Ind.	.025
Min. Current	5 Microamperes	5 Microamperes	.50
Min. Voltage	15 Millivolts	15 Millivolts	.50

Contact Resistance: 25 milliohms maximum in accordance with MIL-S-22885 for switch contacts and one Ohm maximum in accordance with MIL-S-22885 for lamp contacts.

Electrical Life: 25,000 cycles at rated loads in accordance with MIL-S-22885.

Lamp Type: T-1 sub-midget flange base with special pushbutton modifications (24 mA) in accordance with MS3338.

(5) Mechanical

(a) Acutation Force: Unsealed 1.5 ± .5 pounds (standard).

(b) Actuation Travel: .062 ± .025 inches.

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(c)	Pushbutton Extraction Force:	one to four pounds.
(d)	Mechanical Life:	100,000 cycles minimum.
(e)	Weight:	1.3 ounces maximum (one pole switch with standard pushbutton).

(6) Environmental

(a)	Temperature:	(Low) -55°C (High) +85°C.
(b)	Explosion:	In accordance with MIL-STD-810, Method 511.
(c)	Humidity	In accordance with MIL-STD-202, Method 106.
(d)	Shock:	75 G's, 6 milliseconds, half sine.
(e)	Vibration:	10-2000 Hz @ 0.06 DA or 15G (peak) whichever is less. Basic Switch contacts in accordance with MIL-S-8805/101.

d. Toggle switches. The DICASS C&D Buoy switches shall be toggle switches in accordance with MS24655.

e. Circuit breaker/switch. The ON/OFF switch shall be a circuit breaker in accordance with M39019/02-0317.

f. Delay board A9A1. The front panel shall contain circuitry to delay the 28 VDC power within the CG and prevent unwanted activation of AN/ARC-159(V)1 key lines during CG power up. The 28 VDC power activation shall be delayed 280 to 425 ms.

g. Power boost board A9A2. The front panel shall contain circuitry to maintain the 5 VDC logic supply for 50 ms ± 5 ms in the event of a momentary input power drop-out.

3.5.5.2.2 Module interconnect printed wiring board A8. The module interconnect printed wiring board A8 shall meet the following requirements:

a. Function. The module interconnect printed wiring board shall be a non-repairable assembly that provides interconnecting wiring between the printed circuit modules, front panel assembly, power supply, rear mounted interface connectors and terminals for connection to transistors on the heat sink.

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b. Connectors. The module interconnect printed wiring board shall have the following connectors:

Connector Designation	Connector Type	Connector Function
A8J1	AMP 3-87215-3	rear mounted connector wiring
A8J2	Airborne WTB66SED9J	front panel assembly wiring
XA1	Airborne WGE92SED9GY	front panel interface module
XA2	Airborne WGE92SED9GY	dual synthesizer module
XA3	Airborne WGE92SED9GY	CPU module
XA4	Airborne WGE92SED9GY	I/O interface module
XA5	Airborne WGE92SED9GY	power supply module
XA6	Airborne WGE92SED9GY	future expansion
E1-E18	Cambion 140-1785-03-01	power supply

c. Material. The module interconnect printed wiring board shall be constructed of thin laminate copper clad sheet TYPE TFL-GF per MIL-P-55617.

d. Interconnections. The module interconnect printed wiring board shall provide the interconnections necessary for all modules to be powered and to receive data from, and transmit data to, each other, to the front panel assembly and to the rear mounted interface connectors. The interconnections shall be as illustrated on Figure 26.

e. Form factor. The form factor of the module interconnect printed wiring board shall be as illustrated on Figure 27.

3.5.5.2.3 Rear mounted interface connectors A7. The C-10743 shall have the following external interface connectors:

Reference Designation	Receptacle Type	Function
J1	MS3110E14-5P	Power (400 Hz & Panel Lighting)
J2	MS3110E16-26S	AN/ARC-159(V)1 Interface
J3	MS3110E14-15S	AN/ARR-75 Interface/28 VDC Power

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Reference Designation	Receptacle Type	Function
J4	MS3111OE10-6S	T-1295A/AKT-22(V) Interface
J5	MS3110E14-19P	Remote Input and Status Output

Connections to external circuits shall be in accordance with Table V.

3.5.5.2.4 Power supply heat sink A10. The power supply heat sink shall be used to provide proper cooling to a diode bridge rectifier and three series pass transistors. The form factor of the power supply heat sink shall be as illustrated on Figure 28.

3.5.5.2.5 Transformer. The line transformer shall step down the 115 Volt, 400 Hz, primary power input to the AC voltages required by the low-voltage power supply. The dimensions shall be as illustrated on Figure 29.

a. Input voltage. The input voltage shall be single phase 115 VAC, 400 Hz in accordance with MIL-STD-704.

b. Output voltages and current rating. The voltage and current ratings of the secondary windings shall be as specified herein with an input of 115 VAC ± 1V, 400 Hz ± 1 Hz.

Full Load Output Voltage	Full Load Output Current
10.0 Vrms ± 5 percent	7.5 A rms
30.0 Vrms ± 5 percent	0.25 A rms
27.0 Vrms ± 5 percent	2.0 A rms

c. Load regulation. The no-load to full-load voltage regulation shall be five percent or better.

d. Efficiency. The total output power from the four secondary outputs at full-load shall be not less than 90 percent of the primary input power.

e. Electrostatic shielding. The electrostatic shield shall provide a minimum voltage ratio of five to one at 20 KHz when tested in accordance with MIL-T-27.

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f. Dielectric withstanding voltage. The dielectric withstanding voltage (at sea level) between the winding and the core shall be:

(1) Primary Windings 600 Vrms

(2) Secondary Windings 300 Vrms

g. Weight. The weight of the line transformer shall be not greater than 19.0 ounces.

3.5.5.3 ELectronic circuit modules. The electronics of the control generator shall be contained in five plug-in printed circuit card modules installed vertically in the control generator chassis. Sidewall mounted guard rails shall hold the cards in place and ensure their correct insertion. Keyed PC connectors shall prevent accidental insertion of the incorrect board.

a. Form factor. The form factor of the electronic circuit modules shall be as illustrated on Figure 30.

b. Data bus. Data transfer to and from the CPU module (see Figure 22) shall be conducted via a 12 bit bidirectional data bus (see Figure 23). Data latches shall be used to gate data to and from the data bus and from devices other than the P.I.E's. Data latches shall be controlled by the CPU via the P.I.E's. (see 3.5.5.1). The data bus shall be in accordance with the following:

(1) Signal Levels -
Logic High = 3.5V minimum
Logic Low = 0.4V maximum

(2) Drive Capacity -
Source 160 mA minimum
Sink 1350 mA minimum

c. Input/output (I/O) control. All input/output operations of the CG shall be controlled by the CPU via the P.I.E's. Hardware codes associated with the I/O functions shall be in accordance with Table VI.

d. Test points. Each circuit module shall have test points for operational checkout, maintenance testing, and troubleshooting. Test points on each board shall be in accordance with Table VII.

e. Signal definitions. Signal definitions for each circuit module shall be in accordance with Table VIII.

f. Circuit module function. The boards and their respective electronics shall be in accordance with the following.

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(1) Module A1. This module (Front panel interface board) shall be located closest to the front of the chassis and shall provide the control generator front panel interface functions with the CPU (module A3). Circuitry shall also be contained which illuminates the control annunciators on the front panel, and selects between local and remote operations (see Figure 31). Should remote operating be provide at some future date, the functions of the A1 module would be commanded via the remote input/output connections. These functions are not defined on the input/output connections at this time.

(a) Input power. 5 VDC at .1A max

(b) Lamp drivers. Drivers shall be used to interface between the circuitry of module A1 and the annunciator lamps of the front panel.

(c) Front panel switch codes. The Binary code used to indicate front panel switch closures shall be in accordance with Table IX.

(d) Electrical connections. Connections to external circuits shall be in accordance with Table XI.

(2) Module A2. This module (Dual synthesizer board) shall contain two identical frequency synthesizers which shall generate audio frequencies for sonobuoy command tones, trigger tones and address signals (see Figure 32).

(a) Input power.
+5 VDC at .5 A max
12 VDC at 50 mA max
-12 VDC at 50 mA max

(b) Frequency synthesis. Frequency synthesis shall be based upon a phase lock loop design with a reference frequency of 12.87 kHz supplied from the A3 module. The output frequency of the synthesizers shall be controllable over the range of 6.050 kHz to 38.0 kHz minimum. The data codes are generated by addressing a PROM look-up table to obtain the desired output frequency and shall be of the binary form:

$$X\ X\ B\ A\ N_7\ N_6\ N_5\ N_4\ N_3\ N_2\ N_1\ N_0$$

where

$$F_{OUT} = \frac{12.8 * 10^3\ (257 - N)}{16*2\ EXP\ [A+(B*2)+2]}$$ and X = don't care

If both A and B are set to one the output frequency shall be equal to the input frequency at FMIN/16.

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(c) Frequency stability. Stability of the synthesizer output frequency shall be in accordance with MIL-S-81895.

(d) Digital to analog conversion. Digital to analog conversion shall be employed to convert the squarewave output of the frequency synthesizers to a stepped sinewave. The amplitude of the sinewave shall be 10 Vpp ± 1.0 Vpp tolerance.

(e) Electrical connections. Connections to external circuits shall be in accordance with Table XI.

(3) Module A3. The A3 module (CPU board) shall contain the microprocessor or Central Processing Unit (CPU). The CPU shall process data within the CG and provide program control, timing and data for module A2 (frequency synthesizer), and I/O functions. Data flow to and from the microprocessor shall be via bidirectional bus drivers under microprocessor control (see Figures 33 and 34).

(a) Input power. 5 VDC at 0.5 A max

(b) Random Access Memory. Random Access Memory (RAM) shall be employed as scratch pad data memory for the CPU. Total RAM shall be 256 X 12 (three 256 X 4 devices).

(c) Read Only Memory. Fixed software for the CPU shall be stored in Erasable-Programmable-Read-Only Memory (ROM). Total ROM shall be 4K X 12 (two 4K X 8 devices).

(d) Crystal oscillator. A 1.28 MHz crystal oscillator shall provide the master time base for the CPU. The oscillator output shall have a 50 percent duty cycle and frequency stability of .01 percent. The oscillator shall be divided down to generate the 12.8 kHz 20 percent duty cycle reference for the frequency synthesizers.

(e) Electrical connections. Connections to external circuits shall be in accordance with Table X.

(4) Module A4. The I/O interface assembly shall provide the circuitry necessary to receive channel information from two AN/ARR-75 receivers, control two AN/ARC-159(V)1 transceivers, and to send sonobuoy related signals to the AN/ARC-159(V)1's and T-1295A/AKT-22(V) (see Figure 35).

(a) Input power.

+5 VDC at 75 mA max
+12 VDC at 75 mA max
-12 VDC at 75 mA max
+28 VDC at 100 mA max

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(b) ARR-75 interface. Module A4 shall provide signal conditioning and interface circuitry between module A3 and the ARR-75 channel data. Data latches used in the ARR-75 interface shall be controlled by the A2 board P.I.E. via the status of the 'ARR-75 CH' line command. Data codes from the ARR-75 for each of the sonobuoy channels will be a five bit binary code in accordance with Table XII.

(c) ARC-159(V)1 interface. Circuitry shall be provided to interface the CG to the AN/ARC-159(V)1. The interface circuitry shall accept differential clock data from the selected UHF radio and monitor the state of UHF key lines. The interface circuitry shall be capable of controlling selection of the proper UHF transceiver and exercising the following functions, in accordance with MIL-R-81877.

1 ON/OFF

2 Remote Take Control

3 Key

4 Differential Data

Circuitry shall be provided to determine the status of the UHF transceiver key lines. The key line status shall be available at the key sense outputs in the form of two D.C. levels, one to indicate BUSY and another to indicate NOT BUSY.

(d) AN/ARC-159(V)1 audio interface. Circuitry shall be provided to control the signal flow of the synthesizer outputs from module A2 to the AN/ARC-159(V)1. Signal flow and summing shall be controlled by analog switches. The audio interface shall be in accordance with MIL-R-81877.

(e) T-1295A interface. The A4 module shall provide a trigger tone signal of 610 + 120 mVRMS to the T-1295A/AKT-22(V) simultaneously with ping tone transmissions to the selected AN/ARC-159(V)1. (See 3.5.5.1.b.)

(f) Electrical connections. Connections to external circuits shall be in accordance with Table XI.

(5) Module A5. This module (Power Supply Board) shall provide filtering and regulation of the +5 VDC, +12 VDC and -12 VDC power to the control generator internal circuitry (see Figure 36).

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(a) Output power

1 +5 ± .3 VDC at 2 Amps minimum regulated
2 +12 ± 0.7 VDC at 0.20 Amps minimum regulated
3 -12 ± 0.7 VDC at 0.20 Amps minimum regulated

(b) Indicator dimming

1 Input voltage: 0 to 28 VDC from aircraft
2 Output voltages:
  a 0 to 4.8 VDC circuit regulated but variable for the transmit indicator
  b 7 to 28 VDC regulated but variable for lamp indicators

(c) Edge lighting. Input power - 0 to 28 VDC from aircraft dimming bus 0.6 A maximum.

(d) External connections. Connections to external circuits shall be in accordance with Table XI.

(6) Module A6. The A6 module (FM Sweep) shall be a microprocessor-controlled synthesizer that provides FM ping frequency sweep generation capability for use against specific target countermeasures (See Figure 38).

(a) Input power requirements

1 +5 VDC 325 MA (Max)
2 Digital Ground
3 +12 VDC 10 MA (Max)
4 -12 VDC 10 MA (Max)
5 Analog Gound

(b) Data inputs. The following existing inputs shall be applied to the FM sweep printed circuit card (See Figure 39).

1 12.8 kHz clock (Pin 4)
2 12 bit data bus (Pins 14 through 23, 26, 30)
3 PIE PORT 5
  a Flag 2 (Pin 9)
  b Flag 1 (Pin 34)
  c Sense 2 (Pin 38)
4 PIE PORT 6
  a Flag 4 (Pin 27)
  b Write 2 (Pin 55)
5 Reset (Pin 32)

(c) Data outputs. The following outputs from the FM sweep printed circuit card shall be provided (See Figure 39).

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1 Lamp Drivers
- a Buoy C CW (Pin 86)
- b Buoy C FM (Pin 87)
- c Buoy C FM (Pin 88)
- d Buoy D CW (Pin 89)
- e Buoy D FM (Pin 90)
- f Buoy D FM (Pin 91)

2 Synthesizer Inputs
- a FM in 2 (Pin 65)
- b FM in 1 (Pin 64)

(d) Test Points. All test points which interface to CMOS logic shall be buffered through TTL logic circuitry. Sufficient test points shall be incorporated on the upper edge of the printed circuit board to permit signal measurement for testing and fault isolation.

(e) FM modulation. The FM modulation capability shall include provisions for the manual selection of the present CW ping mode or FM-Slide up-or-down modes. Independent control of each of the two selected DICASS sonobuoys shall possible to permit a mix of CW, FM-Up or FM-Down modes to be selected. In addition to the generation of the selected modulations, the timing signals which are multiplexed with the echo signals to be sent to the acoustic processor shall be supplemented by anticipate signals alerting the operator/processor of the nature of the next ping. Thus just before a ping is generated by the C-10743( )/AKT-22(V), the operator or processor is specifically alerted if the selected ping modulation is to be FM-Up or FM-Down. Refer to Figure 40 for the timing diagram.

(f) FM sweep. The FM sweep shall be a precision function consisting of 400 discrete frequency steps which are generated under microprocessor control. The shape of the FM sweep function shall be determined by information stored in a preprogrammed read only memory. Consequently, any linear or non-linear FM sweep shape can be implemented simply by the field replacement of a preprogrammed PROM integrated circuit. The firmware shall be organized into a 400-step look-up to avoid mathematical restrictions. This enables an arbitrary profile to be generated. Two sweep profiles shall be provided. These are: 1) a linear sweep up and 2) a linear sweep down. The FM sweep shall be a 1.0 second duration signal. During this period, exactly 400 discrete frequency steps shall be generated by the microprocessor. When FM up is desired, a sliding tone

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shall start 200 ± 1 Hz below the DICASS channel center frequency (6.5, 7.5, 8.5, or 9.5 kHz) and end one second later 200 ± Hz above the selected center frequency. Similarly, when FM sweep down is desired, the tone shall start 200 Hz ± 1 Hz above the DICASS channel center frequency and end one second later at 200 Hz ± 1 Hz below the center frequency. Transitions to successive one Hz separated frequency levels shall be programmed at equal 0.0025 second intervals, thus creating a sliding tone which is a linear variation of frequency with respect to time, as illustrated in Figure 41.

3.5.5.4 Antenna control panel C-XXXXX/AKT-22(V). The antenna control panel shall enable the entering of the helicopter to ship bearing by the senso and steer the AS-3033( )/AKT-22(V) antenna when installed in SH-2F aircraft.

a. Inputs. The inputs shall be as follows:

(1) 28 VDC 5A

(2) Steering and reference synchro signals

(3) Control discrete

(a) Directional mode

(b) Omnidirectional mode

b. Outputs. The output shall be a fault discrete.

c. Controls. As a minimum, the antenna control panel shall have the following controls.

(1) Directional/Omnidirectional switch

(2) Directional control positioner

d. Indicator. The antenna control panel shall have a fault discrete indicator.

e. Total Weight. The total weight of the antenna control panel shall be not greater than three pounds.

f. Form Factor. The dimensions of the antenna control panel shall not be greater than 4.0 inches long, 5.75 inches wide and 3.5 inches high.

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3.5.5.5 Software. The control generator shall contain software modules which shall be programmed to accomplish the functional requirements of 3.5.5. The following functions shall be included: (see Figure 37).

a. Initialization/Power Up

b. Built-In-Test

c. Keyboard Polling

d. Input/Output Handlers

e. Key Entry Processing

f. Intermediate and Depot Level Test

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the govenment tc acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are clasified as follows:

a. First Article Inspections (see 4.3)
b. Initial Production Tests (see 4.4)
c. Acceptance Tests (see 4.5)
d. Life Tests (see 4.6)

4.3 First article inspections. When specified in the contract, first article inspection shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract (see 6.3). First article inspections shall be accomplished under the approved test procedure of 4.7.

4.3.1 First article inspection data. The contractor shall make available test data collected in conducting these tests for procuring activity review (see 6.2.2).

4.3.2 Scope of tests. First article inspections shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article inspections shall include environmental tests in accordance with MIL-T-5422, vibration levels in accordance with MIL-E-5400 curve IIb (2g) for the electronic equipment and curve IIIb (5g) for the antenna and actuator. Interference tests shall be in accordance with MIL-STD-461, and reliability qualification tests in accordance with 4.5.3.

4.3.3 First article approval. Approval of the first article sample will be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first

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article sample. Prefabrication of production equipment prior to the approval of the first article sample is at the contractor's own risk. The approved first article sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article sample shall not be considered one of the equipments under the contract.

4.3.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each equipment shall be capable of succsssfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection and, for equipment already accepted by the Government, it shall be the obligation of the contactor to make necessary corrections as approved by the procuring activity.

4.4 Initial production tests. When specified in the contract, one of the first ten production equipments will be selected and sent at the contractor's expense, to a designated Government laboratory for tests (see 6.2.1.1). This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the initial production inspections. The first article sample shall not be selected for this test.

4.4.1 Scope of tests. This equipment may be subjected to any and all tests the procuring acitvity deems necessary to assure that the production sample in design, construction, workmanship, performance and quality meets all requirements of this specification.

4.4.2 Accessory Material. In addition to the complete equipment submitted for initial production inspections, the contractor shall also prepare such necessary material and include data necessary to test the equipment (See 6.2.2).

4.4.3 Initial production sample approval. Approval of the Initial Production Sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be performed on equipment previously accepted when requested by the procuring activity.

4.5 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance inspections. All inspection and testing shall be subject to monitoring and witnessing by the government inspector. The contractor shall make available test reports showing quantitative results for all acceptance tests (See 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory. Acceptance tests shall consist of the following:

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a. Individual Tests (see 4.5.1)
b. Sampling Tests (see 4.5.2)
c. Reliability Assurance Tests (see 4.5.3)
d. Special Tests (see 4.5.4)

4.5.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Run in Test

4.5.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics, and record adequate data, to assure specified equipment operation.

4.5.1.3 Manufacturing run in test. Each equipment shall be operated under the conditions specified below for a minimum period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

a. Temperature	Ambient room
b. Humidity	Ambient room
c. Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant frequencies) and a minimum amplitude of $\pm$ 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of five minutes prior to the beginning of the ten hour period of operation. The equipment shall be operated during this vibration period for the purpose of detecting manufacturing defects. Operation within the specified limits of specified performance is not necessarily required during the vibration period. The direction of vibration shall be vertical to the normal mounting plane. During the ten hour period of operation following the five minute vibration period, the equipment shall be cycled every 2.5 hours through its

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various phases of operation. Should a failure occur, and a failure analysis warrants it, the equipment shall be repaired and the test started over, except that the vibration period need not be repeated when it is certain the failure was not a result of the vibration test. Should repetitive failures occur, corrective action shall be taken to eliminate recurrence of failure from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.5.2 Sampling tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered For Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

One out of the first ten need not be selected and tested if initial production tests are conducted. Sampling Tests are not required when Reliability Assurance Tests are conducted.

4.5.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. The environmental conditions for operational test of each equipment should be based on results of the preproduction, initial production, individual and special tests.

c. Manufacturing run in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.5.3 Reliability assurance tests. Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074. Equipment selected for reliability assurance tests shall have first passed the Individual Test. Failure analysis reports shall be provided for all failures except those occuring during burn-in. Corrective action shall be taken to prevent the recurrence of all failures.

4.5.3.1 Qualification phase. When specified in the contract, (see 6.2.1), and prior to the acceptance of equipments under the contract, a minimum of three equipments shall be tested as specified in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E shall be used. The Acceptance Reject Criteria for Test Plan III shall be used.

4.5.3.2 All equipment test. Each equipment produced, except those submitted for the Reliability Qualification Test, shall be tested for a minimum of 74 hours. The last 24 operating hours shall be failure free. If a failure occurs during the last 24 hour operating period the equipment shall be repaired, proper operation verified and the test continued. The accumulation of failure free time (i.e. last 24 hours) shall start at zero time upon resumption of testing. Test Level E of MIL-STD-781 shall be used.

a. Prior to the 74 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is used, the details thereof must be included in the approved test procedures. The burn-in shall be the same for each equipment with respect to time and severity.

b. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not including burn-in failures or burn-in operating time) shall be totalled and the results compared with the reject line of Test Plan II of MIL-STD-781 (Extend the line as necessary to accommmodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be made available to the procuring activity (see 6.2.2).

c. At any time that the current totals of test hours and test failures plotted on Test Plan II curve shows a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

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4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria and preventive maintenance to be allowed during the test shall be part of the test procedures to be made available and found acceptable to the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.5.4 Special tests. When specified in the contract, (see 6.2.1.1), special tests shall be conducted for purpose of checking the effect of any Class 1 design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On a piece of equipment which had to be modified because of an engineering or material change made after the equipment was manufactured.

b. Whenever failure reports or other information indicate that additional tests are required, as determined by the procuring activity.

4.5.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for procuring activity acceptance prior to conducting the special tests.

4.5.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of the failure.

b. Determine if the failure is an isolated case or design effect.

c. Make available to the procuring activity for acceptance, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test to check all equipment for this requirement until assurance is obtained that the defect has been corrected.

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4.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. A life test procedure and log shall be prepared to record performance of the test with pertinent data sheets applicable thereto. At a minimum the log shall contain:

a. Time and date of start of life test.

b. Elapsed time meter reading at start of test.

c. Elapsed time meter reading at end of test.

d. Time and date of operating data recording.

e. Failure reports where applicable.

f. Serial number and part number of equipment under test.

The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. Equipments which have successfully passed the Initial Production Inspection, Sampling Tests, Reliability Tests, or special tests may be selected for Life Tests. Test Life accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the following:

Quantity of Equipments Offered For Acceptance	Quantity to be Selected For Sampling Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the following simulated service conditions:

a. Temperature	Room ambient
b. Altitude	Normal ground (0-5000 ft.)
c. Humidity	Room ambient
d. AC Voltage	115 ± 5 Volts (at lowest applicable frequency)
e. DC Voltage	27.5 ± 2.0 Volts

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4.6.2 Test periods. The test may run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, as specified in approved life test procedure, the equipment shall be turned on and off several times and cycled through its various modes of operation.

4.6.3 Performance check. At approximately 24-hour intervals during the test a limited performance check shall be made. The performance check proposed by the contractor shall be acceptable to the procuring activity.

4.6.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, a failure report shall be initiated and the defective part shall be replaced and the operation resumed for the balance of test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

A copy of failure reports shall be included with the life test log.

4.6.4.1 Failure report. In the event of a failure, the Government inspector shall be notified as soon as possible. A report shall be made available to the procuring activity upon completion of test (See 6.2.2). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.7 Test procedures. The contractor shall develop procedures to be used for conducting pre-production tests, acceptance tests, and life tests (See 6.2.2). The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to insure compliance with requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of procedures, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

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4.8 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The AN/AKT-22(V)6 Transmitting Set, Telemetric Data, is intended for use as an airborne system to control and condition ASW information gathered on board an airborne platform and relay that information via an FM/FM radio link to surface vessels for data processing (See Figure 1).

6.2 Ordering data.

6.2.1 Acquisition Requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Equipment title and type designation.

c. Selection of applicable levels of packaging or packing.

6.2.1.1 Test and inspection options. These options, if exercised, must be stated as required in the contract.

a. First article inspections (see 4.3).

b. Initial production tests (see 4.4).

c. Qualification phase tests (see 4.5.3.1).

d. Special tests (see 4.5.4).

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6.2.2 Data Requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

a. Applicable for all equipments.

Paragraph Number	Data Requirements Title	Applicable DID	Option
3.2.1	Non-Standard Parts, Approval Requests	DI-MISC-80071	
3.3.2	Reliability Program Plan	DI-R-7079	
3.3.4	Requirements, Rack or Console Mounted Control Drawing	UDI-E-21339	
3.3.8	Request, Assignment of Nomenclature	UDI-E-21582	
3.3.8	Request, Confirmation of Nomenclature	UDI-E-21583	
3.3.8	Request, Approval of Identification Plate Drawings	UDI-E-21584	
3.3.8	Request, Serial Number Assignment	UDI-E-21585	
3.3.8	Request, Assignment of Type Equipment Codes	UDI-E-21586	
3.3.8	Request, Confirmation of Type Equipment Codes	UDI-E-21587	
4.5 and 4.5.3.2.b	Reports, First Article, (Preproduction) Testing	UDI-T-21349	
4.6.4.1	Report, Failure	DI-R-21598	
4.7	Test Procedures	UDI-T-21347	

When approved test procedures from previous contract are available, they may be considered applicable without further procuring activity approval.

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b. Additional data if first article inspection required.

Paragraph Number	Data Requirements Title	Applicable DID	Option
4.3	Report, First Article (Preproduction) Testing	UDI-T-21349	

c. Additional data if initial production tests required.

Paragraph Number	Data Requirements Title	Applicable DID	Option
4.4	Data, Design-Equipment Submitted for Testing	UDI-E-21338	

d. Additional data if qualification phase tests required.

Paragraph Number	Data Requirements Title	Applicable DID	Option
4.7	Procedures, Reliability Tests	DI-R-7035	
4.5.3	Report, Reliability, Test and Demonstration	DI-R-7034	

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First Article. When a first article inspection is required, the item should be a first production item as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3.1 Standard Sample. Standard sample requirements should be in accordance with 4.3.3, first article approval, of this specification.

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6.4 Associated Equipment. The equipment covered by this specification should operate with the following associated equipment:

a. Transceiver AN/ARC-159(V)1
b. Radio Receiving Set AN/ARR-75
c. Intercommunications System AN/AIC-14
d. Telemetric Data Receiving Set AN/SKR-4A
e. Sonobuoy AN/SSQ-41
f. Sonobuoy AN/SSQ-47
g. Sonobuoy AN/SSQ-53
h. Sonobuoy AN/SSQ-57
j. Sonobuoy AN/SSQ-62
k. Test Set, Telemetric Data AN/UKM-2B
m. Test Set, Telemetric Data AN/UKM-3A

6.5 Subject Term (Key Word) Listing.

Aerodynamic loading
Command Signal Generator (CSG)
Control Generator (CG)
Data item description
DICASS
DIFAR
Distribution statement
First article
Form factor
Frangibility
LOFAR
Microprocessor (CPU)
Order of precendence
Parallel Interface Element (P.I.E.)
Phase lock loop
Power boost
Programmable Read Only Memory (PROM)
Random Access Memory (RAM)
Voltage Controlled Oscillator (VCO)

Preparing Activity
NAVY-AS
(Project 5821-N- 175)

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TABLE I. Transmitter - Multiplexer T-1295A Connector Requirements

CONNECTOR	PIN	FUNCTION
J1 (UG-30/U) (MATES WITH UG-21/U)	PIN SHELL	RF SIGNAL INPUT GROUND
J2(BENDIX PT07A-16-23P) (MATES WITH MS3116P-16-26S)	A B C D E F G H J K L M N P R S T U V W X Y Z a b c	CHAN A INPUT CHAN A RETURN CHAN B INPUT CHAN B RETURN CHAN C INPUT CHAN C RETURN CHAN D INPUT CHAN D RETURN TRIG A INPUT TRIG A RETURN TRIG B INPUT TRIG B RETURN TRIG C INPUT TRIG C RETURN TRIG D INPUT TRIG D RETURN ICS INPUT ICS RETURN DIFAR AII INPUT DIFAR AII RETURN DIFAR B INPUT DIFAR B RETURN +28, RETURN } NOT USED CG INPUT COMPOSITE MUX OUTPUT
J3 (BENDIX PT07A-16-23P) (MATES WITH MS3116P-16-23S)	A B C Z D E F	PHASE A, PHASE B, PHASE C, NEUTRAL — 115/200 VAC 400 Hz POWER NOT USED DIFAR MODE CONTROL SPARE

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TABLE I. Transmitter - Multiplexer T-1295A Connector Requirements (Continued)

CONNECTOR	PIN	FUNCTION
J3 (BENDIX PT07A-16-23P) (MATES WITH MS3116P-16-23S)	G	MODE CONTROL RETURN
	H	STANDBY MODE CONTROL
	J	OPERATE MODE CONTROL
	K	SPARE
	L	SPARE
	M	FORWARD POWER MONITOR
	N	REFLECTED POWER MONITOR
	P	POWER MONITOR RETURN
	R	OPERATE LIGHT
	S	STANDBY LIGHT
	T	TRIGGER INPUT (NOT USED)
	U	TRIGGER RETURN (NOT USED)
	V	+28V INPUT FOR CHAN C AND D
	W	SPARE
	X	SPARE
	Y	SPARE
J4 (BENDIX PT07A-12-10S) (MATES WITH MS3116P-12-10P)	A	CHAN SEL 2^4
	B	CHAN SEL 2^3
	C	CHAN SEL 2^2 — REMOTE
	D	CHAN SEL 2^1 — FREQUENCY
	E	CHAN SEL 2^0 — SELECTION
	F	RETURN (SYNTHESIZER GND)
	G	SPARE
	H	SPARE
	J	SPARE
	K	SPARE

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TABLE II. FM Transmitter and Synthesizer Programming Code

CHANNEL NUMBER	RF FREQ(MHz)	BINARY WORD	SYNTH FREQ(MHz)	INPUT CODE 1/				
				A4	A3	A2	A1	A0
41	2212.5 (RED)	1	107.125	0	0	0	0	1
42	2216.5	2	107.325	0	0	0	1	0
43	2220.5 (GREEN)	3	107.525	0	0	0	1	1
44	2224.5	4	107.725	0	0	1	0	0
45	2228.5	5	107.925	0	0	1	0	1
46	2232.5	6	108.125	0	0	1	1	0
47	2236.5	7	108.325	0	0	1	1	1
48	2240.5 (YELLOW)	8	108.525	0	1	0	0	0
49	2244.5	9	108.725	0	1	0	0	1
50	2248.5	10	108.925	0	1	0	1	0
51	2252.5	11	109.125	0	1	0	1	1
52	2256.5 (WHITE)	12	109.325	0	1	1	0	0
53	2260.5	13	109.525	0	1	1	0	1
54	2264.5	14	109.725	0	1	1	1	0
55	2268.5	15	109.925	0	1	1	1	1
56	2272.5	16	110.125	1	0	0	0	0
57	2276.5 (BLACK)	17	110.325	1	0	0	0	1
58	2280.5	18	110.525	1	0	0	1	0
59	2284.5	19	110.725	1	0	0	1	1
60	2288.5	20	110.925	1	0	1	0	0
				ALL OTHER CODES INVALID				

1/
ONE: 3.5V ± 1.0 VDC (I=200ma, max)
ZERO: 0V ± 0.4VDC (I= 6ma, max)

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TABLE III. Telemetry Mount Connector Requirement.

Position	Associated Module	Module Reference Designator	PIN	FUNCTION
J1	LPF-1 (1C)	A1	1	DIFF Input (+)
J2	LPF-2 (2C)	A3	2	Sig & Power Isolated Return
J3	LPF-3 (5C)	A5	3	DIFF Input (-)
J4	LPF-4 (6C)	--	4	N.C.
J5-J8	Trigger Filter	A9	5	SIG Output
J9	LPF-9 (Voice)	A17	6	CHASSIS GND
			7	+28 VDC, Isolated
			8	+28 VDC, except J3 & 4
			9	28V Return
J10	Mixer-Amplifier	A20	1	Input
			2	Sig & Power Return
			3	N.C.
			4	N.C.
			5	Output
			6	CHASSIS GND
			7	+28 VDC
			8	N.C.
			9	N.C.
J11	VCO-1 (1C)	A2	1	Input
J12	VCO-2 (2C)	A4	2	Sig & Power Return
J13	VCO-3 (5C)	A6	3	N.C.
J14	VCO-4 (6C)	A8	4	N.C.
J15-J17	Trigger Filter	A9	5	Output
J18	--	--	6	CHASSIS GND
J19	DIFAR B	A19	7	+28 VDC
			8	N.C.
			9	N.C.
J20	DIFAR Mode Select Module	A18	1	DIFAR AII Input
			2	Audio Return
			3	Audio Return
			4	Band Pass Filter Input (DIFAR B)
			5	Relay Output
			6	Relay Coil Return & Chassis Gnd
			7	+28 VDC
			8	Relay Normal (N.C.) Input (LOFAR)
			9	Relay Coil Supply
J21	--	--	1	28 VDC POWER +
			2	28 VDC POWER -
			3	N.C.
			4	Chan A In
			5	Chan A Return

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TABLE III. Telemetry Mount Connector Requirement. (Continued)

Position	Associated Module	Module Reference Designator	PIN	FUNCTION
J21-Cont.	--	--	6	Chan B In
			7	Chan B Return
			8	Chan C In
			9	Chan C Return
			10	Chan D In
			11	Chan D Return
			12	TRIGGER A In
			13	TRIGGER A Return
			14	TRIGGER B In
			15	TRIGGER B Return
			16	TRIGGER C In
			17	TRIGGER C Return
			18	TRIGGER D In
			19	TRIGGER D Return
			20	ICS INPUT
			21	ICS INPUT Return
			22	DIFAR A II In
			23	DIFAR A II Return
			24	DIFAR B In
			25	DIFAR B Return
J22	--	--	A	N.C.
			B	N.C.
			C	Chan 3 & 4 ACT
			D	DIFAR CONT
			E	MOD Return
			F	N.C.
			H	N.C.

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TABLE IV. Control-Indicator C-10742/AKT-22(V) Connector Requirements

J1		J2	
PIN	FUNCTION	PIN	FUNCTION
1	RCVR A STANDARD OUTPUT	1	POWER RETURN
2	RCVR A OUTPUT RETURN	2	FORWARD POWER
3	RCVR A OUTPUT SHIELD	3	REFLECTED POWER
4	RCVR B STANDARD OUTPUT	4	TRANS (CONT)
5	RCVR B OUTPUT RETURN	5	+28 VDC (STBY)
6	RCVR B OUTPUT SHIELD	6	IND (ON)
7	RCVR C STANDARD OUTPUT	7	IND (STBY)
8	RCVR C OUTPUT RETURN	8	VOICE OUTPUT
9	RCVR C OUTPUT SHIELD	9	VOICE RETURN
10	RCVR D STANDARD OUTPUT	10	ICS INPUT
11	RCVR D OUTPUT RETURN	11	ICS RETURN
12	RCVR D OUTPUT SHIELD	12	ICS OUTPUT AIC-14
13	TRIG RCVR A	13	ICS KEY
14	TRIG RCVR A RETURN	14	+28 VDC INPUT
15	TRIG RCVR A SHIELD	15	DC GROUND
16	TRIG RCVR B	16	PANEL LIGHTS
17	TRIG RCVR B RETURN	17	UP IND ANT
18	TRIG RCVR B SHIELD	18	DOWN IND ANT
19	TRIG RCVR C	19	ANTENNA UP
20	TRIG RCVR C RETURN	20	ANTENNA DOWN
21	TRIG RCVR C SHIELD	21	ANTENNA UP (DN SIDE OF WHEEL
22	TRIG RCVR D	22	N.C. LOCK)
23	TRIG RCVR D RETURN	23	N.C.
24	TRIG RCVR D SHIELD	24	N.C.
25	N.C.	25	N.C.
26	N.C.	26	DIFAR OFF/ON
27	N.C.	27	+28 VDC (UP SIDE WHEEL LOCK)
		28	N.C.
		29	N.C.
		30	N.C.
		31	N.C.
		32	N.C.
		33	N.C.
		34	N.C.
		35	N.C.
		36	N.C.
		37	N.C.

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TABLE V. C-10743 Connector Interface.

Connector -PIN	Signal	
J1-A	ANN DIM	From 28 VDC Aircraft Light Bus in accordance with MIL-STD 704.
J1-E	ANN DIM RTN	
J1-B	115 VAC GND	From 400 Hz Aircraft Power in accordance with MIL-STD-704.
J1-C	115 VAC IN	
J1-D	115 VAC RTN	
J2-A	DIFF DATA -UHF 1	To ARC-159 in accordance with MIL-R-81877
J2-R	DIFF DATA +UHF 1	
J2-C	DIFF DATA -UHF 2	To ARC-159 in accordance with MIL-R-81877
J2-B	DIFF DATA +UHF 2	
J2-D	SPARE	
J2-Y	DIFF CLK 1 IN -	From ARC-159 in accordance with MIL-R-81877
J2-Z	DIFF CLK 1 IN +	
J2-F	DIFF CLK 2 IN -	From ARC-159 in accordance with MIL-R-81877
J2-E	DIFF CLK 2 IN +	
J2-G	GND	
J2-H	TRANSMIT	Logic "1" when transmit and Logic "0" when not. In accordance with MIL-R-81877
J2-J	DICASS 1	Audio input to ARC-159 in accordance with MIL-R-81877
J2-K	DICASS 2	
J2-L	ON/OFF 1	GND/28 VDC in accordance with MIL-R-81877
J2-N	ON/OFF 2	
J2-P	KEY 1	In accordance with MIL-R-81877
J2-M	KEY 2	
J2-S	SPARE	
J2-T	SPARE	
J2-U	SPARE	
J2-V	SPARE	
J2-W	SPARE	
J2-X	SPARE	
J2-b	RMT 1	in accordance with MIL-R-81877
J2-a	RMT 2	
J2-c	NC	

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TABLE V. C-10743 Connector Interface (Continued)

Connector -PIN	Signal
J3-A	RCVR "D" A
J3-N	RCVR "D" B
J3-B	RCVR "D" C
J3-C	RCVR "D" D
J3-D	RCVR "D" E
	From ARR-75 Channel Selector Switches.
J3-R	RCVR "C" A
J3-J	RCVR "C" B
J3-K	RCVR "C" C
J3-L	RCVR "C" D
J3-M	RCVR "C" E
	From ARR-75 Channel Selector Switches.
J3-E	SPARE
J3-F	SPARE
J3-H	SPARE
J3-G	GND
J3-P	28 VDC (Aircraft DC Power)
J4-A	TRIGGER OUT RETURN
J4-F	TRIGGER OUT
	Trigger tone to T-1295A mixing resistor.
J4-B	+5 VDC
J4-C	+12 VDC
J4-D	-12 VDC
	Power Supply monitor points
J4-E	SPARE
J5-A	IN 1
J5-B	IN 2
J5-C	IN 3
J5-D	IN 4
J5-E	IN 5
J5-F	IN 6
	Remote Control Input (Future) TTL Compatible

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TABLE V. C-10743 Connector Interface (Continued)

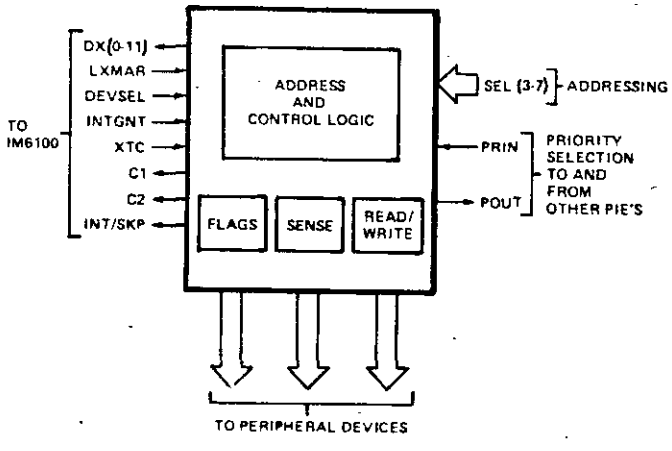
Connector -PIN	Signal	
J5-L	OUT 1	Remote Control Output (Future) TTL Compatible
J5-M	OUT 2	
J5-N	OUT 3	
J5-P	OUT 4	
J5-R	OUT 5	
J5-S	OUT 6	
J5-G	RTN	
J5-H	RMT DATA AVAIL	Control Lines for Remote Control (Future) TTL Compatible
J5-U	RMT DATA ACK	
J5-T	CMD SIG GEN DATA AVAIL	
J5-J	CMD SIG GEN DATA ACK	
J5-K	RMT SENSE	
J5-V	CMD SIG GEN GND	

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TABLE VI. C-10743 Hardware Control Line Signals

P.I.E. A1-U7 (I/O PORT ADDRESS 5) 1/	
LINE	FUNCTION
READ 1	PUSH BUTTON STROBE
SENSE 1	FRONT PANEL DATA AVAILABLE
READ 2	TOGGLE SWITCHES STROBE
FLAG 4	NOT USED
FLAG 3	CMD GATE
FLAG 2	NOT USED
FLAG 1	TRIG GATE
SENSE 4	KEY 2 SENSE
SENSE 3	KEY 1 SENSE
SENSE 2	NOT USED
WRITE 2	(LIGHT CONTROL) BIT 0 UHF 2 GREEN BIT 1 TRANSMIT BIT 2 LOCAL GREEN/REM AMBER BIT 3 LOCAL AMBER/REM GREEN BIT 4 SCU AMBER BIT 5 SCU GREEN BIT 6 DEPTH AMBER BIT 7 DEPTH GREEN BIT 8 ARM D AMBER BIT 9 ARM D GREEN BIT 10 ARM C AMBER BIT 11 ARM C GREEN
WRITE 1	(LIGHT CONTROL) BIT 0 UHF 2 AMBER BIT 1 UHF 1 GREEN BIT 2 UHF 1 AMBER BIT 3 BUSY AMBER BIT 4 BIT AMBER BIT 5 BIT GREEN BIT 6 0.1 BIT 7 0.5 BIT 8 1.0 BIT 9 16 BIT 10 8 BIT 11 MNL

FUNCTIONAL BLOCK DIAGRAM



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TABLE VI C-10743 Hardware Control Line Signals (Continued)

P.I.E. A1-U16 (I/O PORT ADDRESS 6) 1/	
LINE	FUNCTION
READ 1	IN STROBE
SENSE 1	REMOTE DATA AVAILABLE
SENSE 2	DCSG DATA ACKNOWLEDGE
SENSE 3	REMOTE SENSE
SENSE 4	NOT USED
WRITE 1	OUT STROBE
READ 2	NOT USED
WRITE 2	NOT USED
FLAG 2	LOOP AROUND (REMOTE) OPEN
FLAG 3	DCSG DATA AVAILABLE
FLAG 1	REMOTE ACKNOWLEDGE
FLAG 4	NOT USED
P.I.E. A2-U2 (I/O PORT ADDRESS 21) 1/	
LINE	FUNCTION
FLAG 1	KEY 1
FLAG 2	UHF 1
FLAG 3	KEY 2
FLAG 4	UHF 2
SENSE 3	MAINTENANCE MODE TEST POINT E2
SENSE 4	NOT USED
READ 1	ARR-75 CH
READ 2	NOT USED
SENSE 1	SYNTHESIZER 1 LOOP TEST
SENSE 2	SYNTHESIZER 2 LOOP TEST
WRITE 1	WRITE TO SYNTHESIZER 1
WRITE 2	WRITE TO SYNTHESIZER 2

1/ COMPUTER INPUT/OUTPUT ADDRESSING

The computer communicates with three P.I.E. devices by addressing them individually. An individual P.I.E. device address location is referred to as a "port". Port 5 refers to P.I.E. AI-U7. Port 6 refers to P.I.E. AI-U16. Port 21 refers to P.I.E. A2-U2.

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TABLE VII.- C-10743 Electronic Circuit Module Test Points

A1 TEST POINTS	
1	OUT STB - PORT 6, WRITE 1
2	INSTB - PORT 6, READ 1
3	LO OPEN - PORT 6, FLAG 2
4	ANN 2 CLK - PORT 5, WRITE 2
5	ANN 1 CLK - PORT 5, WRITE 1
6	DXB 6
7	DXB 7
8	DXB 8
9	DXB 9
10	DXB 10
11	DXB 11
12	GND
13	TOGSTB - PORT 5, READ 2
14	FRONT PANEL - SAMPLING FREQ
15	DXB Ø
16	DXB 1
17	DXB 2
18	DXB 3
19	DXB 4
20	DXB 5

A2 TEST POINTS	
1	F OUT 2
2	F OUT 1
3	GND
4	F1
5	TST 1 - PORT 21, SENSE 1
6	F2
7	PORT 21, WRITE 1
8	DEVSELB
9	TST 2 - PORT 21, SENSE 2
10	PORT 21, WRITE 2
11	C2
12	SKP
13	C1

MAINTENANCE MODE CONNECTIONS

E1	GND
E2	SENSE 3

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TABLE VII.- C-10743 Electronic Circuit Module Test Points (Continued)

A3 TEST POINTS	
1	RAM SELECT
2	XTCB
3	LXMARB
4	GND
5	RUN/HLT
6	XTA
7	INTGNTB
8	12.8 Hz 1

A4 TEST POINTS	
1	SERIAL CLK
2	DIFF CLK 1
3	DIFF CLK 2
4	ARR75 CH
5	KEY 1 SENSE
6	KEY 2 SENSE
7	SERIAL DATA OUT
8	KEY 2
9	REM 2
10	KEY 1
11	REM 1
12	UHF 2
13	UHF 1
14	DICASS 2
15	DICASS 1
16	ADDRESS CMD
17	GND

A5 TEST POINTS	
1	-12 VDC
2	+12 VDC
3	TRANSMIT ANUC
4	ANUC DIM
5	+5 VDC
6	SIG GND
7	+28V ANUC

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TABLE VIII. - C-10743 Signal Definitions

Control and Reference Lines	
Signal Name	**Definition**
12.8 KHz 1 12.8 KHz 2 12.8 KHz	Stable 20% Duty Cycle Reference frequencies derived from 1.28 MHz Master Oscillator
$\overline{C0}$	Tied High
$\overline{C1}$ $\overline{C2}$	Control line inputs from peripheral devices during I/O transfers.

Control Lines				
$\overline{C0}$	$\overline{C1}$	$\overline{C2}$	**Operation**	**Description**
H	H	H	DEV ← AC 1/	The content of the AC is sent to the device.
H	L	H	AC ← AC V DEV	Data is received from a device, OR'ed with the data in the AC and the result is stored in the AC.
H	H	L	PC ← PC + DEV 2/	Data from the device is added to the contents of the PC. This is referred to as a RELATIVE JUMP.
H	L	L	PC ← DEV	Data is received from a device and loaded into the PC. This is referred to as an ABSOLUTE JUMP.

1/ AC = CPU Accumulator 2/ PC = CPU Program Counter

DEVSELB	Device Select for I/O Transfers
DXB_0 - DXB_{11}	Data X-Multiplexed data in, data out and address lines
FOUT 1 FOUT 2	Output from A2 Module Synthesizer
INHCTRL	Control line for CPU Bus Driver
INTGNTB	Peripheral Device Interrupt Grant
INTREQ	Peripheral Device or Watch Dog Timer Interrupt Request
LXMARB LXMARBD	Load External Memory Address Register Used to store memory and peripheral address externally
MEMSELB	Memory Select for Memory Transfers

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TABLE VIII. - C-10743 Signal Definitions (Continued)

Signal Name	Definition
RESET	Clears CPU accumulator, loads 7777_8 in the CPU program counter, Halts CPU and clears front panel indicators.
RUN/HALT	Pulsing the RUN/HALT line causes the CPU to alternately run and halt by changing the state of the CPU's internal RUN/HALT flip flop.
SKP	Peripheral devices assert this line low to generate interrupt requests and to signal the CPU when sense flip flops are set during SKIP instructions. The CPU skips the next sequential instruction if this line is asserted during an I/O instruction.
XTCB XTCBD	External coded minor cycle timing - used in conjunction with DEVSEL and MEMSEL to specify read or write operations.
XTA	External coded minor cycle timing - signifies input transfers to CPU.
XTB	External coded minor cycle timing - signifies output transfers from the CPU.
FRONT PANEL INTERFACE LINES	
ARM C GREEN ARM C AMBER ARM D GREEN ARM D AMBER BIT AMBER BIT GREEN BUSY AMBER DEPTH AMBER/D DEPTH GREEN/M LOCAL AMBER/REMOTE GREEN LOCAL GREEN/REMOTE AMBER SCUTTLE GREEN SCUTTLE AMBER UHF 1 AMBER UHF 1 GREEN UHF 2 AMBER UHF 2 GREEN MNL 8 16 0.1 0.5 1.0	Front Panel indicator control lines. Lamp is ON when respective line is grounded through lamp driver (Darlington Pair) of A1 circuit module.
TRANSMIT	Transmit indicator control line lamp is ON when line is at a HI logic level.
C ON D ON	Bouy C and Buoy D switch input when in the ON position.
ROW Y1 - Y4 COL X1 - X4	Front panel switch matrix sense lines.

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TABLE VIII. - C-10743 Signal Definitions (Continued)

I/O CONTROL LINES

Signal Name	Definition
ARR-75 CH	ARR-75 Channel Gate When active, the 5 bit binary code representing ARR-75 Channel C & D are gated onto the Data Bus.
CMD GATE	Command Gate along with trigger gate indicates where the current tone being generated is an Address Ping or Command and controls routing of the tones
DICASS 1 DICASS 2	Audio output to UHF transceivers 1 and 2 respectively. In accordance with MIL-R-81877.
DIFF CLOCK 1 IN - DIFF CLOCK 1 IN + DIFF CLOCK 2 IN - DIFF CLOCK 2 IN +	Differential Clock Input from AN/ARC-159 #1 and #2 respectively per MIL-R-81877 used to synchronize differential data with UHF clock
DIFF DATA + UHF 1 DIFF DATA - UHF 1 DIFF DATA + UHF 2 DIFF DATA - UHF 2	Differential Data output from CSG to control UHF tuning per MIL-R-81877
KEY 1 KEY 2	Remote key lines to UHF transceivers per MIL-R-81877
KEY 1 SENSE KEY 2 SENSE	UHF key sense line indicates current status of UHF key lines (Keyed/Not Keyed)
ON/OFF 1 ON/OFF 2	Remote ON/OFF Control lines for UHF transceivers per MIL-R-81877
RCVR C A - RCVR C E RCVR D A - RCVR D E	5 line input from ARR-75 which indicate selected channel for received C & D respectively
REM 1 REM 2	Remote take control lines to UHF transceivers per MIL-R-81877
TRIG GATE	See CMD Gate
TRIGGER OUT	Trigger output line to T-1295A transmitter multiplexer

TRIG GATE	CMD GATE	TONE TYPE
H	H	Command
H	L	Address
L	H	Ping
L	L	N/A

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TABLE VIII.- C-10743 Signal Definitions (Continued)

Signal Name	Definition
UHF 1 UHF 2	Internal UHF selection lines - controls which UHF transceiver will be controlled by REM and ON/OFF lines and receive differential data and audio tones.
UHF 1 KEY UHF 2 KEY	"Key Radio" command from A2 board, when active the appropriate transceiver will be keyed by the A4 board
UNUSED LINES (Nomenclature Suggested for Expansion Spares)	
CCW	Channel C CW lamp control - Future
CFM↑	Channel C FM Up lamp control - Future
CFM↓	Channel C FM Down lamp control - Future
DCSGDACK	Command Signal Generator Data Acknowledge - Future
DCSGDAV	Command Signal Generator Data Available - Future
DCW	Channel D CW lamp control - Future
DFM↑	Channel D FM Up lamp control - Future
DFM↓	Channel D FM Down lamp control - Future
FM IN 1 FM IN 2	FM tone input to synthesizer - Future
IN 1 - IN 6	Remote Terminal Input Lines - Future
PORT 5 FLAG 2 PORT 5 FLAG 4	Unused P.I.E. control and input lines
PORT 5 SENSE 2 PORT 6 FLAG 4 PORT 6 SENSE 4 PORT 6 WRITE 2 PORT 21 READ 2 PORT 21 SENSE 4	
REM DACK	Remote terminal Data Acknowledge - Future
REM DAV	Remote terminal Data Available - Future
REM SENSE	Remote terminal Sense Line - Future
OUT 1 - OUT 6	Remote terminal Output Lines - Future
SPARE X	Spare (Unused) lines - Future

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TABLE IX. Front Panel Key Input Codes

SWITCH DEPRESSED (ROW, COL FROM FIGURE 23)	OUTPUT CODE					
	DX6	DX7	DX8	DX9	DX10	DX11
ARM D (Y1, X1)	X	X	0	0	0	0
ARM C (Y1, X2)	X	X	1	0	0	0
0.1/0.5/1.0 (Y1, X3)	X	X	0	1	0	0
UHF1/UHF2 (U1, X4)	X	X	1	1	0	0
LOCAL/REMOTE (Y2, X1)	X	X	0	0	1	0
BIT/BUSY (Y2, X2)	X	X	1	0	1	0
MNL/8/16 (Y2, X3)	X	X	0	1	1	0
SCU (Y2, X4)	X	X	1	1	1	0
SPARE (Y3, X1)	X	X	0	0	0	1
DEPTH (Y3, X2)	X	X	1	0	0	1
NOT USED (Y3, X3)	X	X	0	1	0	1
NOT USED (Y3, X4)	X	X	1	1	0	1
NOT USED (Y4, X1)	X	X	0	0	1	1
NOT USED (Y4, X2)	X	X	1	0	1	1
PING C (Y4, X3)	X	X	0	1	1	1
PING D (Y4, X4)	X	X	1	1	1	1

	DX6	DX7	DX8	DX9	DX10	DX11
BUOY C ON	1	X	X	X	X	X
BUOY D ON	X	1	X	X	X	X

X=DON'T CARE

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TABLE X. Front Panel Wire List

FROM	TO	FUNCTION	FROM	TO	FUNCTION
T1-3	E1	XFMR SHIELD	P2-20	S7-2	0.1
P2-3	CB1-LINE (2)	115 VAC IN	P2-19	S7-1	0.5
CB1-LOAD (1)	T1-1	115 VAC IN (SW)	P2-25	S7-4	1.0
T1-2	P2-34	115 VAC RETURN	P2-56	S11-	LOC GR/RMT AM
J1-4	P2-6	EDGE LIGHT RTN	P2-55	S11-3	LOC AM/RMT GR
J1-1	P2-65	EDGE LIGHT HI	P2-18	S5-2	BIT GR
J1-2	P2-64	TRANSMIT ANN	P2-17	S5-3	BIT AM
J1-3	J1-4	TRANSMIT ANN RTN	P2-58	S5-4	BUSY AM
P2-40	S10-1 (NO)	COL X1	P2-60	S8-2	ARM C GR
S10-1 (NO)	S11-1 (NO)	COL X1	S8-2	S8-3	ARM C GR
S11-1 (NO)	S12-1 (NO)	COL X1	P2-27	S8-1	ARM C AM
P2-41	S8-1 (NO)	COL X2	S8-1	S8-4	ARM C AM
S8-1 (NO)	S5-1 (NO)	COL X2	P2-24	S10-2	ARM D GR
S5-1 (NO)	S4-1 (NO)	COL X2	S10-2	S10-3	ARM D GR
P2-42	S7-1 (NO)	COL X3	P2-23	S10-1	ARM D AM
S7-1 (NO)	S6-1 (NO)	COL X3	S10-1	S10-4	ARM D AM
S6-1 (NO)	S2-1	COL X3	P2-22	S9-2	UHF 1 GR
P2-8	S9-1 (NO)	COL X4	P2-21	S9-3	UHF 1 AM
S9-1 (NO)	S3-1 (NO)	COL X4	P2-48	S9-1	UHF 2 GR
S3-1 (NO)	S1-1	COL X4	P2-51	S9-4	UHF 2 AM
P2-39	S9-3 (C)	ROW Y1	P2-54	S3-2	SCU GR
S9-3 (C)	S7-3 (C)	ROW Y1	S3-2	S3-3	SCU GR
S7-3 (C)	S8-3 (C)	ROW Y1	P2-59	S3-1	SCU AM
S8-3 (C)	S10-3 (C)	ROW Y1	S3-1	S3-4	SCU AM
P2-9	S3-3 (C)	ROW Y2	P2-53	S4-2	DEPTH GR/M
P2-52	S4-1	DEPTH AM/D	S4-2	S4-3	DEPTH GR/M
S4-1	S4-4	DEPTH AM/D	S3-3 (C)	S6-3 (C)	ROW Y2
P2-29	S5-	CCW	S6-3 (C)	S5-3 (C)	ROW Y2
P2-16	S5-	CFM	S5-3 (C)	S11-3 (C)	ROW Y2
P2-15	S5-	CFM	P2-10	S4-3 (C)	ROW Y3
P2-14	S3-	D CW	S4-3 (C)	S12-3 (C)	ROW Y3
P2-13	S3-	DFM	P2-11	S1-2	ROW Y4
P2-12	S3-	DFM	S1-2	S2-2	ROW Y4
P2-28	CB1-C	+28V	P2-43	S1-3	D ON
P2-37	A1-E2	+28V	P2-38	S2-3	C ON
P2-5	T1-4	10 VAC HI	P2-7	S11-C	+28V ANN
P2-33	T1-5	10 VAC LO	S11-C	S10-C	+28V ANN
P2-63	T1-9	27 VAC HI	S10-C	S9-C	+28V ANN
P2-61	T-10	27 VAC LO	S9-C	S8-C	+28V ANN
P2-62	T1-6	30 VAC HI	S8-C	S7-C	+28V ANN
P2-66	T1-7	30 VAC CT	S7-C	S6-C	+28V ANN
P2-32	T1-8	30 VAC LO	S6-C	S5-C	+28V ANN
J1-3	A9A1-E3	GND	S5-C	S4-C	+28V ANN
CB1-NO	A9A1-E1	+28V SW	S4-C	S3-C	+28V ANN
P2-1	A9A1-E5	+5V	S3-C	S12-C	+28V ANN
P2-36	A9A1-E4	-12V	P2-57	S6-3	MNL
P2-50	SG-1	8			
P2-49	S6-4	16			

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TABLE X Front Panel Wire List.(continued)

FUNCTION	FROM	TO
XFMR SHIELD	T1-3	E1
115 VAC IN	P2-3	CB1-LINE (2)
115 VAC IN (SW)	CB1-LOAD (1)	T1-1
115 VAC RETURN	T1-2	P2-36
EDGE LIGHT RTN	J1-4	P2-6
EDGE LIGHT HI	J1-1	P2-65
TRANSMIT ANN	J1-2	P2-64
TRANSMIT ANN RTN	J1-3	J1-4
COL X1	P2-40	S10-1 (NO)
COL X1	S10-1 (NO)	S11-1 (NO)
COL X1	S11-1 (NO)	S12-1 (NO)
COL X2	P2-41	S8-1 (NO)
COL X2	S8-1 (NO)	S5-1 (NO)
COL X2	S5-1 (NO)	S4-1 (NO)
COL X3	P2-42	S7-1 (NO)
COL X3	S7-1 (NO)	S6-1 (NO)
COL X3	S6-1 (NO)	S2-1
COL X4	P2-8	S9-1 (NO)
COL X4	S9-1 (NO)	S3-1 (NO)
COL X4	S3-1 (NO)	S1-1
ROW Y1	P2-39	S9-3 (C)
ROW Y1	S9-3 (C)	S7-3 (C)
ROW Y1	S7-3 (C)	S8-3 (C)
ROW Y1	S8-3 (C)	S10-3 (C)
ROW Y2	P2-9	S3-3 (C)
DEPTH AM/D	P2-52	S4-1
DEPTH AM/D	S4-1	S4-4
CCW	P2-29	S5-
CFM	P2-16	S5-
CFM	P2-15	S5-
D CW	P2-14	S3-
DFM	P2-13	S3-
DFM	P2-12	S3-
+28V	P2-28	CB1-C
+28V	P2-37	A1-E2
10 VAC HI	P2-5	T1-4
10 VAC LO	P2-33	T1-5
27 VAC HI	P2-63	T1-9
27 VAC LO	P2-61	T-10
30 VAC HI	P2-62	T1-6
30 VAC CT	P2-66	T1-7
30 VAC LO	P2-32	T1-8
GND	J1-3	A9A1-E3
+28V SW	CB1-NO	A9A1-E1
+5V	P2-1	A9A1-E5
-12V	P2-36	A9A1-E4
8	P2-50	SG-1
16	P2-49	S6-4

FUNCTION	FROM	TO
0.1	P2-20	S7-2
0.5	P2-19	S7-1
1.0	P2-25	S7-4
LOC GR/RMT AM	P2-56	S11-
LOC AM/RMT GR	P2-55	S11-3
BIT GR	P2-18	S5-2
BIT AM	P2-17	S5-3
BUSY AM	P2-58	S5-4
ARM C GR	P2-60	S8-2
ARM C GR	S8-2	S8-3
ARM C AM	P2-27	S8-1
ARM C AM	S8-1	S8-4
ARM D GR	P2-24	S10-2
ARM D GR	S10-2	S10-3
ARM D AM	P2-23	S10-1
ARM D AM	S10-1	S10-4
UHF 1 GR	P2-22	S9-2
UHF 1 AM	P2-21	S9-3
UHF 2 GR	P2-48	S9-1
UHF 2 AM	P2-51	S9-4
SCU GR	P2-54	S3-2
SCU GR	S3-2	S3-3
SCU AM	P2-59	S3-1
SCU AM	S3-1	S3-4
DEPTH GR/M	P2-53	S4-2
DEPTH GR/M	S4-2	S4-3
ROW Y2	S3-3 (C)	S6-3 (C)
ROW Y2	S6-3 (C)	S5-3 (C)
ROW Y2	S5-3 (C)	S11-3 (C)
ROW Y3	P2-10	S4-3 (C)
ROW Y3	S4-3 (C)	S12-3 (C)
ROW Y4	P2-11	S1-2
ROW Y4	S1-2	S2-2
D ON	P2-43	S1-3
C ON	P2-38	S2-3
+28V ANN	P2-7	S11-C
+28V ANN	S11-C	S10-C
+28V ANN	S10-C	S9-C
+28V ANN	S9-C	S8-C
+28V ANN	S8-C	S7-C
+28V ANN	S7-C	S6-C
+28V ANN	S6-C	S5-C
+28V ANN	S5-C	S4-C
+28V ANN	S4-C	S3-C
+28V ANN	S3-C	S12-C
MNL	P2-57	S6-3

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TABLE XI. Electronic Circuit Module External Connections. 1./

XA1 MODULE			
PIN	SIGNAL	PIN	SIGNAL
1	+5 VDC	56	ARM C AMBER
2	+5 VDC	57	IN 3
3	ROW Y2	58	IN 5
4	ROW Y3	59	OUT 5
5	ROW Y4	60	IN 6
6	LXMARB	61	ANALOG GROUND
7	XTCB	62	DIGITAL GROUND
8	DEVSELB	63	DIGITAL GROUND
9	PORT 5, FLAG 2	64	C ON
10	TRIG GATE	65	ROW Y1
11	C1	66	COL X1
12	C2	67	COL X2
13	SKP	68	COL X3
14	DXB 1	69	D ON
15	DXB 0	70	DCSG DACK
16	DXB 2	71	REM DAV
17	DXB 3	72	UHF 2 ENABLE
18	DXB 6	73	TRANSMIT
19	DXB 7	74	GROUND
20	DXB 5	75	UHF 2 GREEN
21	DXB 9	76	16
22	DXB 10	77	8
23	DXB 11	78	UHF 2 AMBER
24	IN 4	79	DEPTH AMBER/D
25	OUT 4	80	DEPTH GREEN/M
26	DXB 8	81	SCUTTLE GREEN
27	PORT 6, FLAG 4	82	LOCAL AMBER/REMOTE GREEN
28	OUT 2	83	LOCAL GREEN/REMOTE AMBER
29	OUT 3	84	MNL
30	DXB 4	85	BUSY AMBER
31	+12 VDC	86	SCUTTLE AMBER
32	RESET	87	ARM C GREEN
33	COL X4	88	IN 2
34	PORT 5, FLAG 4	89	IN 1
35	CMD GATE	90	OUT 1
36	LXMARBD	91	OUT 6
37	REM SENSE	92	-12 VDC
38	PORT 5, SENSE 2		
39	KEY 1 SENSE		
40	KEY 2 SENSE		
41	XTCBD		
42	PORT 6, SENSE 4		
43	DCSG DAV		
44	REM DACK		
45	SPARE 3		
46	BIT AMBER		
47	BIT GREEN		
48	0.5		
49	0.1		
50	UHF 1 AMBER		
51	UHF 1 GREEN		
52	ARM D AMBER		
53	ARM D GREEN		
54	1.0		
55	PORT 6, WRITE 2		

1./ Reference:Figure 26 (From - To) List

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TABLE XI. Electronic Circuit Module External Connections (Continued)

XA2 MODULE			
PIN	SIGNAL	PIN	SIGNAL
1	+5 VDC	47	JUMPERED
2	+5 VDC	48	
3	12.8 KHZ 1	49	JUMPERED
4	12.8 KHZ 2	50	
5		51	
6		52	
7		53	
8	$\overline{\text{DEVSELB}}$	54	
9	UHF 2	55	
10	$\overline{\text{KEY 2}}$	56	
11	$\overline{\text{C1}}$	57	
12	$\overline{\text{C2}}$	58	
13	$\overline{\text{SKP}}$	59	
14	DXB 1	60	
15	DXB Ø	61	ANALOG GROUND
16	DXB 2	62	DIGITAL GROUND
17	DXB 3	63	DIGITAL GROUND
18	DXB 6	64	FM IN 1
19	DXB 7	65	FM IN 2
20	DXB 5	66	
21	DXB 9	67	
22	DXB 10	68	
23	DXB 11	69	
24		70	
25		71	
26	DXB 8	72	$\overline{\text{UHF2 ENABLE}}$
27		73	TRANSMIT
28	F OUT 1	74	
29	F OUT 2	75	
30	DXB 4	76	
31	+12VDC	77	
32	PORT 21, SENSE 4	78	
33	PORT 21, READ 2	79	
34	ARR-75 CHANNEL	80	
35	$\overline{\text{KEY 1}}$	81	
36	LXMARBD	82	
37	UHF 1	83	
38		84	
39		85	
40		86	
41	XTCBD	87	
42		88	
43		89	
44		90	
45		91	
46		92	-12 VDC

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TABLE XI. Electronic Circuit Module External Connections (Continued)

XA3 MODULE			
PIN	SIGNAL	PIN	SIGNAL
1	+5 VDC	47	
2	+5 VDC	48	
3	12.8 KHZ 1	49	
4	12.8 KHZ 2	50	
5	INHCTRL	51	
6	LXMARB	52	
7	XTCB	53	
8	DEVSELB	54	
9	INTGNTB	55	
10	C0	56	
11	C1	57	
12	C2	58	
13	SKP	59	
14	DXB 1	60	
15	DXB Ø	61	ANALOG GROUND
16	DXB 2	62	DIGITAL GROUND
17	DXB 3	63	DIGITAL GROUND
18	DXB 6	64	
19	DXB 7	65	
20	DXB 5	66	
21	DXB 9	67	
22	DXB 10	68	
23	DXB 11	69	
24		70	
25		71	
26	DXB 8	72	UHF2 ENABLE
27	MEMSELB	73	TRANSMIT
28	XTA	74	
29	XTB	75	
30	DXB 4	76	
31	+12V	77	
32	RESET	78	
33		79	
34	INT REQ	80	
35	MEMSELB	81	
36		82	
37		83	
38	RUN/HLT	84	
39	INT REQ	85	
40	XTA	86	
41	XTB	87	
42		88	
43	128 KHZ	89	
44		90	
45		91	
46		92	-12V

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TABLE XI. Electronic Circuit Module External Connections (Continued)

XA4 MODULE			
PIN	SIGNAL	PIN	SIGNAL
1	+5VDC	47	RCVR CE
2	+5VDC	48	RCVR DA
3	DICASS 2	49	RCVR DB
4	FOUT 1	50	RCVR DC
5	FOUT 2	51	RCVR DD
6	ON/OFF 1	52	RCVR DE
7	KEY 1	53	SPARE 11
8	KEY 2	54	SPARE 15
9	UHF 2	55	SPARE 14
10	TRIG GATE	56	
11	ON/OFF 2	57	DIFF CLOCK 1 IN -
12	REM 2	58	DIFF CLOCK 2 IN -
13	RCVR CC	59	DIFF CLOCK 1 IN +
14	DXB 1	60	DIFF CLOCK 2 IN +
15	DXB Ø	61	ANALOG GRND
16	DXB 2	62	DIGITAL GRND
17	DXB 3	63	DIGITAL GRND
18	DXB 6	64	UHF 1 KEY
19	DXB 7	65	UHF 2 KEY
20	DXB 5	66	
21	DXB 9	67	
22	DXB 10	68	
23	DXB 11	69	
24		70	
25	DIFF DATA + UHF 2	71	
26	DXB 8	72	UHF2 ENABLE
27	DIFF DATA - UHF 1	73	TRANSMIT
28	DIFF DATA - UHF 2	74	
29	DIFF DATA + UHF 1	75	
30	DXB 4	76	
31	+12VDC	77	
32	+28VDC SW	78	
33	DICASS 1	79	
34	ARR-75 CH	80	
35	CMD GATE	81	
36	TRIGGER OUT	82	
37	UHF 1	83	
38	REM 1	84	
39	KEY 1 SENSE	85	
40	KEY 2 SENSE	86	
41		87	
42		88	
43		89	
44	RCVR CB	90	
45	RCVR CA	91	
46	RCVR CD	92	-12VDC

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TABLE XI. Electronic Circuit Module External Connections (Continued)

XA5 MODULE			
PIN	SIGNAL	PIN	SIGNAL
1	+5 VDC	48	
2	+5 VDC	49	
3		50	
4	EMITTER REG OUT (Q1)	51	
5	BASE REG CTRL (Q1)	52	
6	EMITTER REG OUT (Q2)	53	
7	BASE REG CTRL (Q2)	54	
8	TRANSMIT ANUNC	55	
9	TRANSMIT ANUNC	56	V REF
10	+28V ANUNC	57	+20 VDC
11	ANUNC DIM	58	
12	SIGNAL GROUND	59	
13	SIGNAL GROUND	60	
14		61	ANALOG GROUND
15	EMITTER REG OUT (Q3)	62	DIGITAL GROUND
16	BASE REG CTRL (Q3)	63	DIGITAL GROUND
17	COLLECTOR SUP SRCE (Q3)	64	
18		65	
19	+11 VDC	66	
20	+11 VDC	67	
21	GROUND	68	
22	GROUND	69	
23	27 VAC HI	70	
24	27 VAC LO	71	
25	30 VAC HI	72	$\overline{\text{UHF-2 ENABLE}}$
26	30 VAC CT	73	TRANSMIT
27	30 VAC LO	74	
28	30 VAC LO	75	
29		76	
30		77	
31	+12 VDC	78	
32		79	
33		80	
34		81	
35		82	
36		83	
37		84	
38		85	
39		86	
40		87	
41		88	
42		89	
43		90	
44		91	
45		92	-12 VDC
46			
47			

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TABLE XII. AN/ARR-75 Channel Codes

RCVR CHANNEL 1/	RCVR C AND D				
BIT	0	0	0	0	0
17	0	0	0	0	1
18	0	0	0	1	0
19	0	0	0	1	1
20	0	0	1	0	0
21	0	0	1	0	1
22	0	0	1	1	0
23	0	0	1	1	1
24	0	1	0	0	0
25	0	1	0	0	1
26	0	1	0	1	0
27	0	1	0	1	1
28	0	1	1	0	0
29	0	1	1	0	1
30	0	1	1	1	0
31	0	1	1	1	1
1	1	0	0	0	0
2	1	0	0	0	1
3	1	0	0	1	0
4	1	0	0	1	1
5	1	0	1	0	0
6	1	0	1	0	1
7	1	0	1	1	0
8	1	0	1	1	1
9	1	1	0	0	0
10	1	1	0	0	1
11	1	1	0	1	0
12	1	1	0	1	1
13	1	1	1	0	0
14	1	1	1	0	1
15	1	1	1	1	0
16	1	1	1	1	1

1/ : "0" = -12 VDC from ARR-75
"1" = open circuit from ARR-75

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
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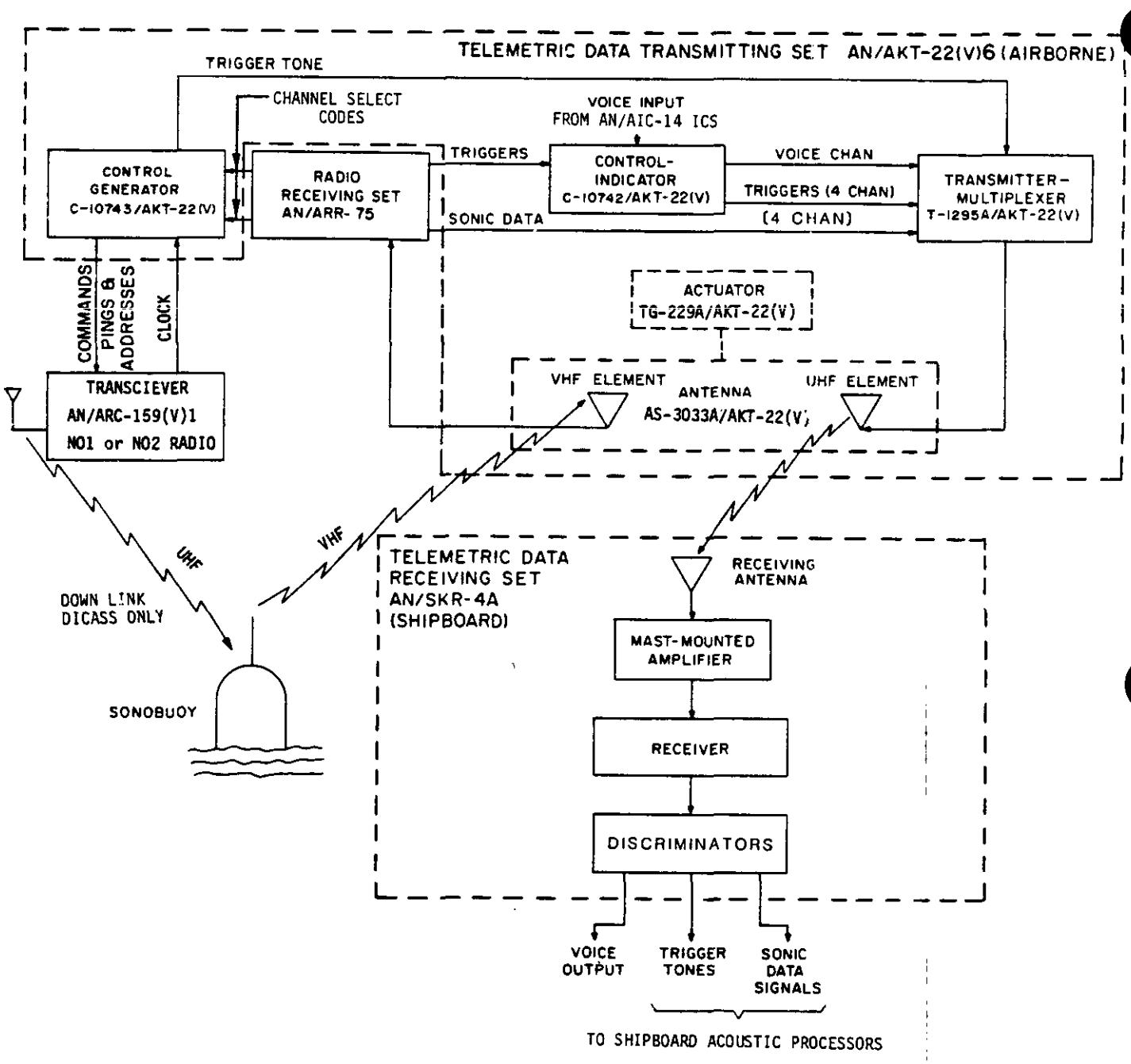
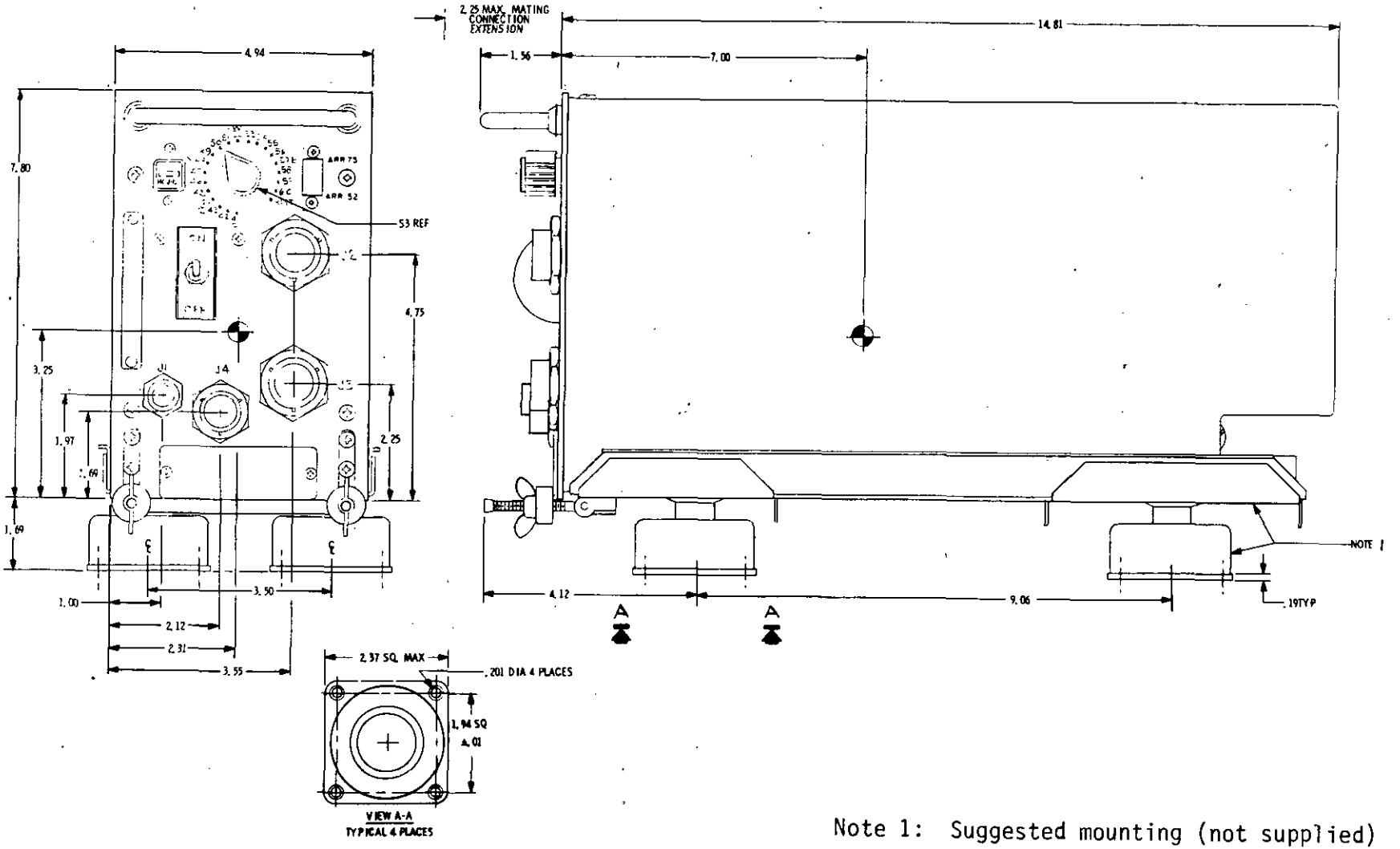


FIGURE 1 Telemetric Data Transmitting Set AN/AKT-22(V)6.

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Note 1: Suggested mounting (not supplied)

FIGURE 2 Transmitter-Multiplexer T-1295A/AKT-22(V) Form Factor

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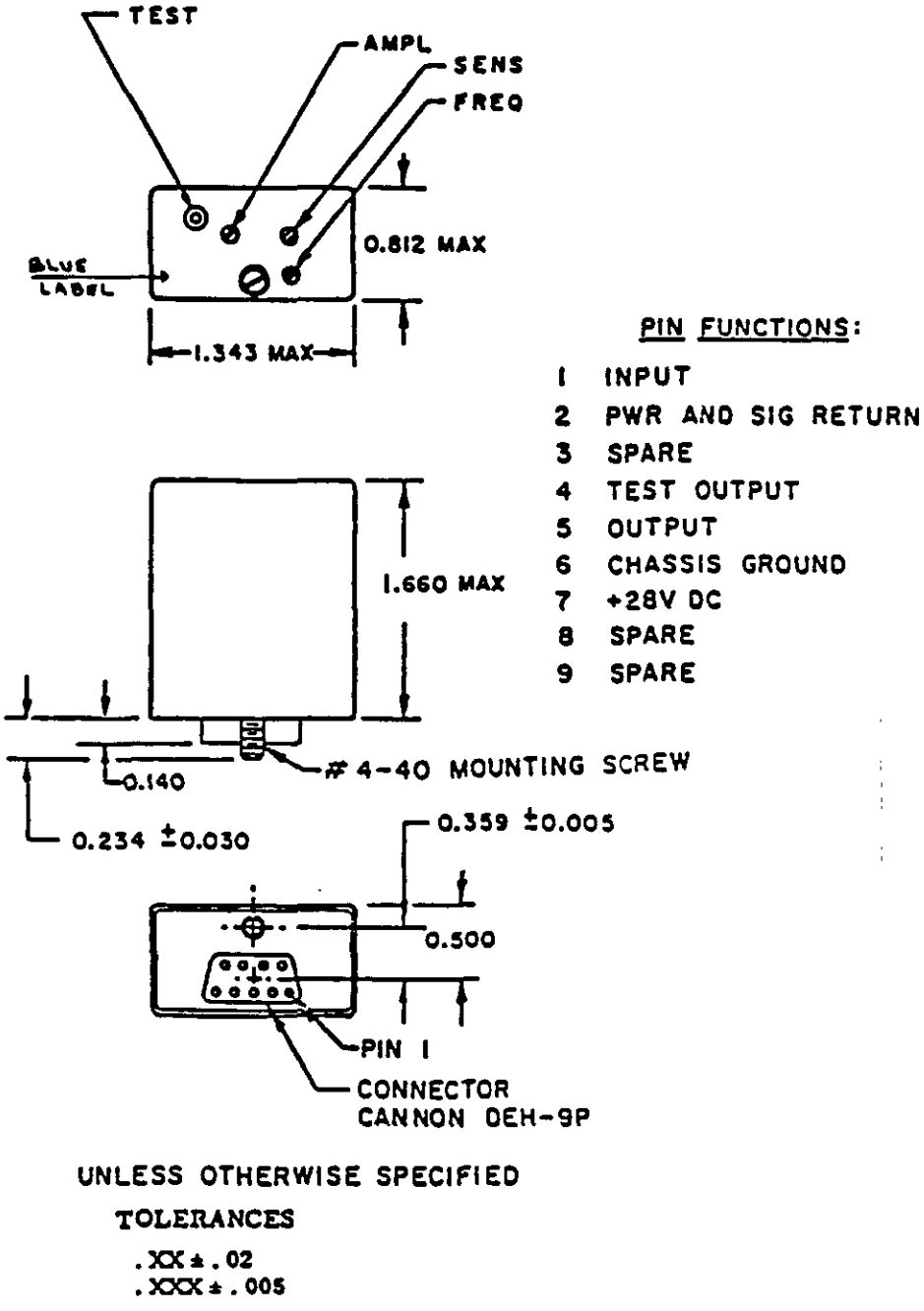
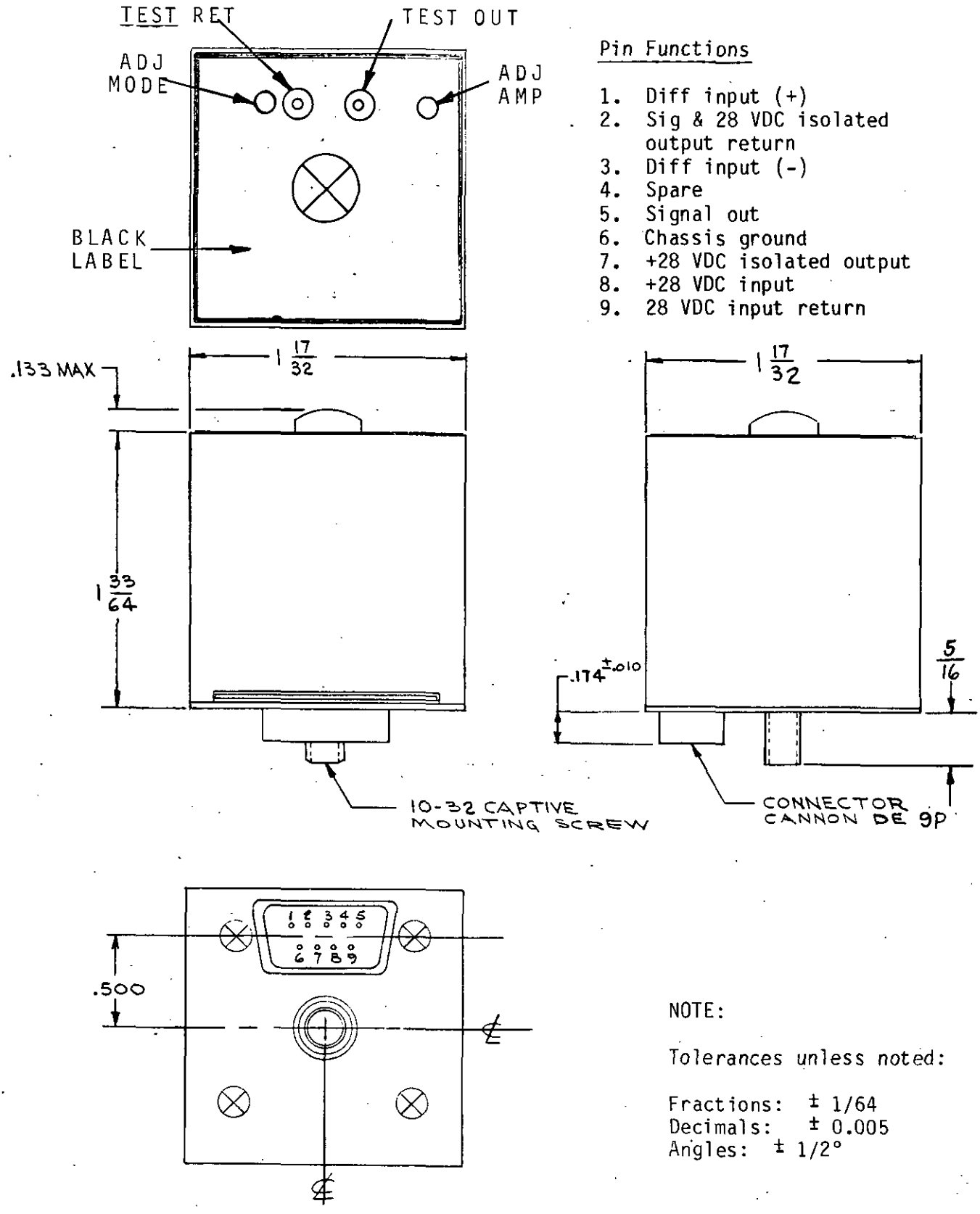


FIGURE 3 Voltage Controlled Oscillators Form Factor and Connector Requirements.

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Pin Functions

1. Diff input (+)
2. Sig & 28 VDC isolated output return
3. Diff input (-)
4. Spare
5. Signal out
6. Chassis ground
7. +28 VDC isolated output
8. +28 VDC input
9. 28 VDC input return

NOTE:

Tolerances unless noted:

Fractions: ± 1/64
Decimals: ± 0.005
Angles: ± 1/2°

FIGURE 4 Low-Pass Filter, Form Factor and Connector Requirements.

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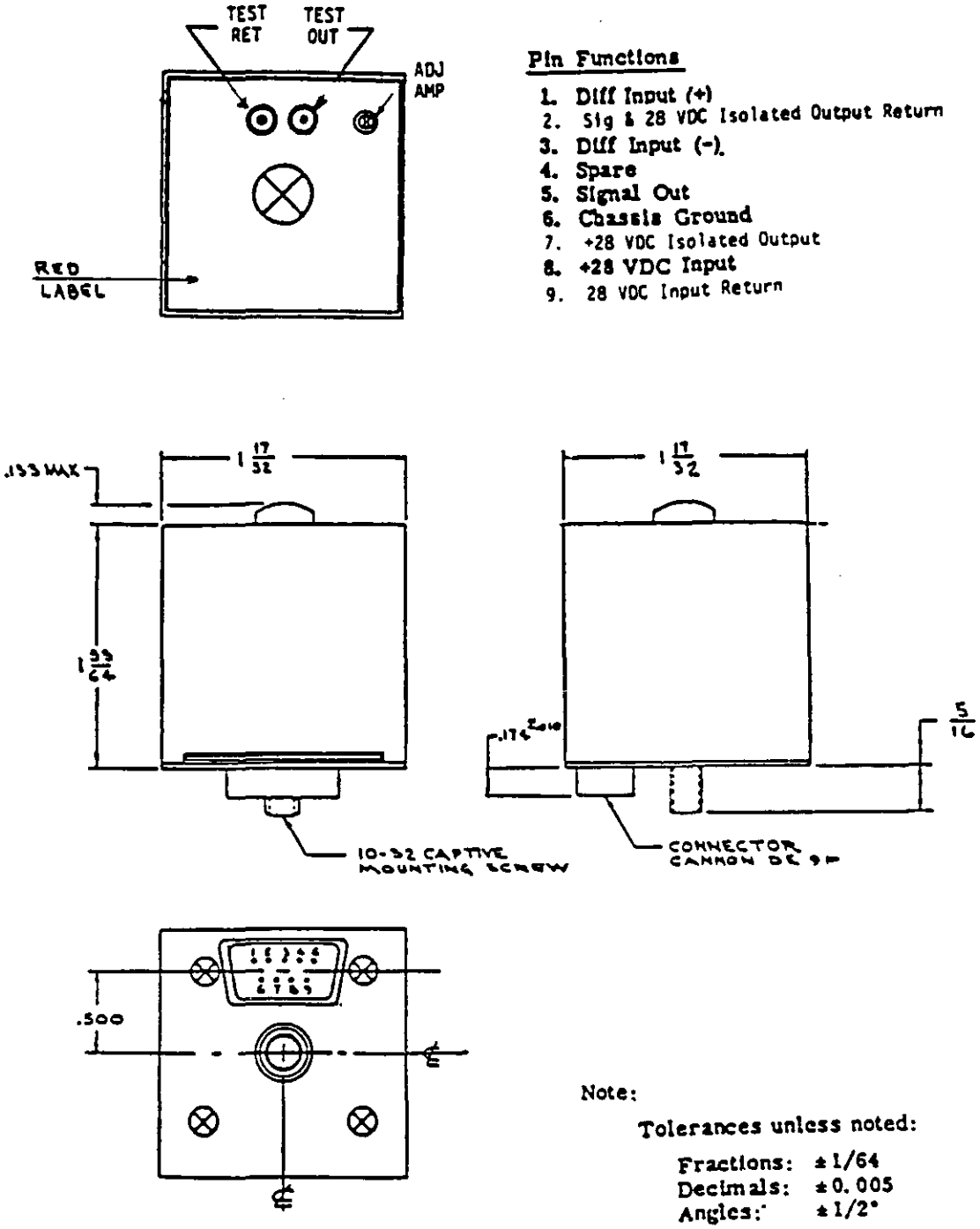
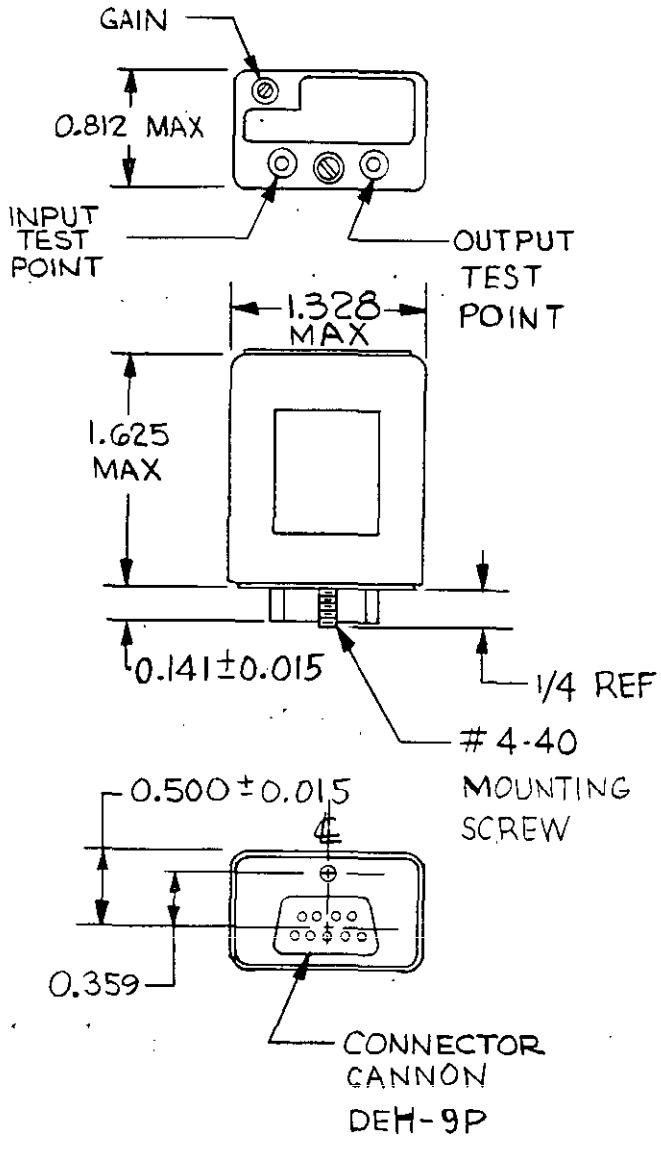


FIGURE 5 Voice Low-Pass Filter, Form Factor and Connector Requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
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Connector Pin Functions:

Pin	Function
1	Input
2	Power & Signal Return
3	Spare
4	Spare
5	Output
6	Chassis Ground
7	+28 VDC
8	Spare
9	Spare

NOTES:
1. ALL DIMENSIONS IN INCHES
2. TOLERANCES ± .005 EXCEPT AS NOTED

Weight: 2.0 ounces (approximately)
Volume: 1.8 cubic inches

FIGURE 6 Mixer Amplifier, Form Factor and Connector Requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
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FUNCTION TABLE

Pin	Function
J1-1	Input A signal
J1-3	Input A return
J2-1	Input B signal
J2-3	Input B return
J3-1	Input C signal
J3-3	Input C return
J4-1	Input D signal
J4-3	Input D return
J5-2	Output return
J5-5	Output signal

NOTES: 1. Connectors J1 through J5 are type 14073-9P.

2. Tolerances unless noted:

Fractions: ± 1/64
Decimals: .xxx ± 0.005
.xx ± 0.01

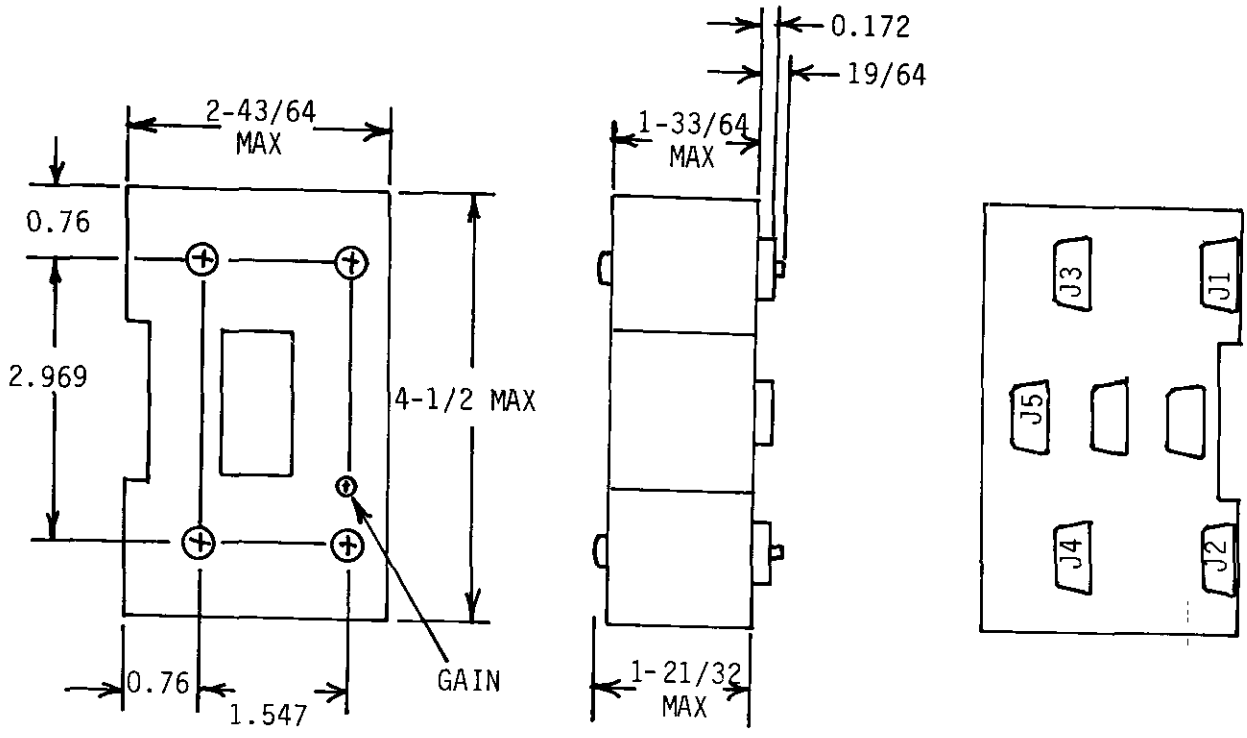
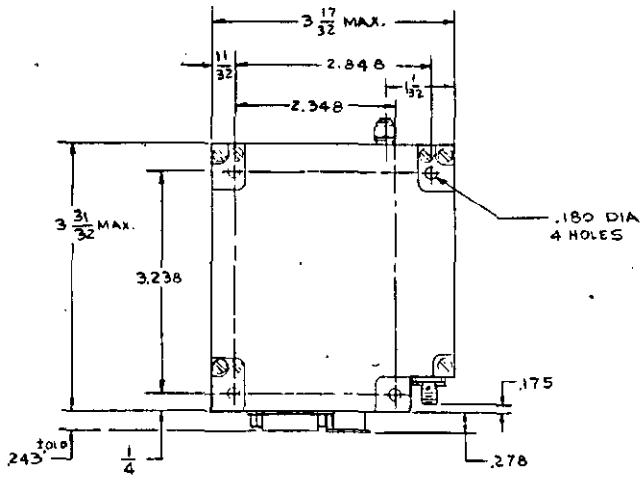
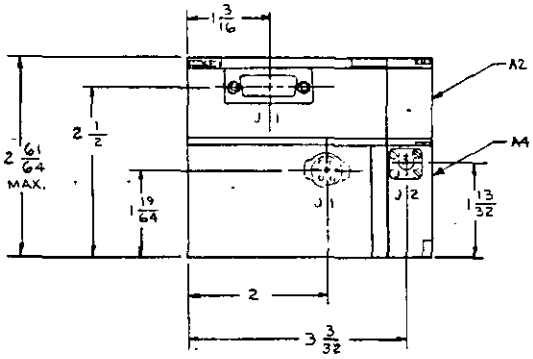


FIGURE 7 Trigger Filter, Form Factor and Connector Requirements

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Connector Pin	Pin Function	
A1J1-A	+28 VDC input	
A1J1-B	Gnd	
A1J1-C	Mod input	
A1J1-D	Mod return input	
A1J1-E	70 MHz cont	
A1J1-F	Spare	
A1J1-G	Spare	
A1J2	RF output	
A1J4	70 MHz input	
A2J1-1	2^4	Binary Program Code Input
A2J1-2	2^3	
A2J1-3	2^2	
A2J1-4	2^1	
A2J1-5	2^0	
A2J1-6	Gnd	
A2J1-7	Spare	
A2J1-8	Spare	
A2J1-9	Spare	





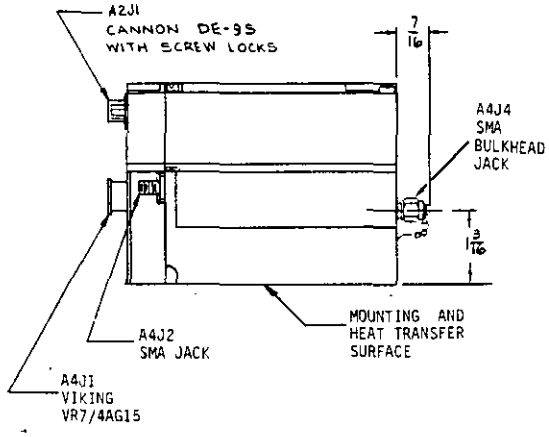
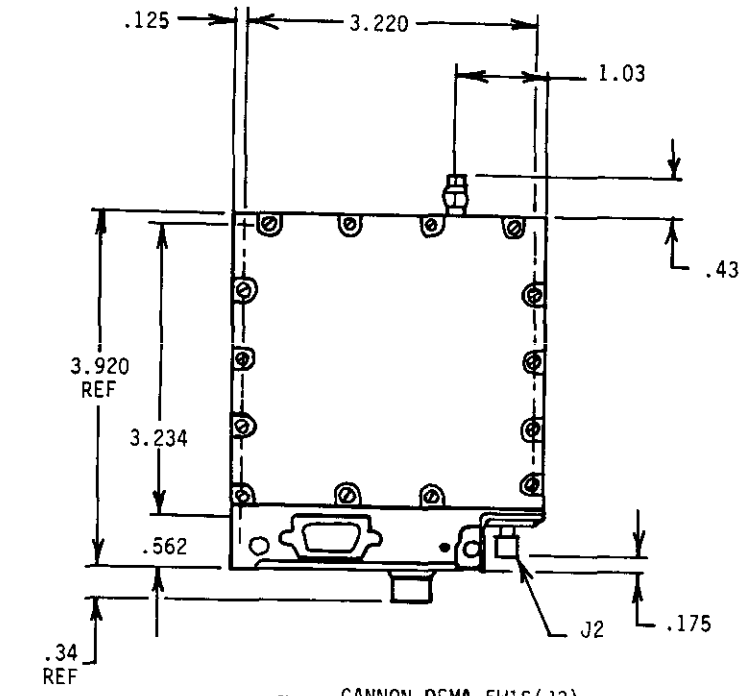
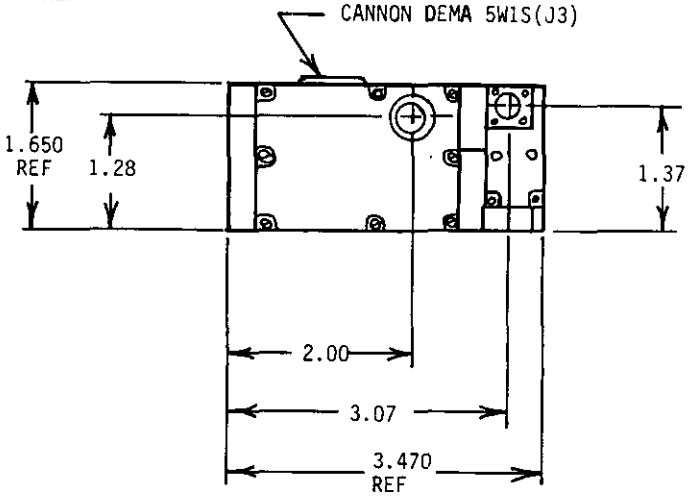


FIGURE 8 20-Channel Transmitter, Form Factor and Connector Requirements.

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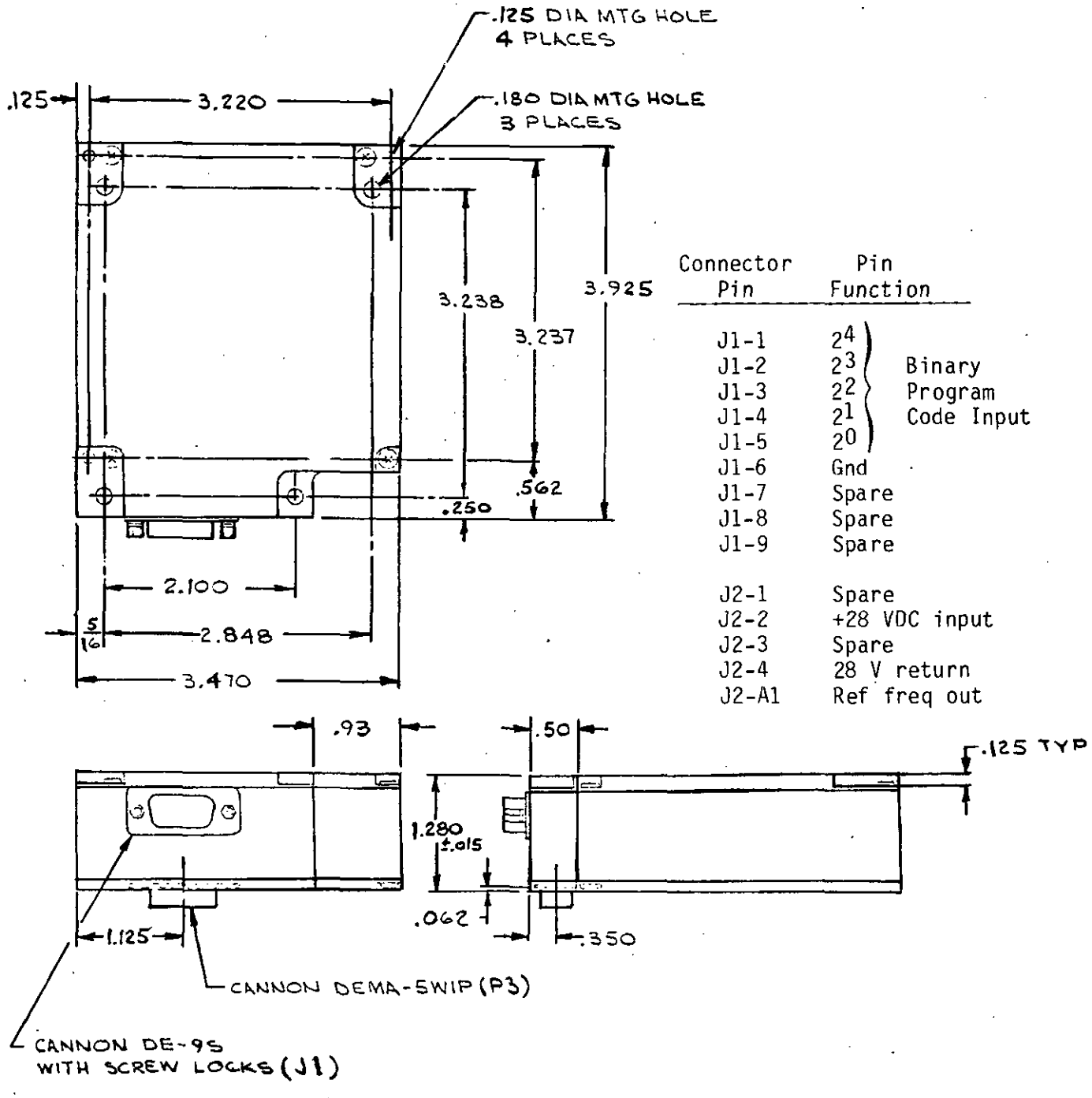




CONNECTOR PIN	PIN FUNCTION
J1-A	+28 VDC input
J1-B	Gnd
J1-C	Mod input
J1-D	Mod return
J1-E	Spare
J1-F	VCO tune
J1-G	Spare
J2	RF output
J3-1	Spare
J3-2	+28 VDC output
J3-3	Spare
J3-4	28 V return
J3-A1	Ref freq input
J4	70 MHz input

FIGURE 9 Transmitter RF Modulator, Form Factor and Connector Requirements

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
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Connector Pin	Pin Function	
J1-1	2^4	Binary Program Code Input
J1-2	2^3	
J1-3	2^2	
J1-4	2^1	
J1-5	2^0	
J1-6	Gnd	
J1-7	Spare	
J1-8	Spare	
J1-9	Spare	
J2-1	Spare	
J2-2	+28 VDC input	
J2-3	Spare	
J2-4	28 V return	
J2-A1	Ref freq out	

FIGURE 10 Transmitter Frequency Synthesizer, Form Factor and Connector Requirements.

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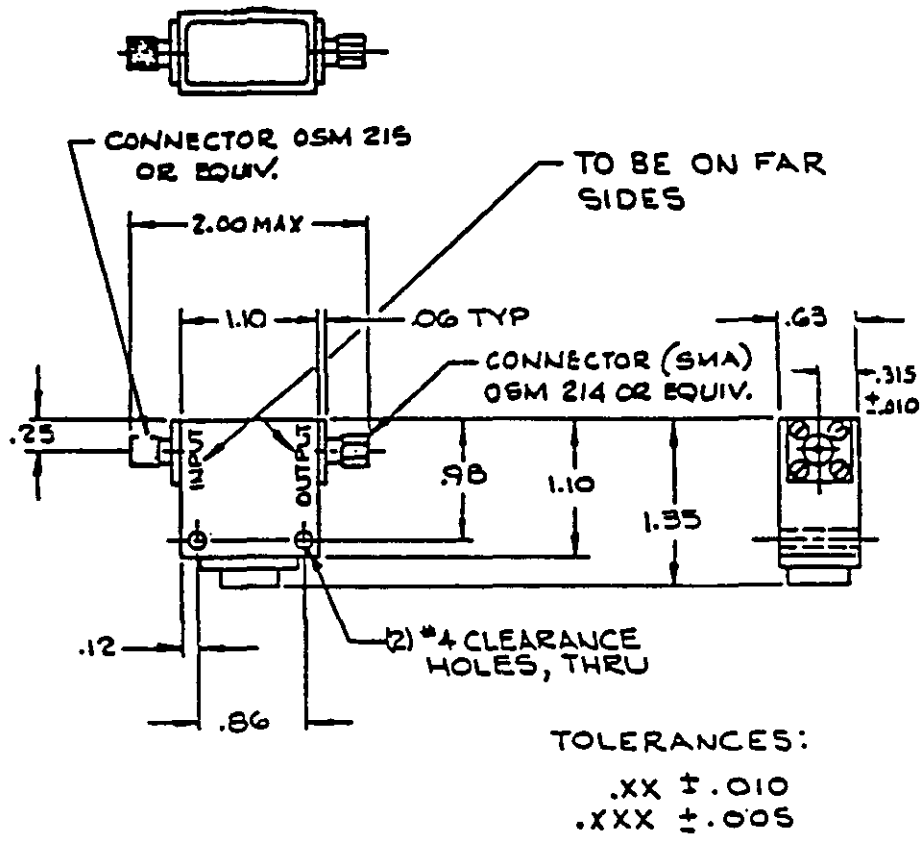


FIGURE 11 Interstage Isolator, Form Factor and Connector Requirements.

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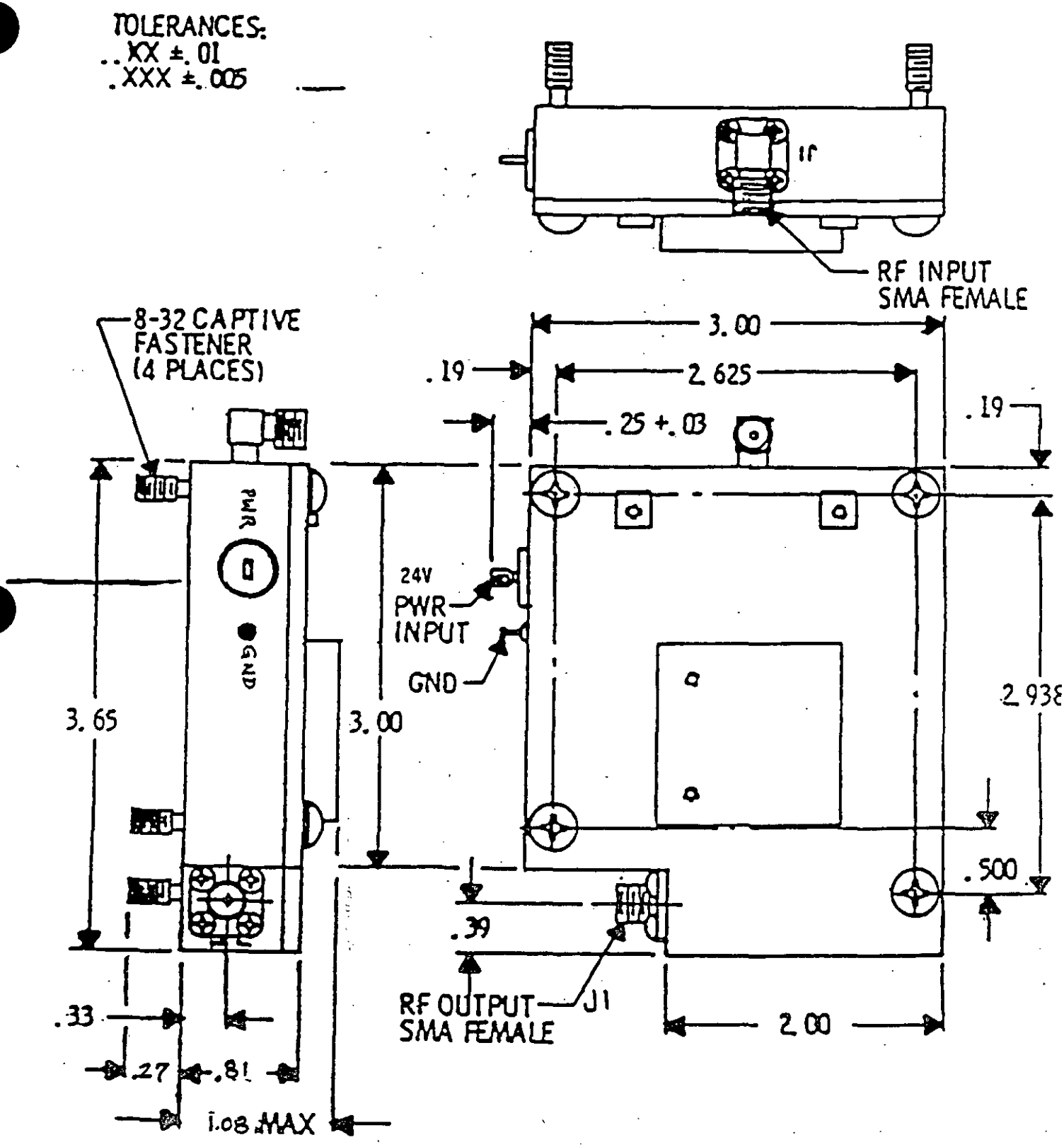


FIGURE 12 Power Amplifier Form Factor.

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Check the source to verify that this is the current version before use.

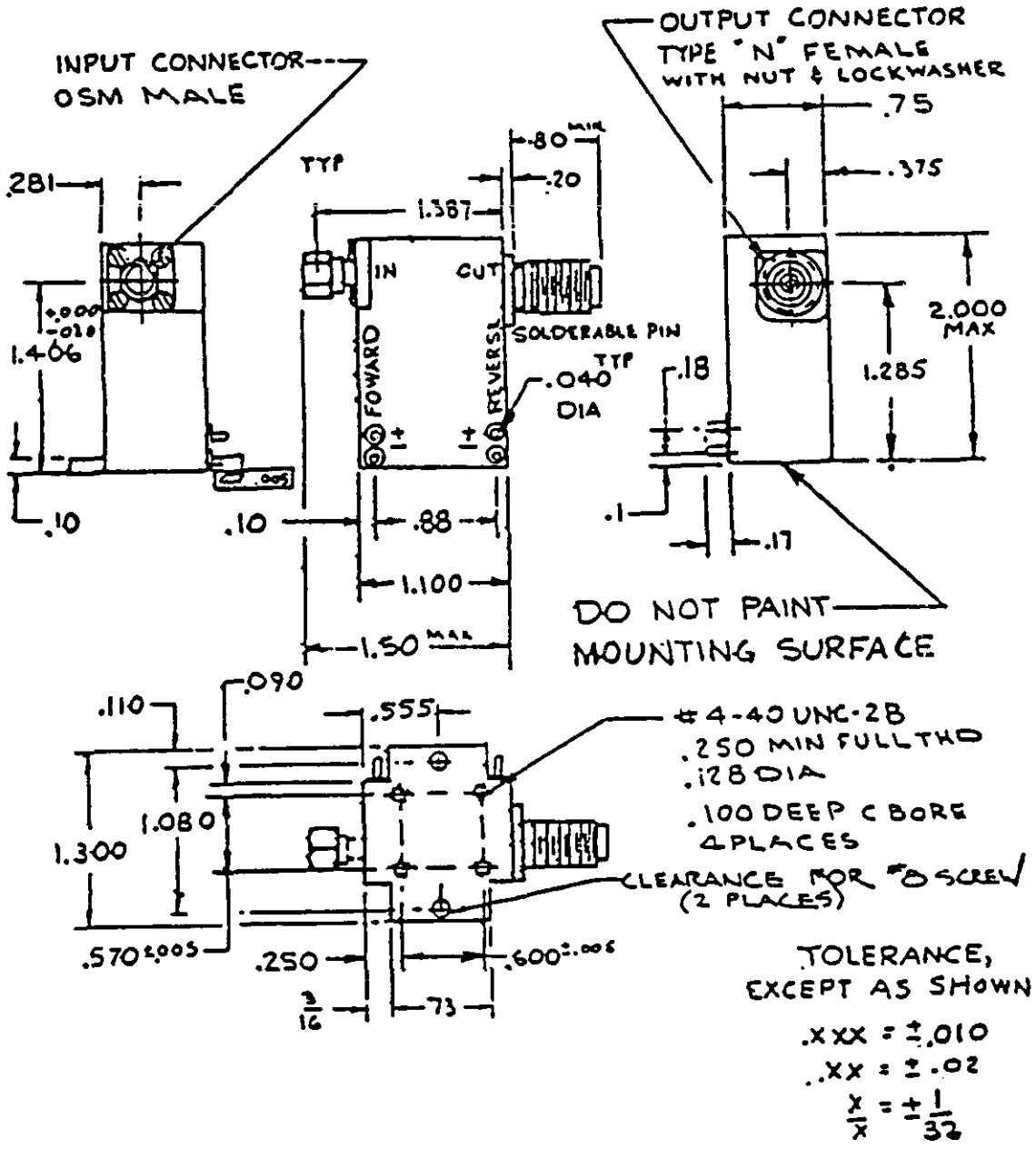
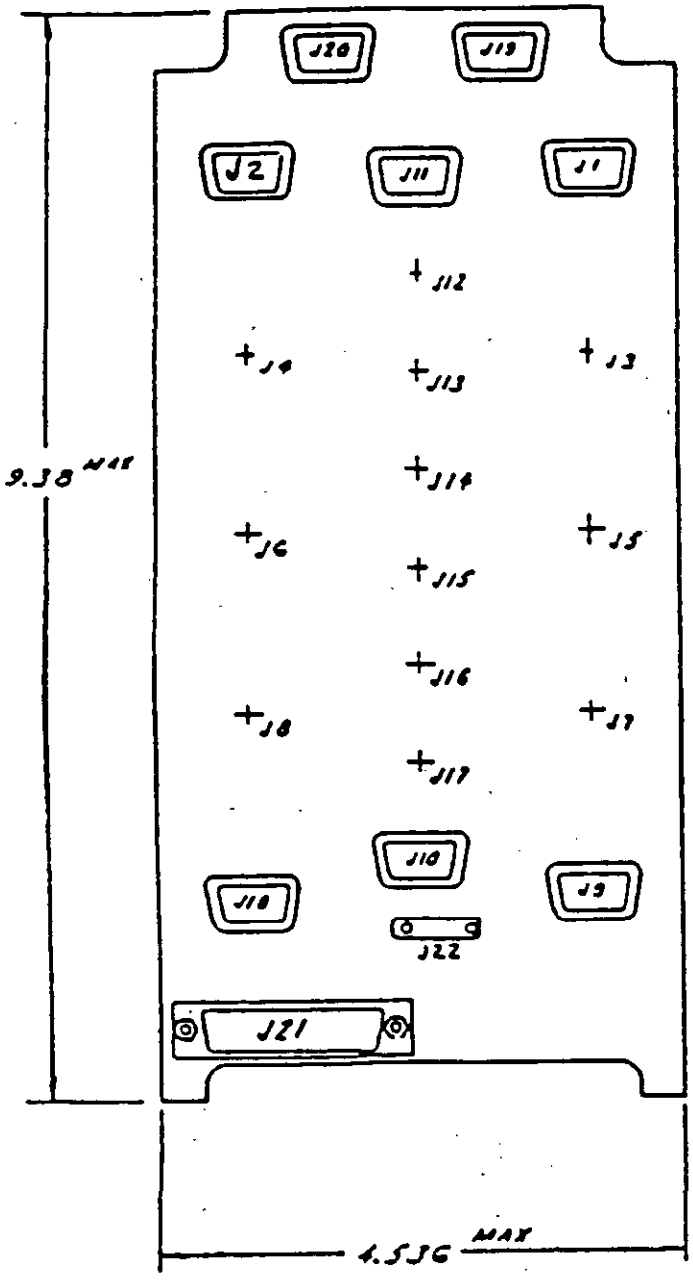
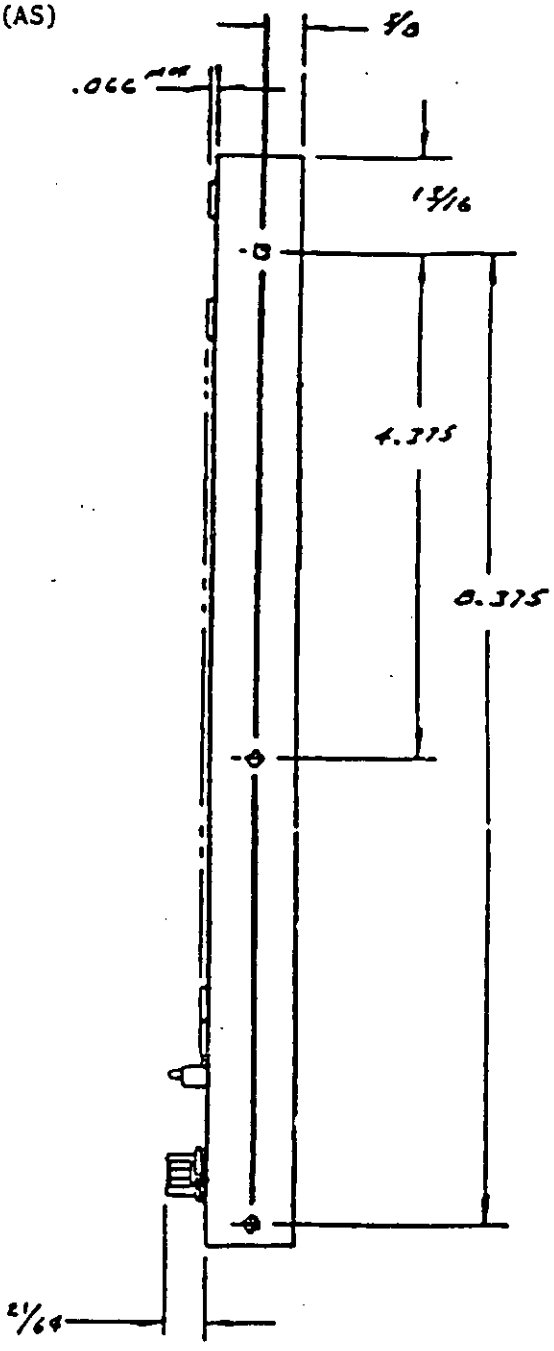


FIGURE 13 Isolator Power Monitor Form Factor.

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NOTE:

J1 through J20 ar
Type 14075-9S; J2
is Type 14073-25

J22 is MMP22-7S

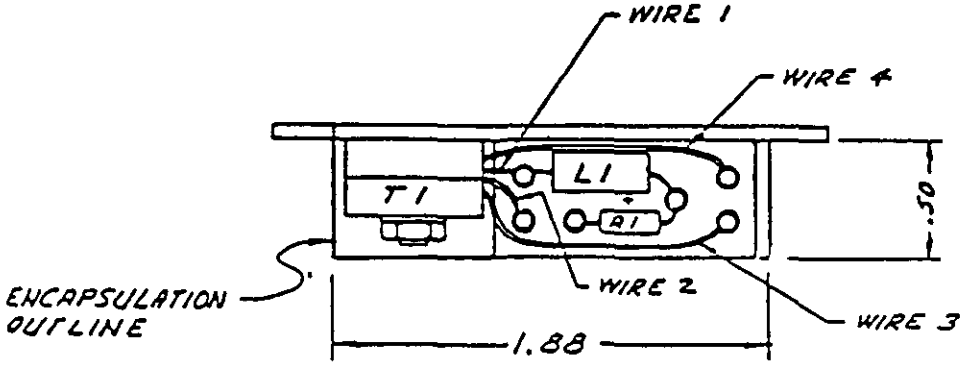
NOTE:

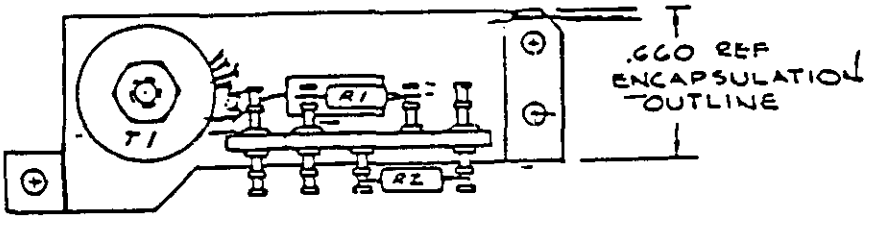
TOLERANCES UNLESS NOTED:

FRACTIONS: ±1/64 DECIMALS: XX: ±0.01 ANGLES: ±1/2°
XXX: ±0.005

FIGURE 14 TELEMETRY MOUNT, FORM FACTOR AND CONNECTOR REQUIREMENTS

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NOTE:

TOLERANCES:
DECIMALS: .XX ±0.02

TERMINAL CONNECTIONS:

E1 - Signal Input
E2 - Isolated Power and Signal Return
E3 - Signal Output
E4 - Output Return (Gnd)

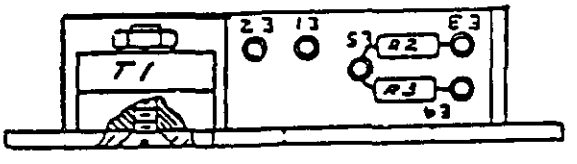
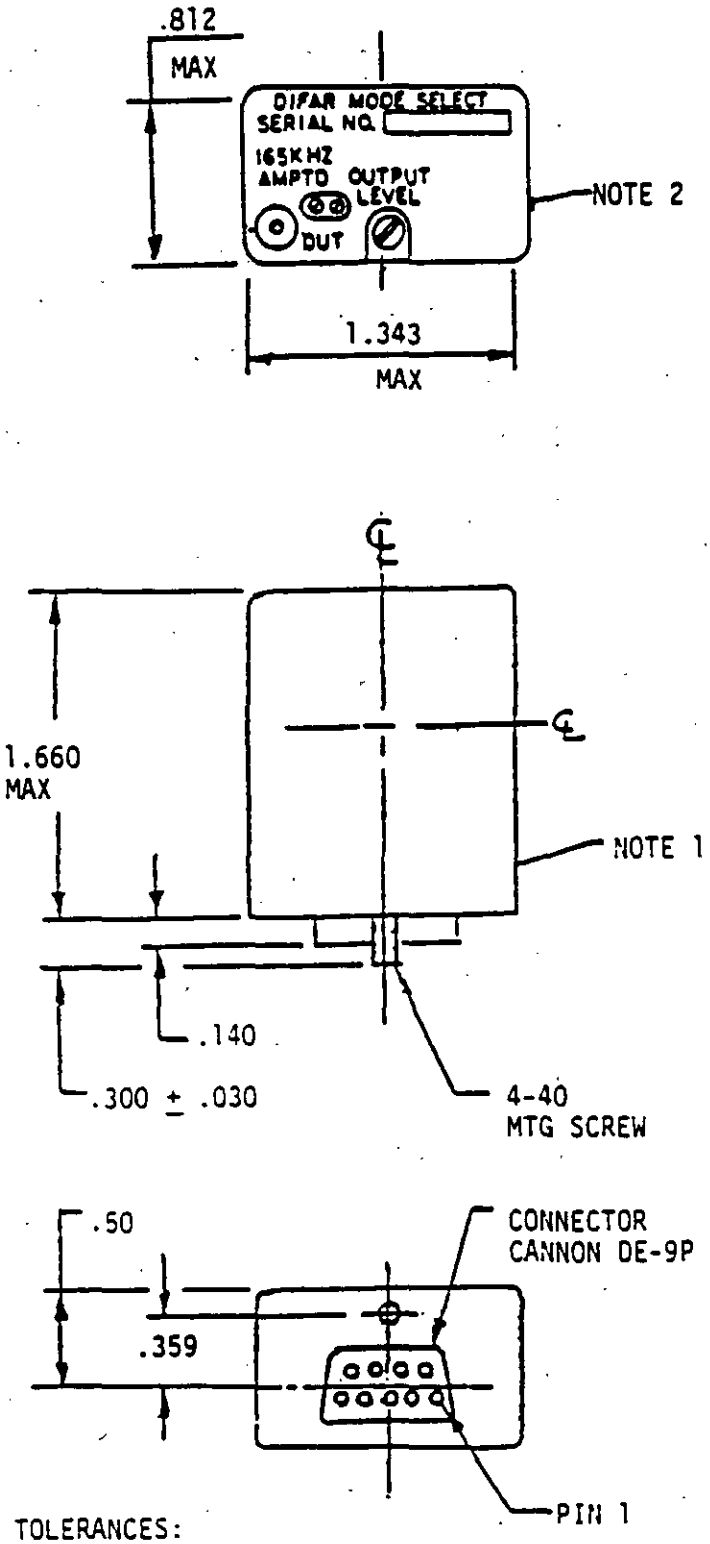


FIGURE 15 Isolation Transformer Assembly Form Factor

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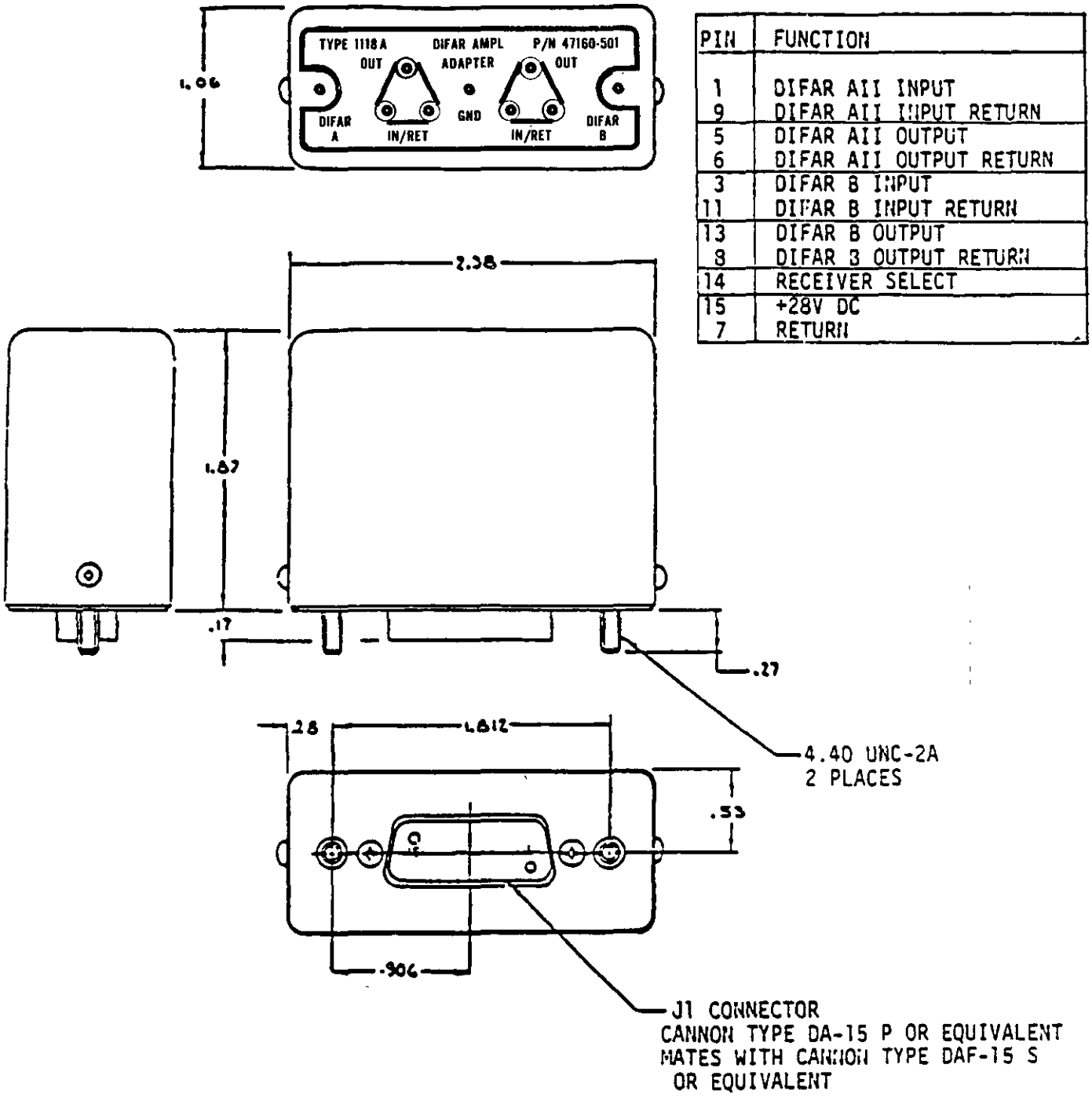
PIN	FUNCTION
1	DIFAR A II INPUT
2	AUDIO RETURN
3	AUDIO RETURN
4	DIFAR B BANDPASS FILTER INPUT
5	RELAY OUTPUT
6	RELAY COIL RETURN AND CHASSIS GND
7	+28 VDC ± 4 VDC
8	LOFAR INPUT
9	+28 VDC ± 4 VDC RELAY COIL SUPPLY (MODE CONTROL)

NOTES:

1. Finish all external surfaces with clear anodize per MIL-A-8625
2. Functions and part no. to be engraved by vendor as shown, locations optional

FIGURE 16 DIFAR Mode Select Module, Form Factor.

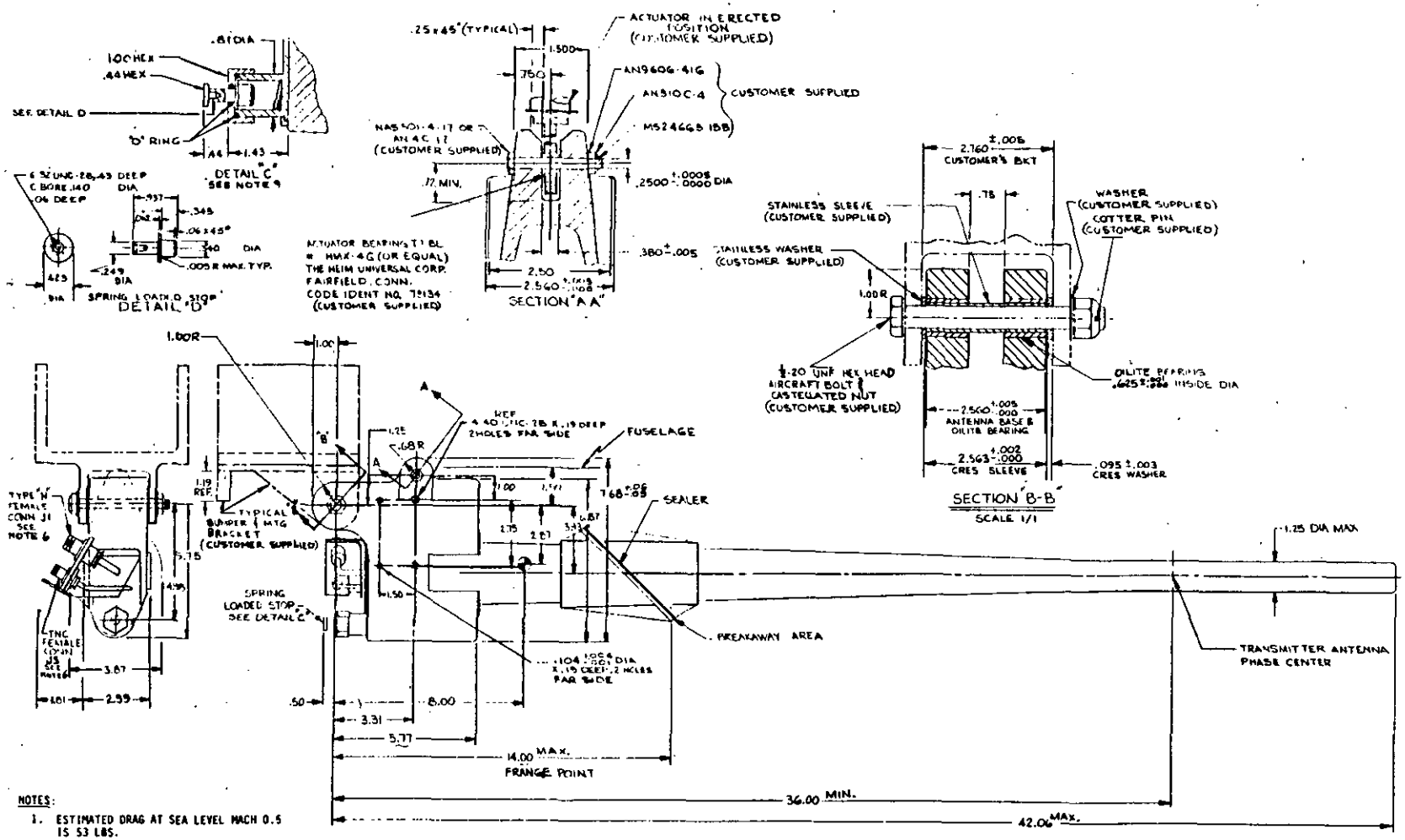
Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.



PIN	FUNCTION
1	DIFAR AII INPUT
9	DIFAR AII INPUT RETURN
5	DIFAR AII OUTPUT
6	DIFAR AII OUTPUT RETURN
3	DIFAR B INPUT
11	DIFAR B INPUT RETURN
13	DIFAR B OUTPUT
8	DIFAR B OUTPUT RETURN
14	RECEIVER SELECT
15	+28V DC
7	RETURN

FIGURE 17 DIFAR Amplifier-Adapter, Form Factor and Connector Requirements.

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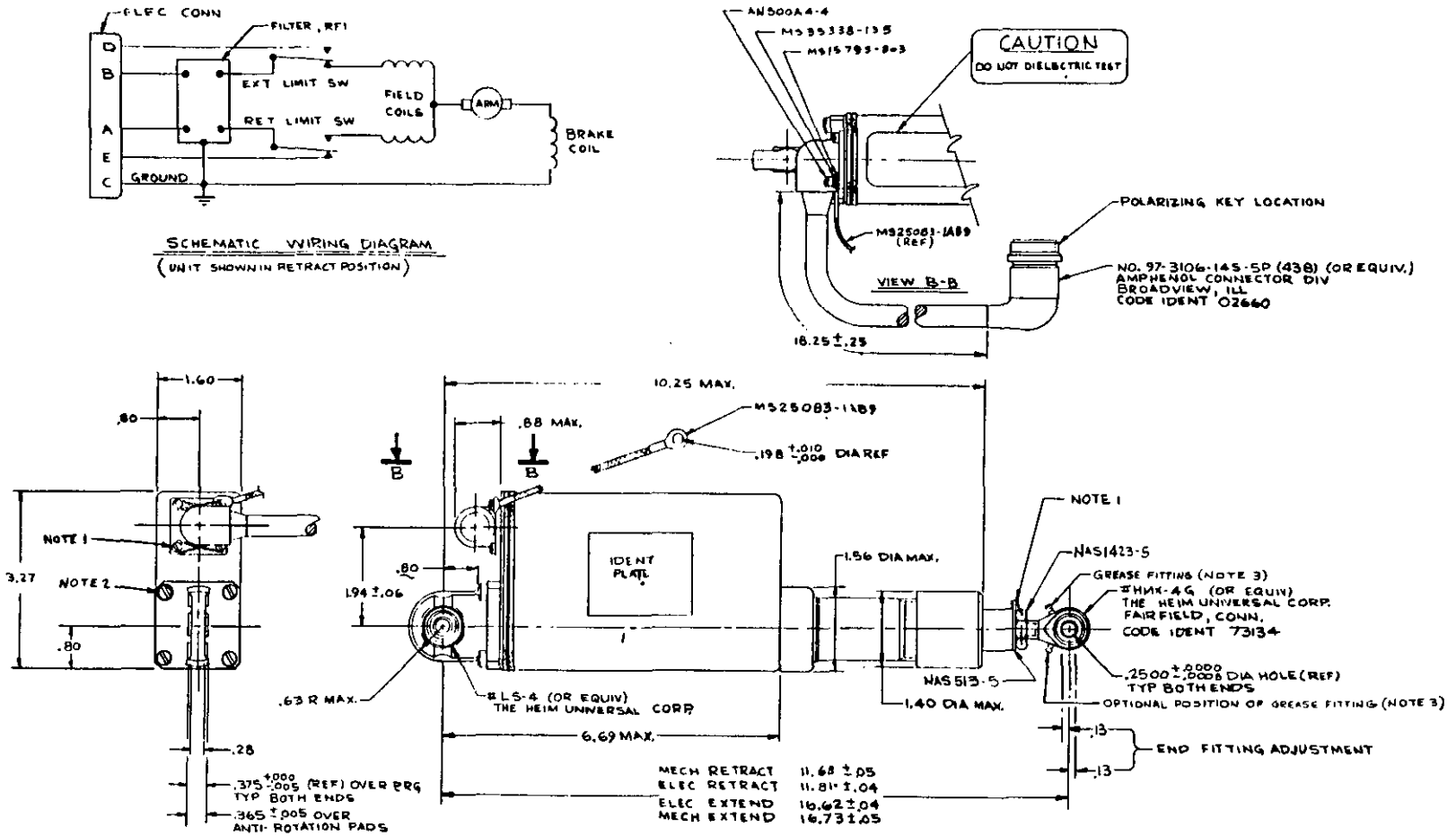


NOTES:

1. ESTIMATED DRAG AT SEA LEVEL MACH 0.5 IS 53 LBS.
2. WEIGHT IS 10.0 LBS. MAX.
3. WORST CASE FLIGHT CONDITION: AIRCRAFT NEGOTIATES A YAW ANGLE OF PLUS OR MINUS 30 DEGREES AT AN AIR SPEED OF 170 KNOTS.
4. DIMENSION SHOWN PERMITS ANTENNA PHASE CENTER TO BE 36" DOWN FROM FUSELAGE IN ERECTED POSITION.
5. CONNECTORS
   J1 - M39012/02-0004 (TYPE N)
   (COMMERCIAL KN19-108)
   J3 - M39012/29-0001 (TYPE TNC)
   (COMMERCIAL - 8483)
6. REF DESIGNATOR J1 & J3 TO BE MARKED ON CONNECTOR MOUNTING BRACKET.

FIGURE 18. Antenna AS-3033A/AKT-22, Form Factor.

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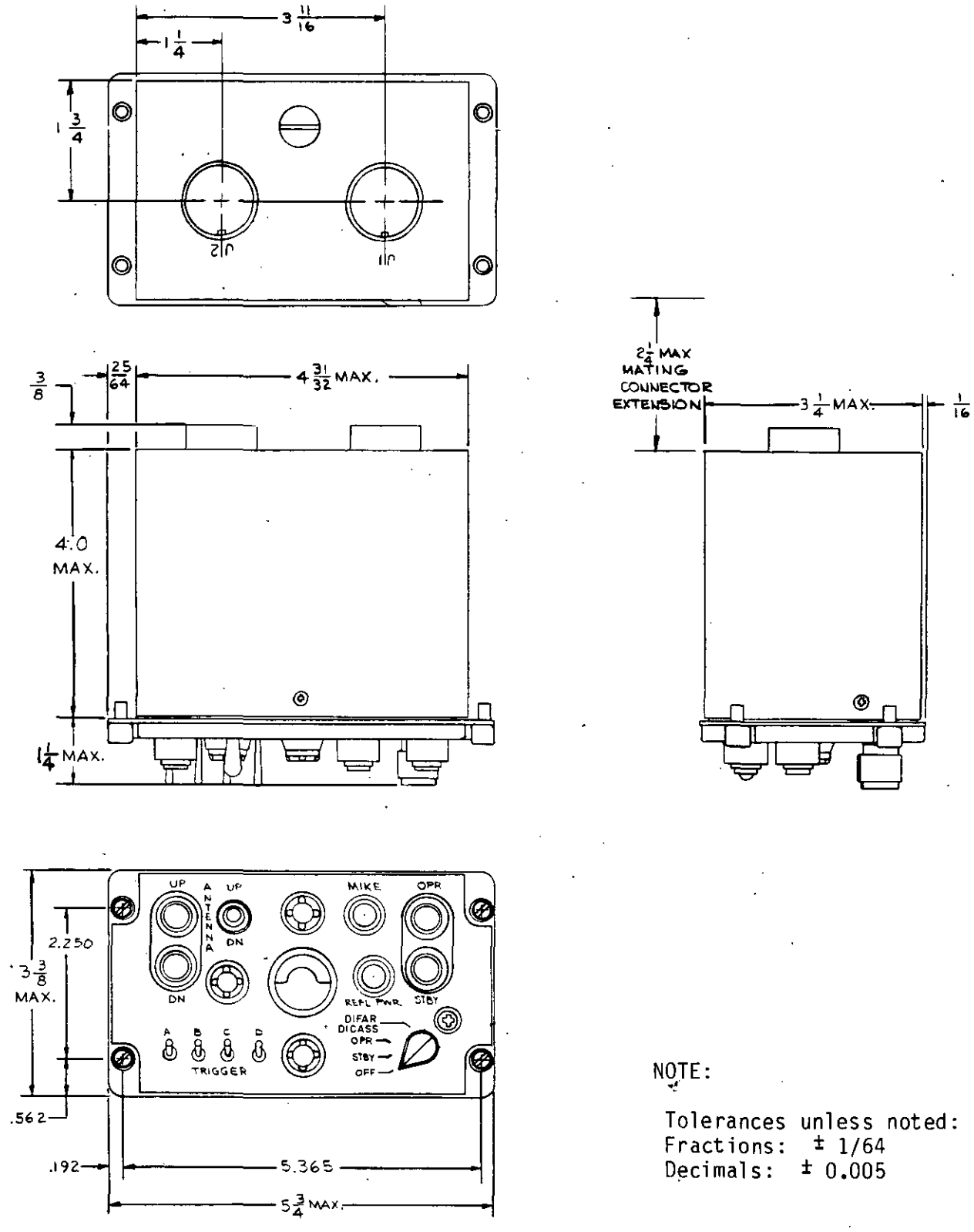


NOTES:

1. USE SAFETY WIRE PER MS 20995 AND APPLY PER DESIGN STANDARD MS 33540.
2. THREAD LOCKING DEVICE TO BE PER MIL-F-18240 CONFIGURATION "B".
3. AFTER LUBRICATION, REPLACE GREASE FITTING WITH SET SCREW MS 51966-25 AND SECURE FOR VIBRATION.
4. RETRACT DIMENSION IS WITH HMX-4G ENGAGED IN ACTUATOR ARM EXACTLY 10 TURNS.

FIGURE 19 Linear Electro-Mechanical Actuator TG-229A/AKT-22(V) Form Factor.

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NOTE:

Tolerances unless noted:
Fractions: ± 1/64
Decimals: ± 0.005

FIGURE 20 Control Indicator C-10742/AKT-22(V) Form Factor.

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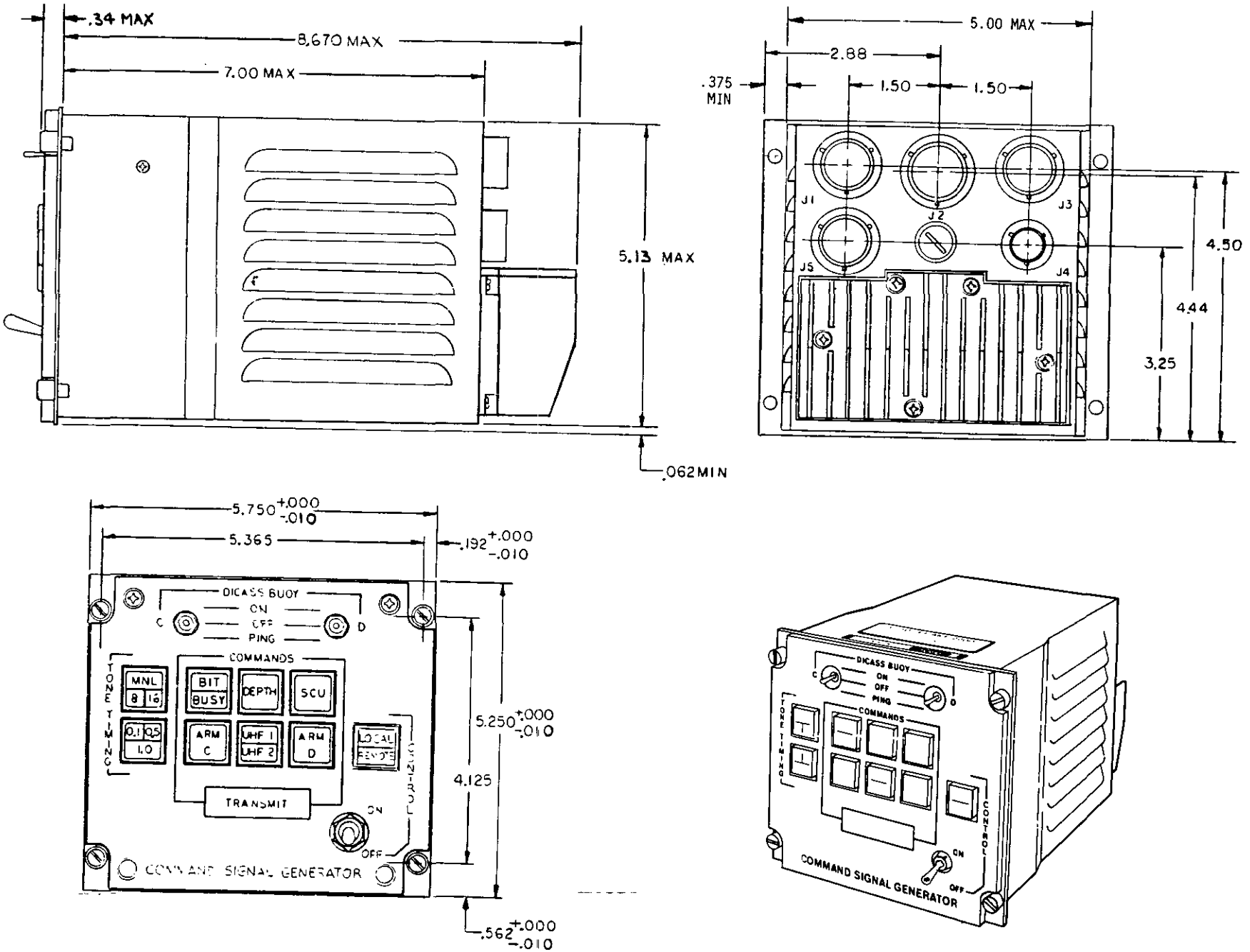


FIGURE 21 Control Generator Form Factor

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Check the source to verify that this is the current version before use.

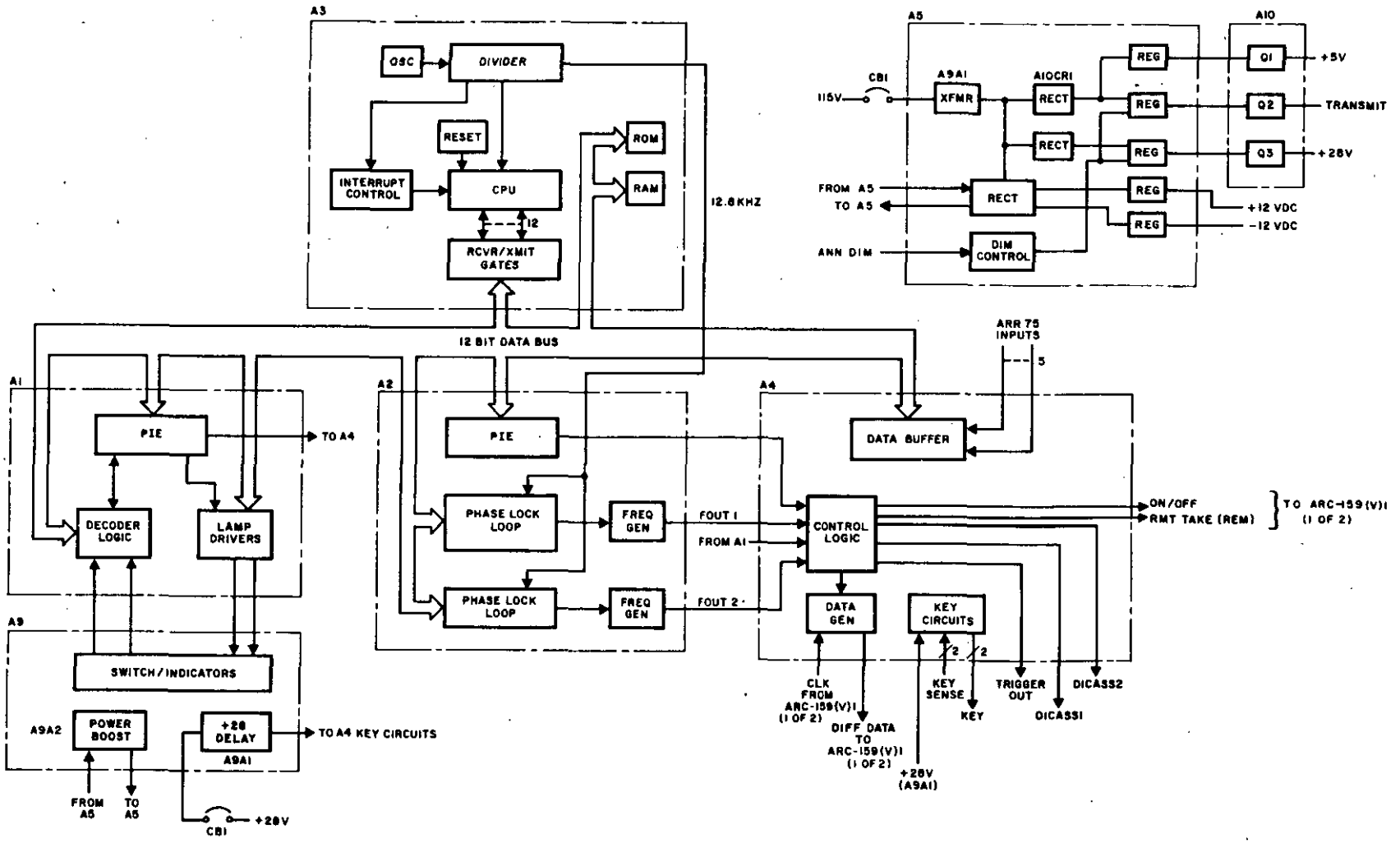


FIGURE 22. Control Generator, Organizational Block Diagram.

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Check the source to verify that this is the current version before use.

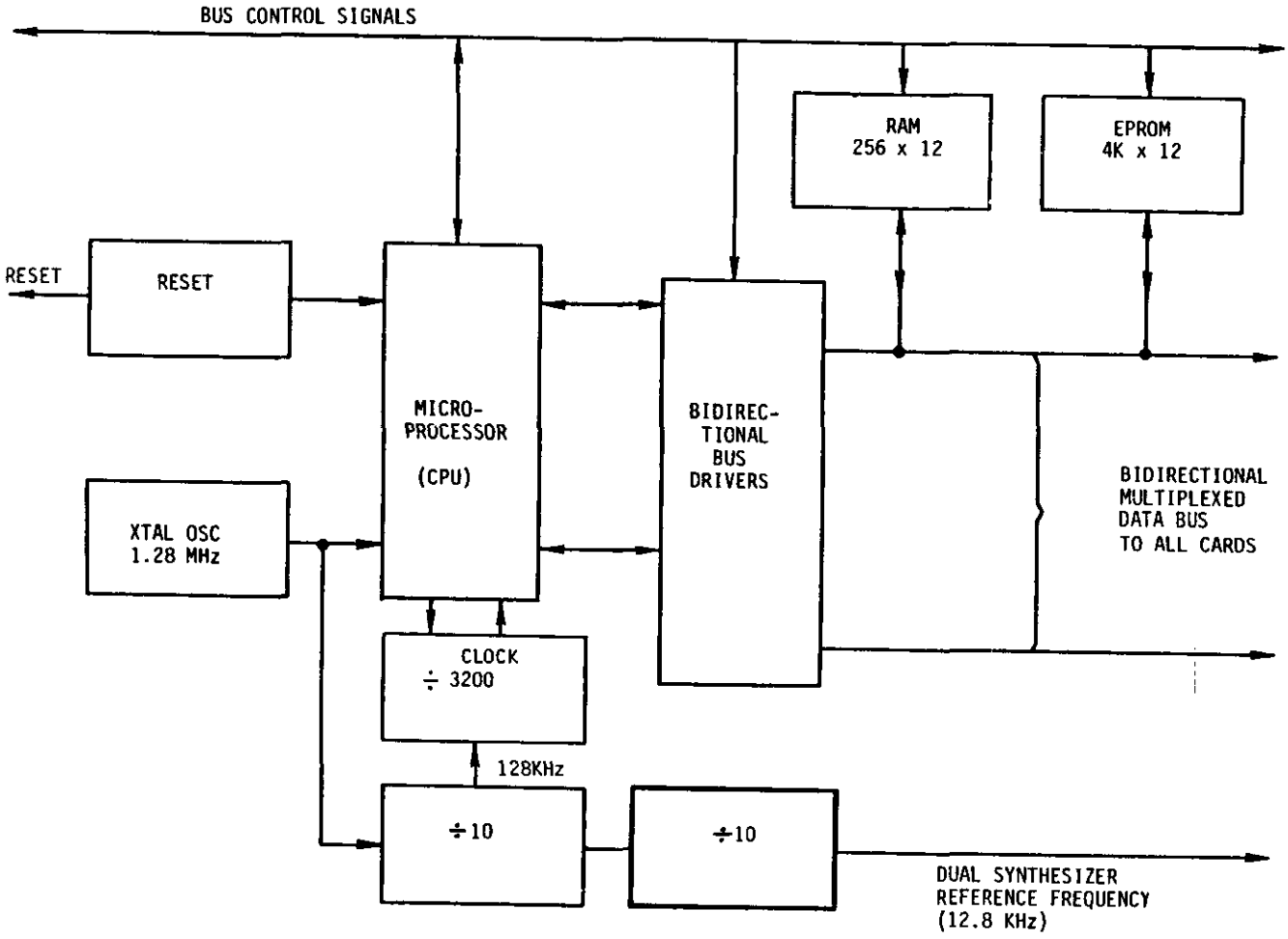


FIGURE 23 Bidirectional Data Bus Flow.

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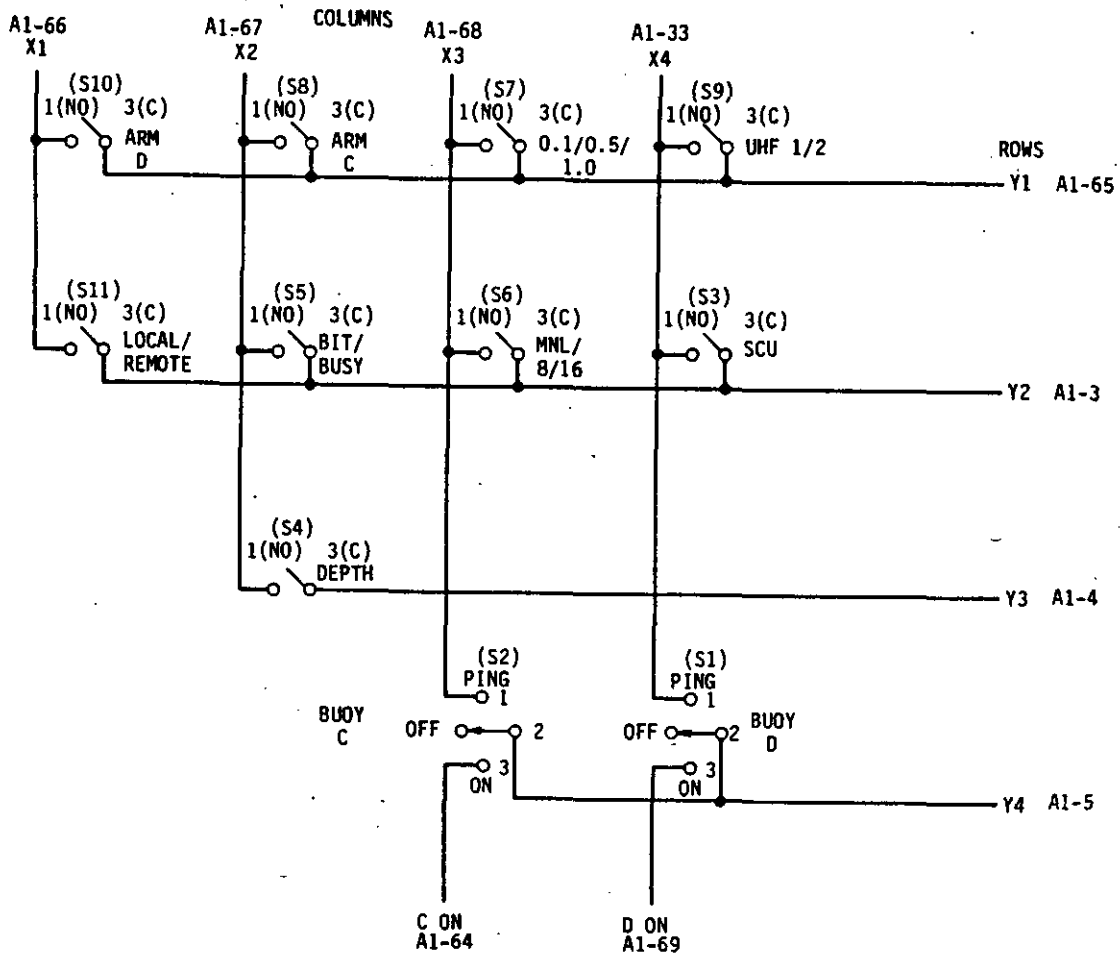
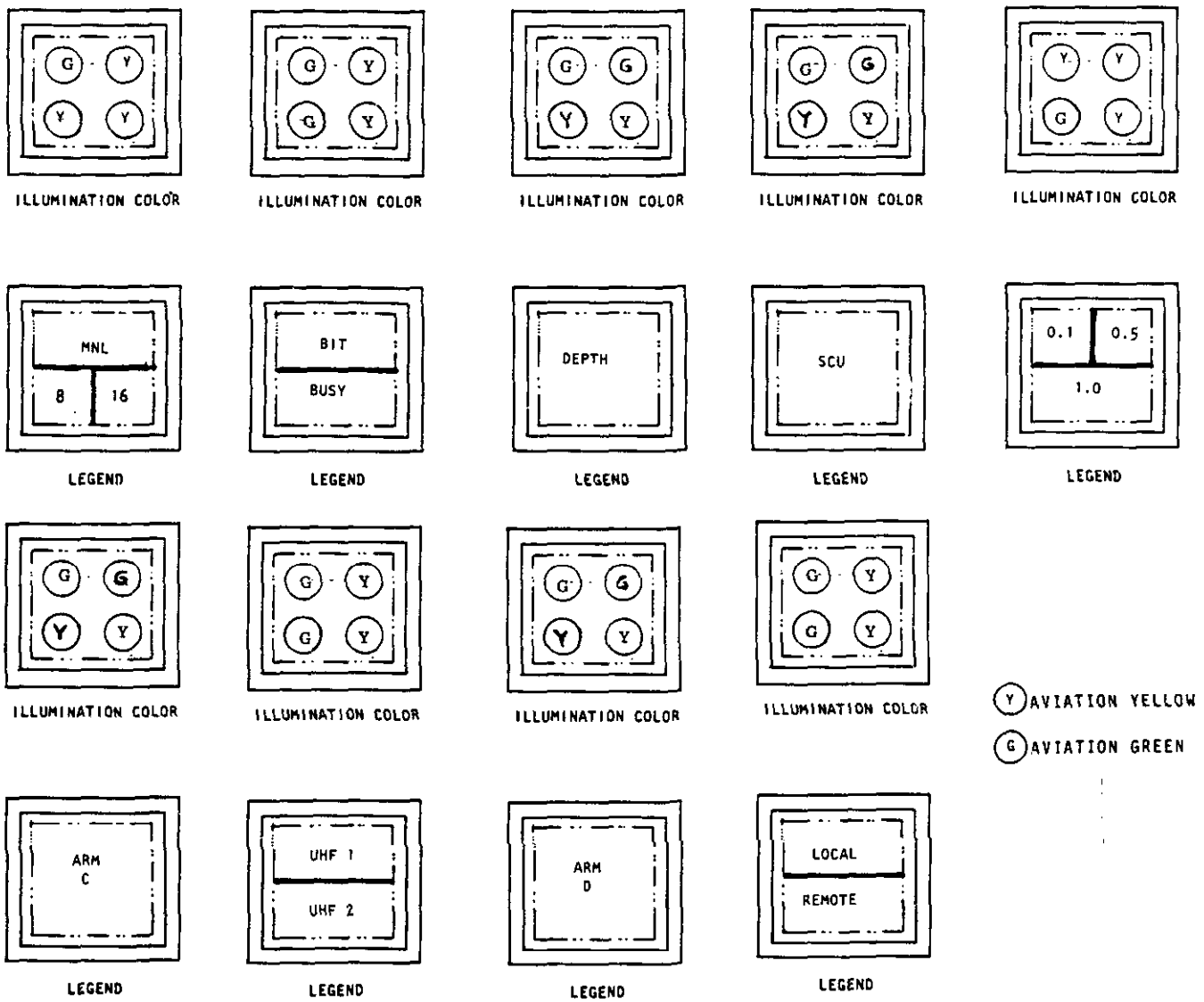


FIGURE 24 Front Panel Switch Connections.

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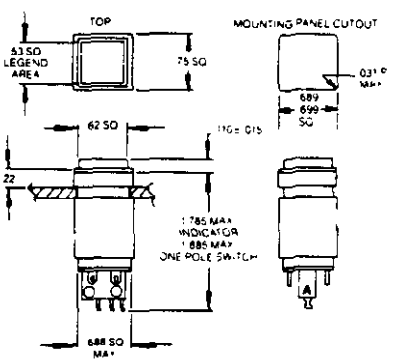
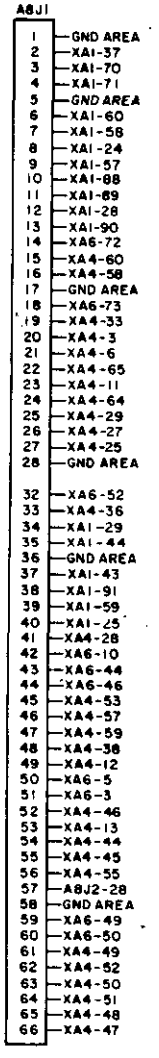
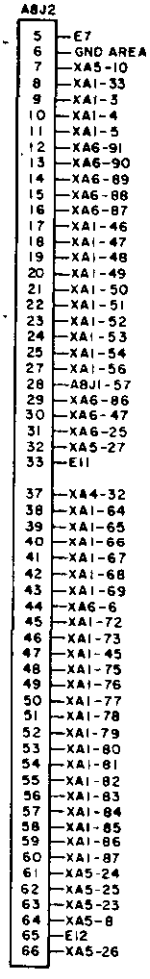


FIGURE 25 Push Button Switches, Form Factor and Legends.

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VOLTAGE / GROUND DISTRIBUTION		
LOCATION		
FROM	TO	VOLTAGE
XA1-1	XA2-1	+5VDC
XA1-2	XA2-2	
XA2-1	XA3-1	
XA2-2	XA3-2	
XA3-1	XA4-1	
XA3-2	XA4-2	
XA4-1	XA5-1	
XA4-2	XA5-2	
XA5-1	XA6-1	
XA5-2	XA6-2	
XA6-1	A8J1-29	
XA6-2	A8J1-29	
XA1-1	A9P2-1	+5VDC
XA1-31	XA2-31	+12V
XA2-31	XA3-31	
XA3-31	XA4-31	
XA4-31	XA5-31	
XA5-31	XA6-31	
XA6-31	A8J1-30	+12V
XA1-92	XA2-92	-12V
XA2-92	XA3-92	
XA3-92	XA4-92	
XA4-92	XA5-92	
XA5-92	XA6-92	
XA6-92	A8J1-31	
XA1-92	A9P2-36	-12V
A8J2-3	E3	115 VAC SPLY
A8J2-34	E1	115 VAC RTN
XA4-32	A9P2-17	+28V
XA5-19	E10	+11VDC
XA5-20	E10	+11VDC

LOCATION		
FROM	TO	GROUND
XA1-74	XA1-62	DIGITAL
XA1-62	XA2-62	
XA2-62	XA3-62	
XA3-62	XA3-5, XA4-62	
XA4-62	XA5-62	
XA5-62	XA6-62	
XA6-62	XA6-74, GND AREA	
XA1-74	XA1-63	DIGITAL
XA1-63	XA2-63	
XA2-63	XA3-63	
XA3-63	XA3-5, XA4-63	
XA4-63	XA5-63	
XA5-63	XA6-63	
XA6-63	XA6-74, GND AREA	
XA1-61	XA2-61	ANALOG
XA2-61	XA3-61	
XA3-61	XA4-61	
XA4-61	XA5-61	
XA5-61	XA6-61	

FIGURE 26 Module Interconnect Printed Wiring Board Interconnection.

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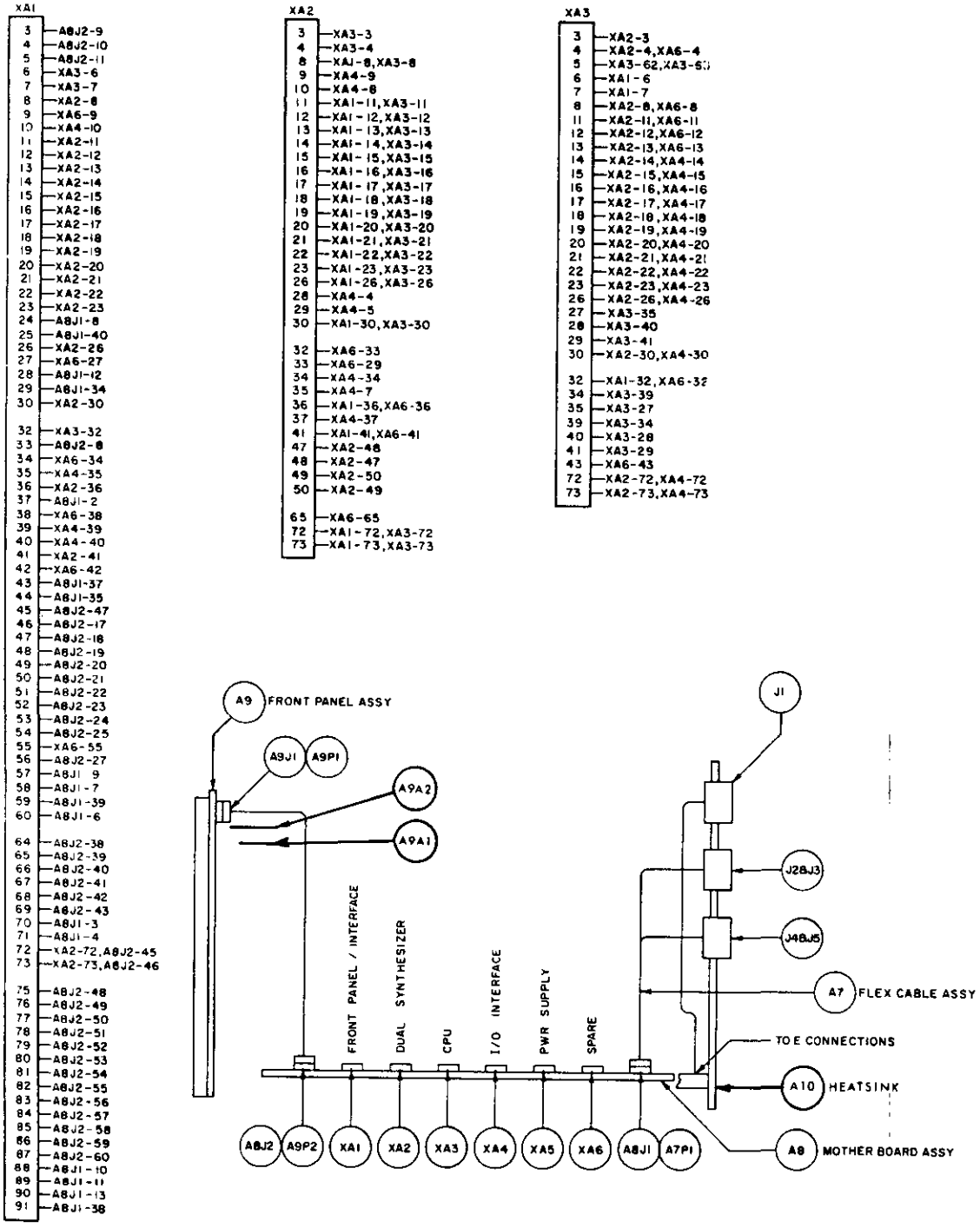


FIGURE 26 Module Interconnect Printed Wiring Board Interconnection. (Continued)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

XA4

3	A8J1-20
4	XA2-28
5	XA2-29
6	A8J1-21
7	XA2-35
8	XA2-10
9	XA2-9
10	XA1-10
11	A8J1-23
12	A8J1-49
13	A8J1-53
14	XA3-14,XA6-14
15	XA3-15,XA6-15
16	XA3-16,XA6-16
17	XA3-17,XA6-17
18	XA3-18,XA6-18
19	XA3-19,XA6-19
20	XA3-20,XA6-20
21	XA3-21,XA6-21
22	XA3-22,XA6-22
23	XA3-23,XA6-23
25	A8J1-27
26	XA3-26,XA6-26
27	A8J1-26
28	A8J1-41
29	A8J1-25
30	XA3-30,XA6-30
32	A8J2-37
33	A8J1-19
34	XA2-34
35	XA1-35
36	A8J1-33
37	XA2-37
38	A8J1-48
39	XA1-39
40	XA1-40
44	A8J1-54
45	A8J1-55
46	A8J1-52
47	A8J1-66
48	A8J1-65
49	A8J1-61
50	A8J1-63
51	A8J1-64
52	A8J1-62
53	A8J1-45
54	XA6-52
55	A8J1-56
57	A8J1-46
58	A8J1-16
59	A8J1-47
60	A8J1-15
64	A8J1-24
65	A8J1-22
72	XA3-72,XA5-72
73	XA3-73,XA5-73

XA5

4	E9
5	E16
6	E6
7	E8
8	A8J2-64,XA5-9
9	XA5-8
10	A8J2-7
11	E5
15	E4
16	E14
17	E13,A9A2E2
20	A9A2E6
21	E15
22	E15
23	A8J2-63
24	A8J2-61
25	A8J2-62
26	A8J2-66
27	A8J2-32
72	XA4-72,XA6-72
73	XA4-73,XA6-73
1	A9A2E5
56	A9A2E4
57	A9A2E1
62	A9A2E3

XA6

3	A8J1-51
4	XA3-4
5	A8J1-50
6	A8J2-44
8	XA3-8
9	XA1-9
10	A8J1-42
11	XA3-11
12	XA3-12
13	XA3-13
14	XA4-14
15	XA4-15
16	XA4-16
17	XA4-17
18	XA4-18
19	XA4-19
20	XA4-20
21	XA4-21
22	XA4-22
23	XA4-23
25	A8J2-31
26	XA4-26
27	XA1-27
29	XA2-33
30	XA4-30
32	XA3-32
33	XA2-32
34	XA1-34
36	XA2-36
38	XA1-38
41	XA2-41
42	XA1-42
43	XA3-43
44	A8J1-43
46	A8J1-44
47	A8J2-30
49	A8J1-59
50	A8J1-60
52	A8J1-32,XA4-54
55	XA1-55
65	XA2-65
72	XA5-72,A8J1-14
73	XA5-73,A8J1-18
86	A8J2-29
87	A8J2-16
88	A8J2-15
89	A8J2-14
90	A8J2-13
91	A8J2-12

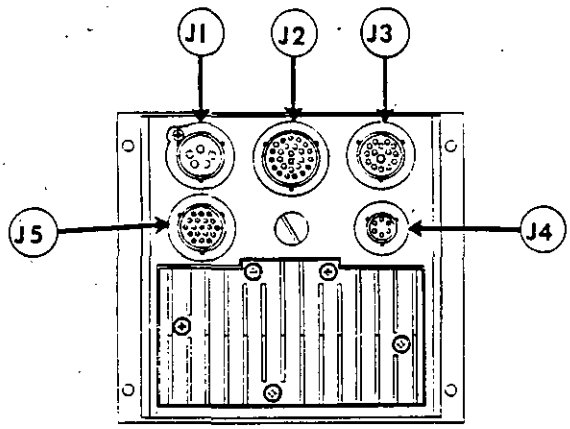


FIGURE 26 Module Interconnect Printed Wiring Board Interconnection. (Continued)

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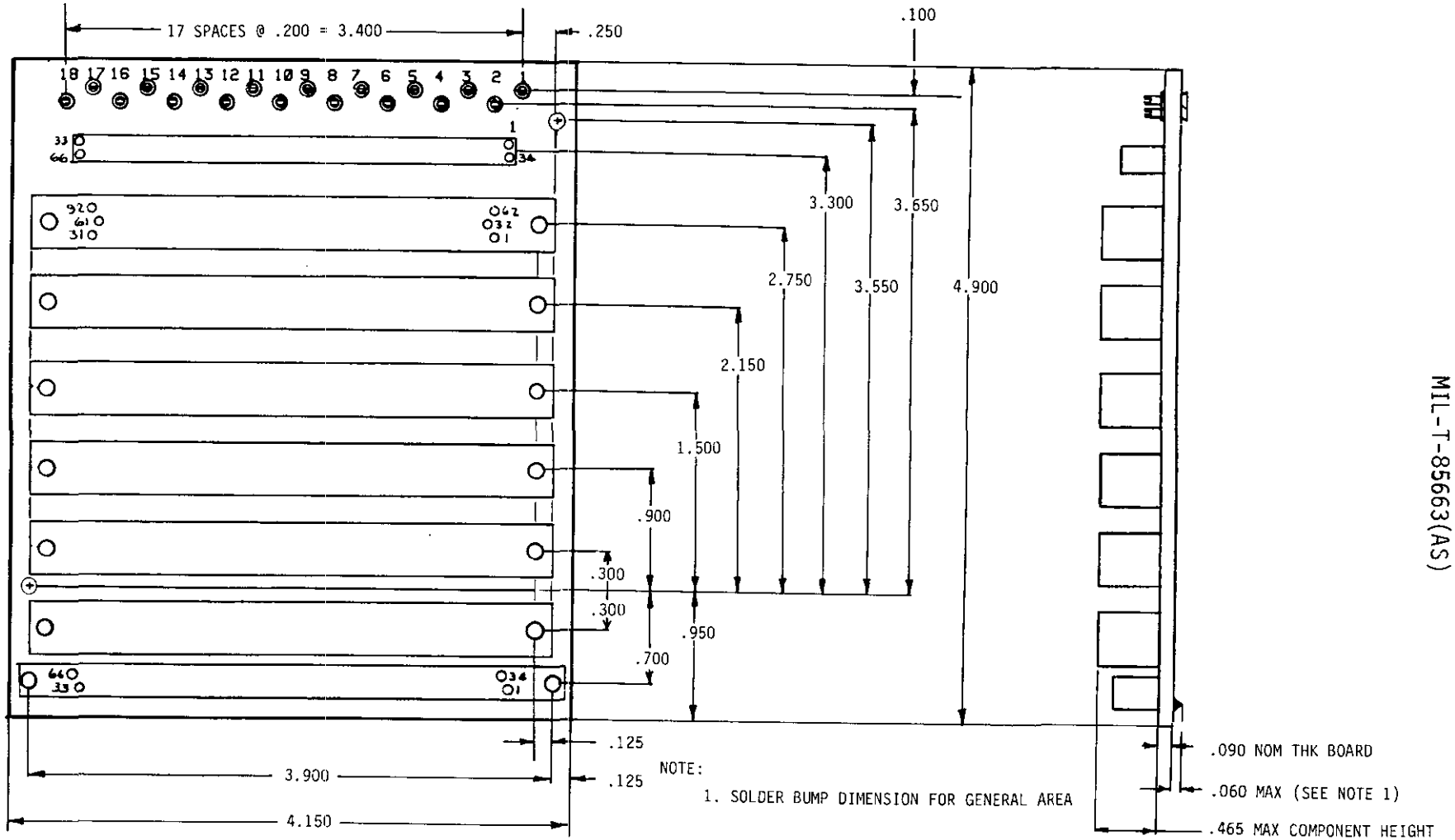


FIGURE 27 Module Interconnect Printed Wiring Board Form Factor

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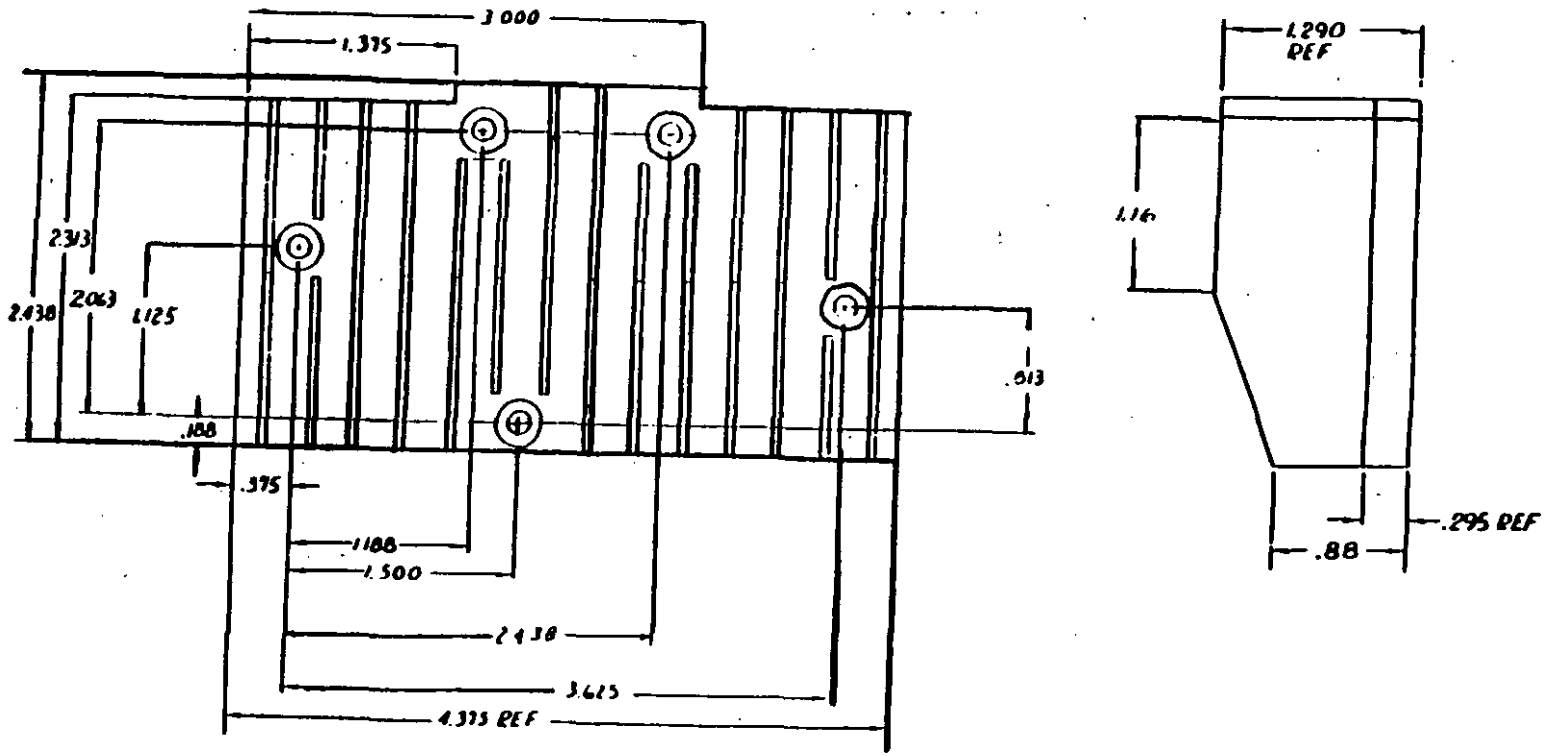
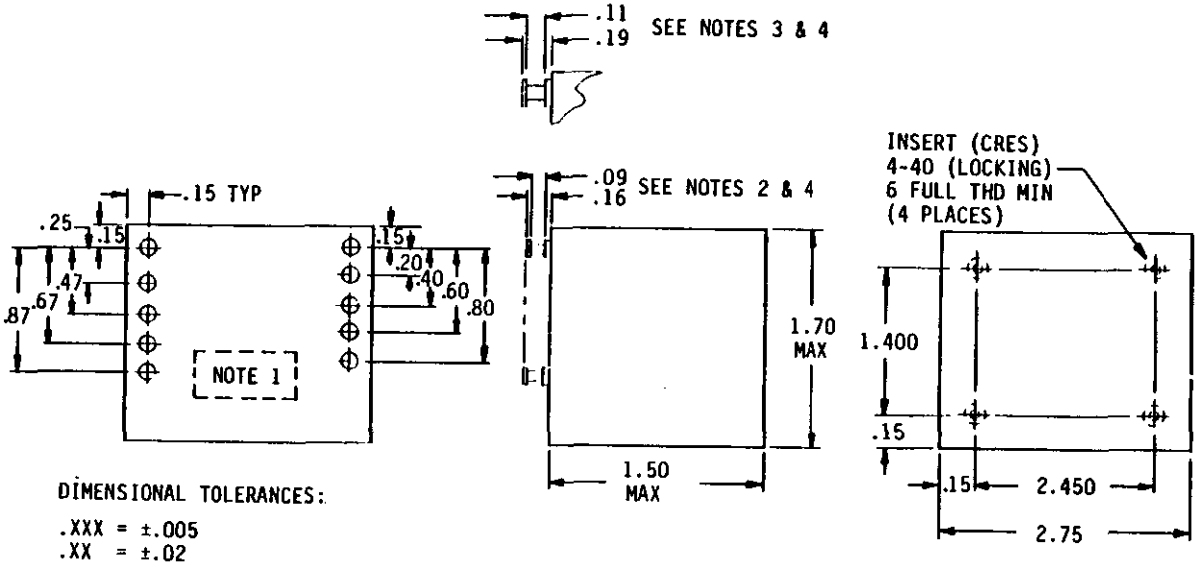


FIGURE 28. Power Supply Heat Sink, Form Factor.

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NOTES:
1. MARK THIS AREA "TDG1015".
2. TERMINALS 1, 2, 3, 6, 7, 8, 9 & 10 TO BE CAMBION 160-1512 OR EQUIVALENT.
3. TERMINALS 4 & 5 TO BE CAMBION 160-1797 OR EQUIVALENT.
4. WIRE AND SOLDER NOT TO EXCEED .032 ABOVE TERMINAL.

FIGURE 29 Power Supply Transformer, Form Factor.

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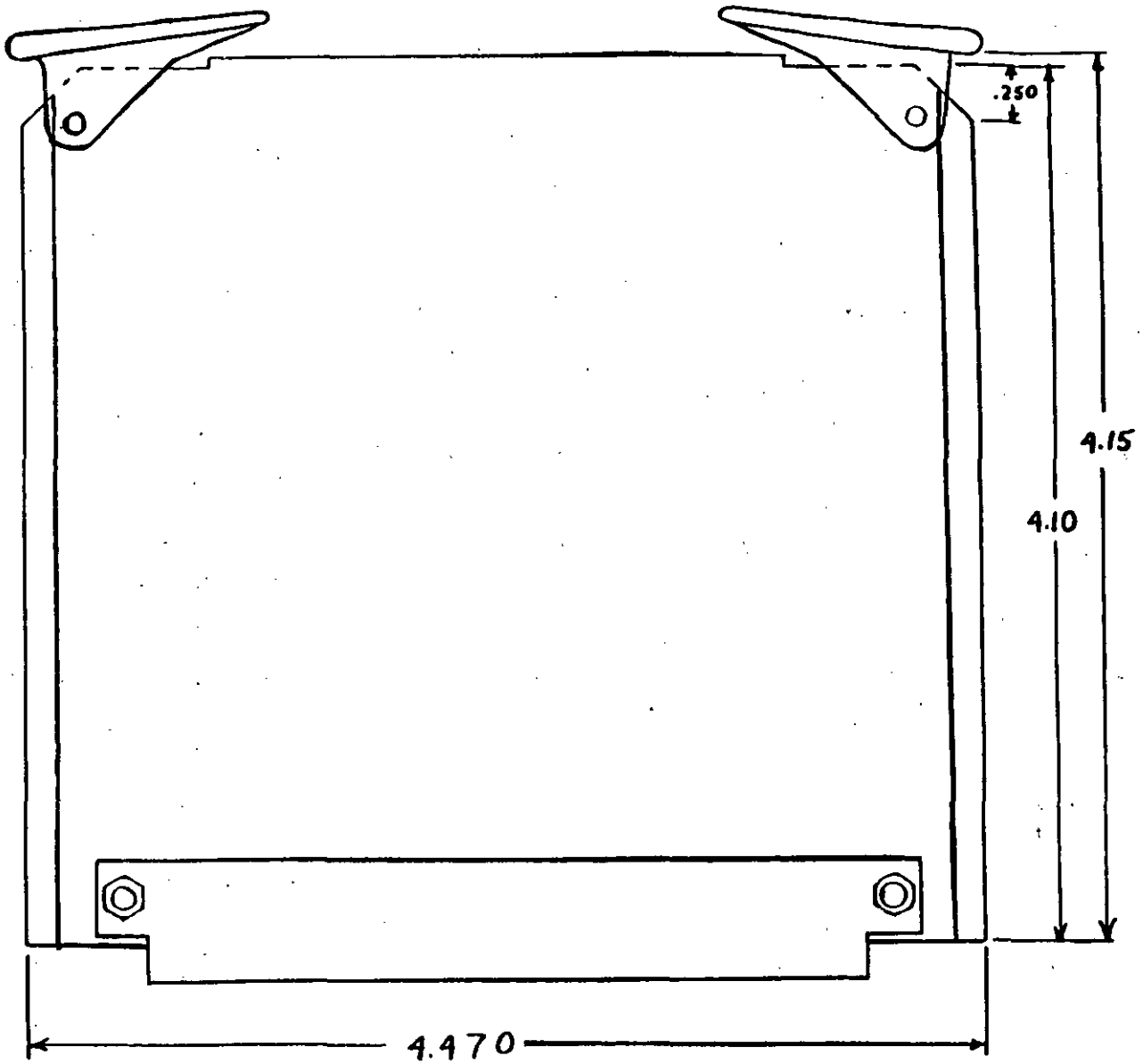


FIGURE 30. Electronic Component Modules, Form Factor.

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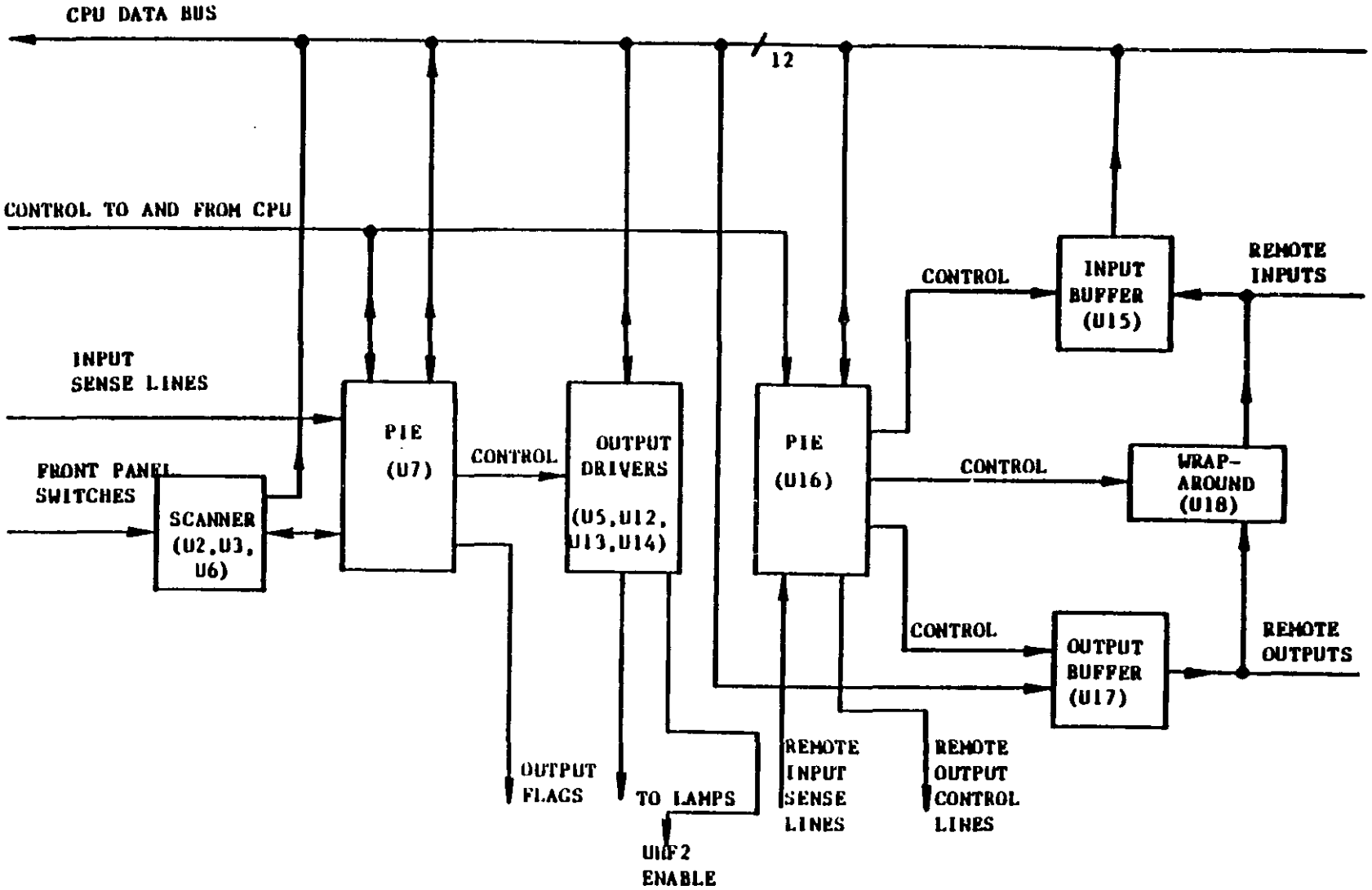


FIGURE 31 Functional Block Diagram of Module A1.

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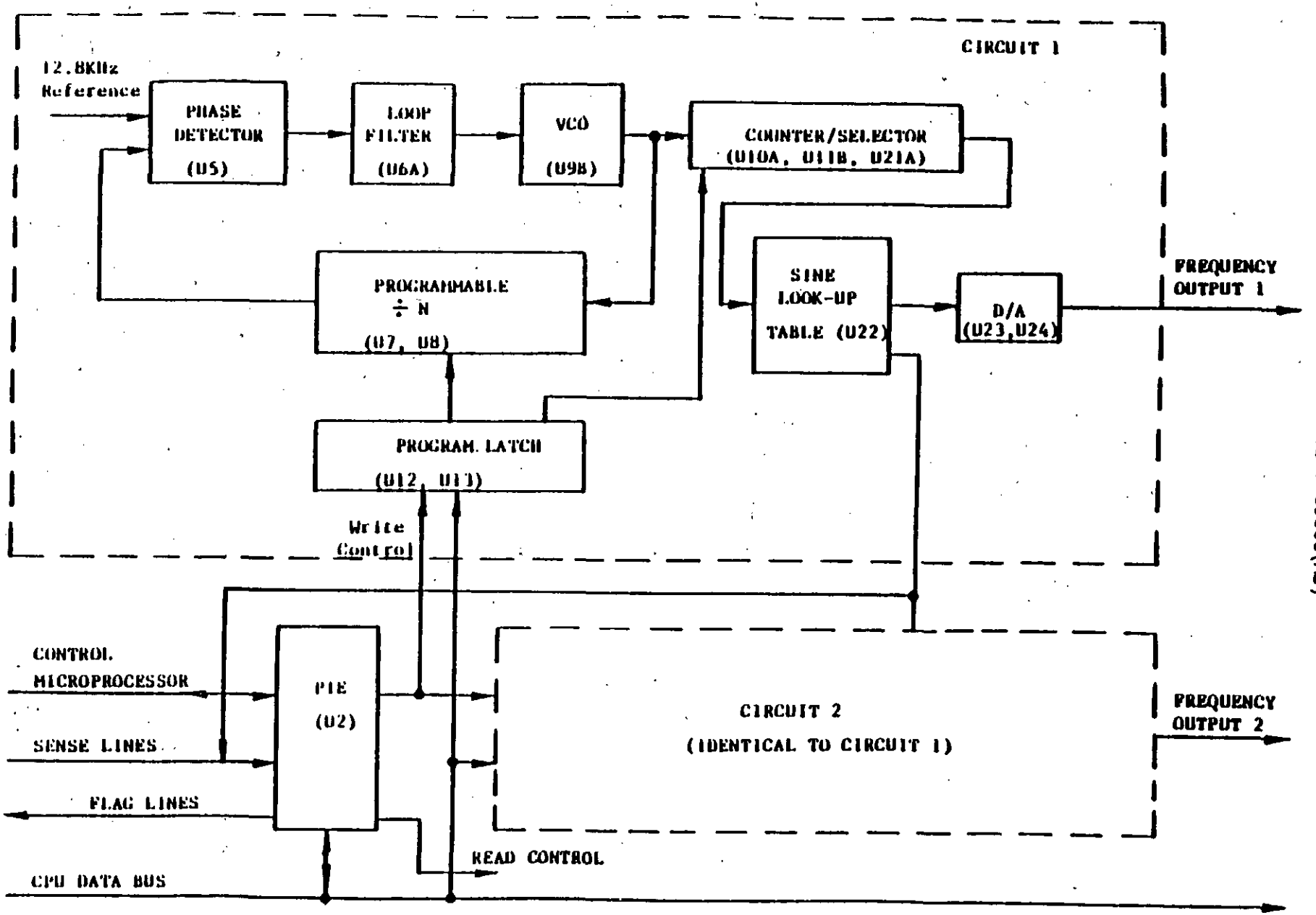


Figure 32 Functional Block Diagram of Module A2.

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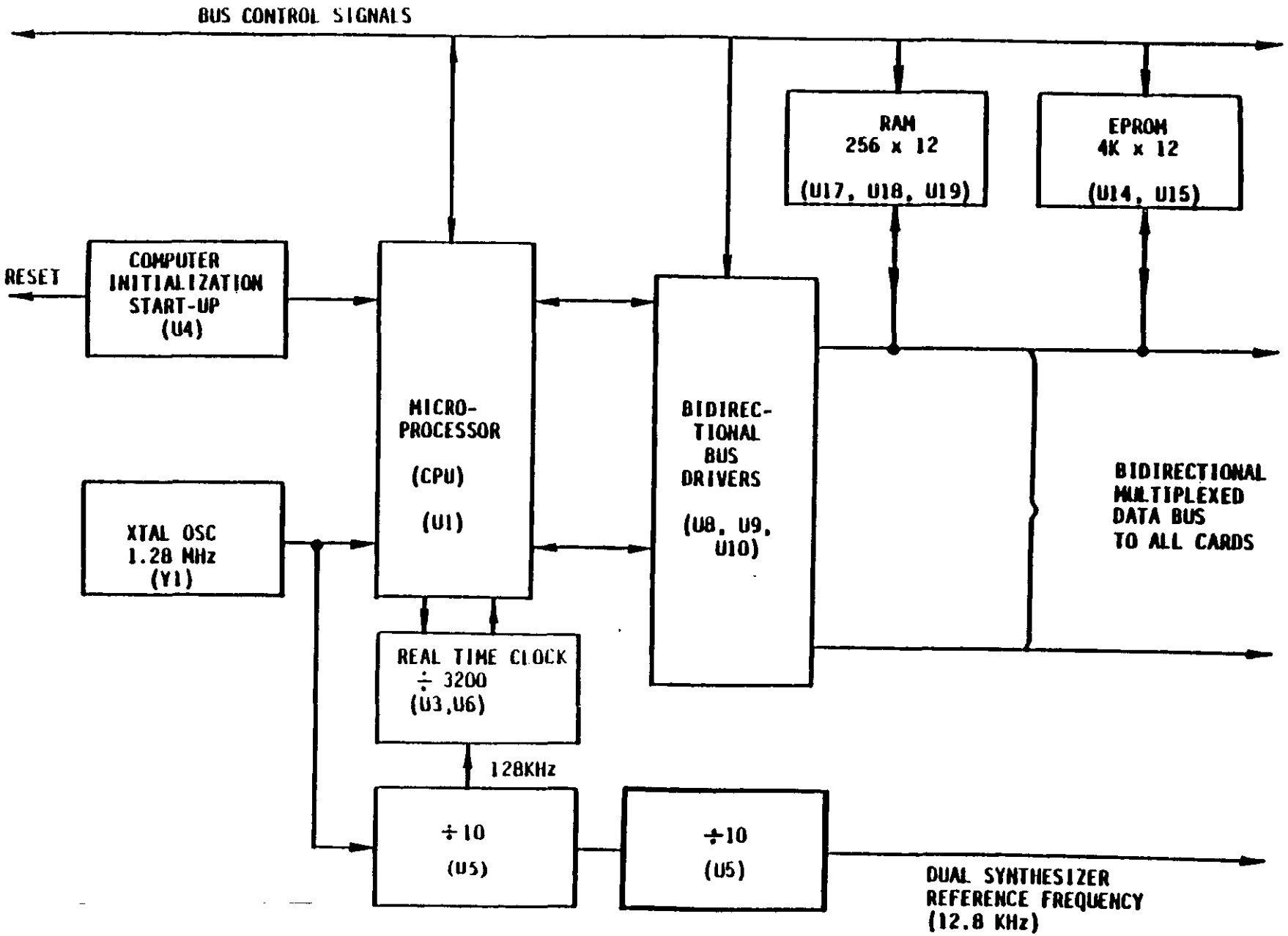


FIGURE 33 Functional Block Diagram of Module A3.

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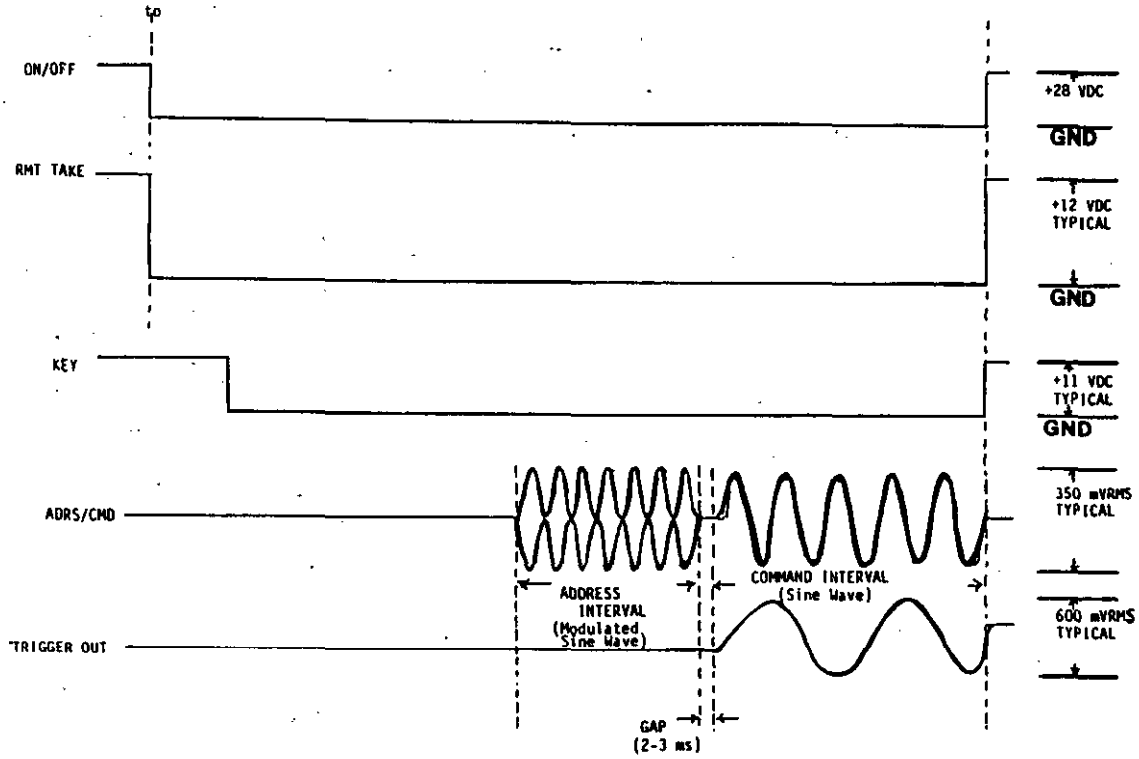


FIGURE 34 CSG Output Timing.

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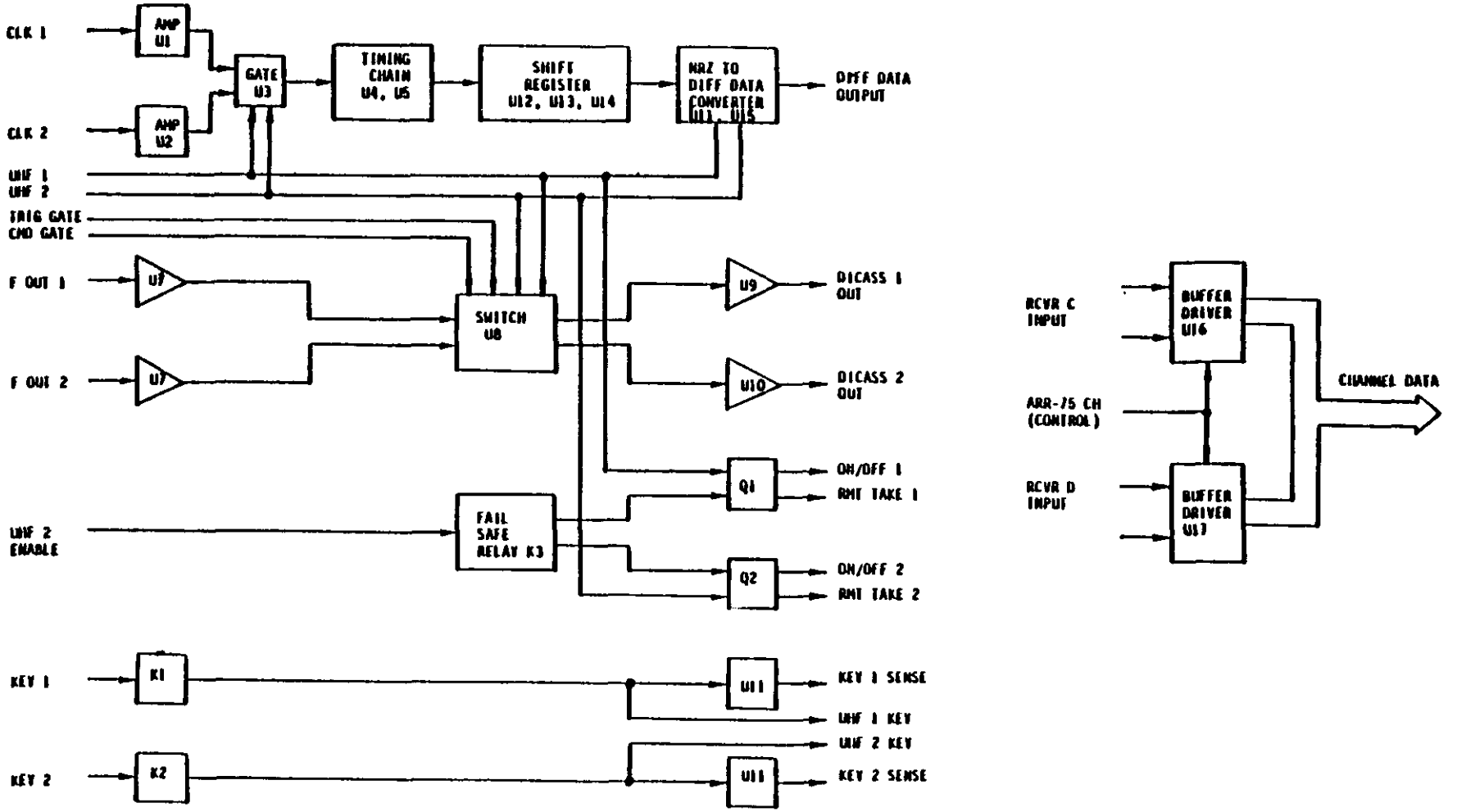
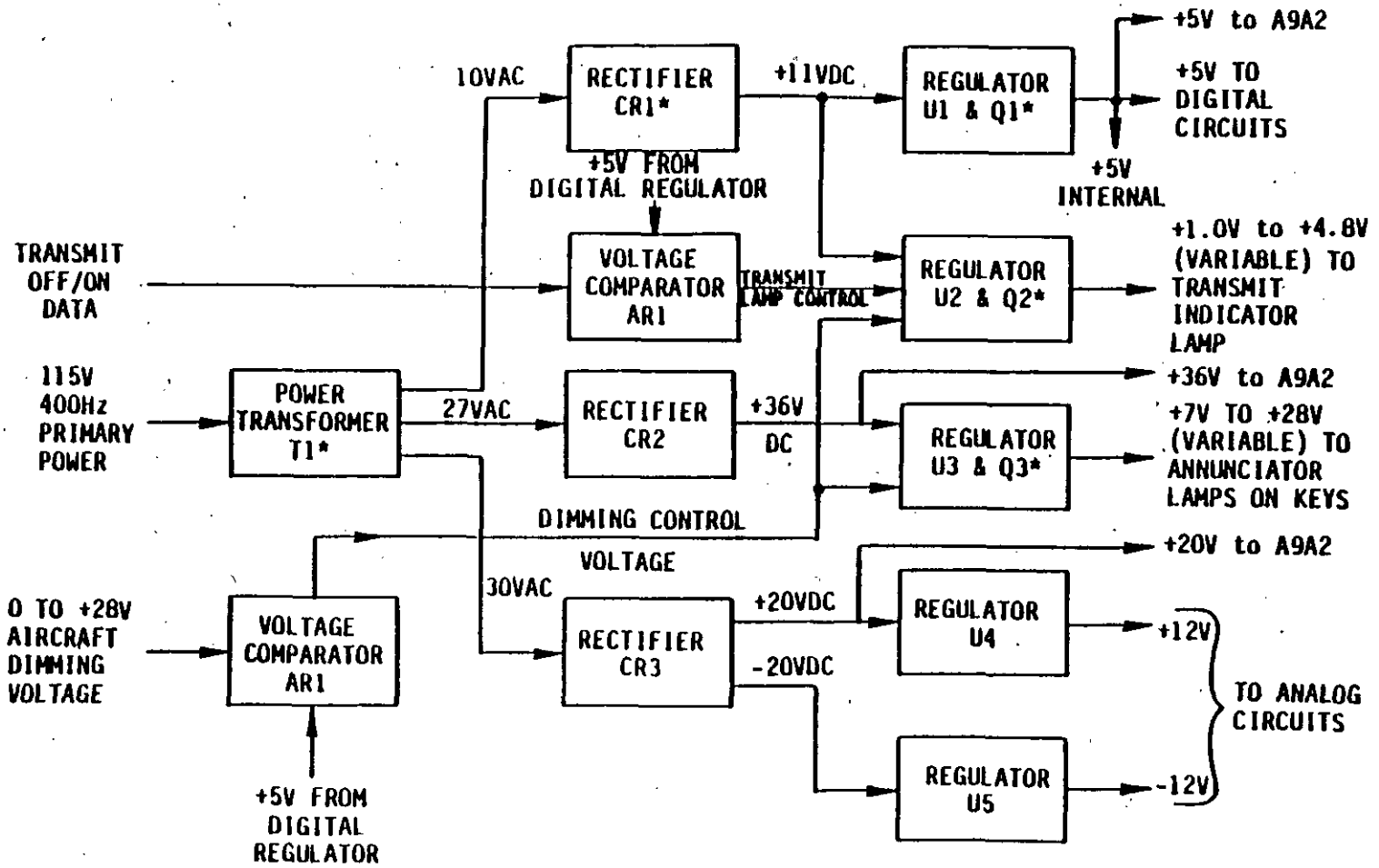


Figure 35 Functional Block Diagram of Module A4.

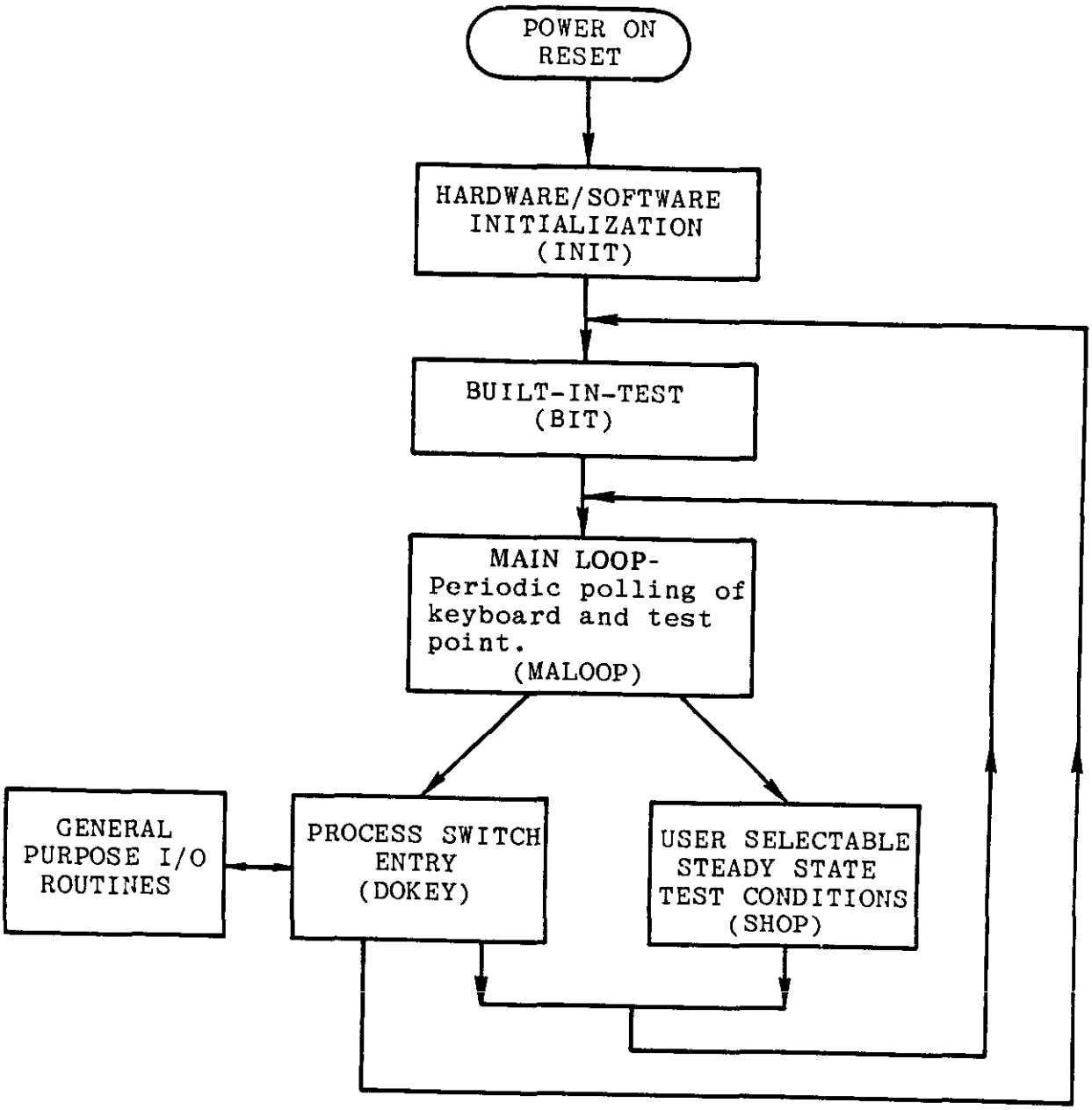
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*Component not on Power Supply Assembly,A5.

Figure 36 Functional Block Diagram of Module A5.

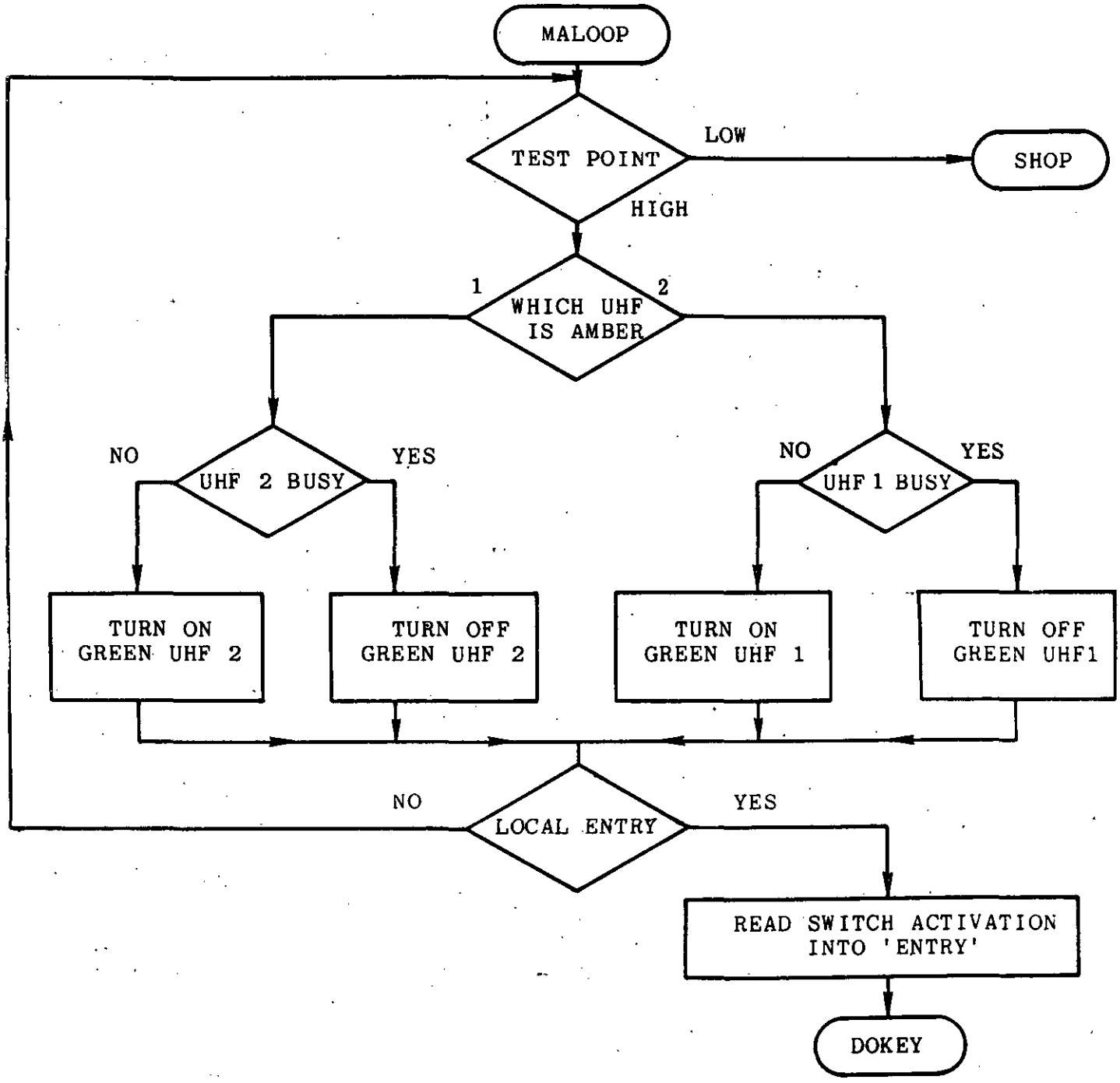
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OVER PROGRAM FLOW

FIGURE 37 CPU Flow Charts.

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MAIN LOOP FLOW

FIGURE 37 CPU Flow Charts. (continued)

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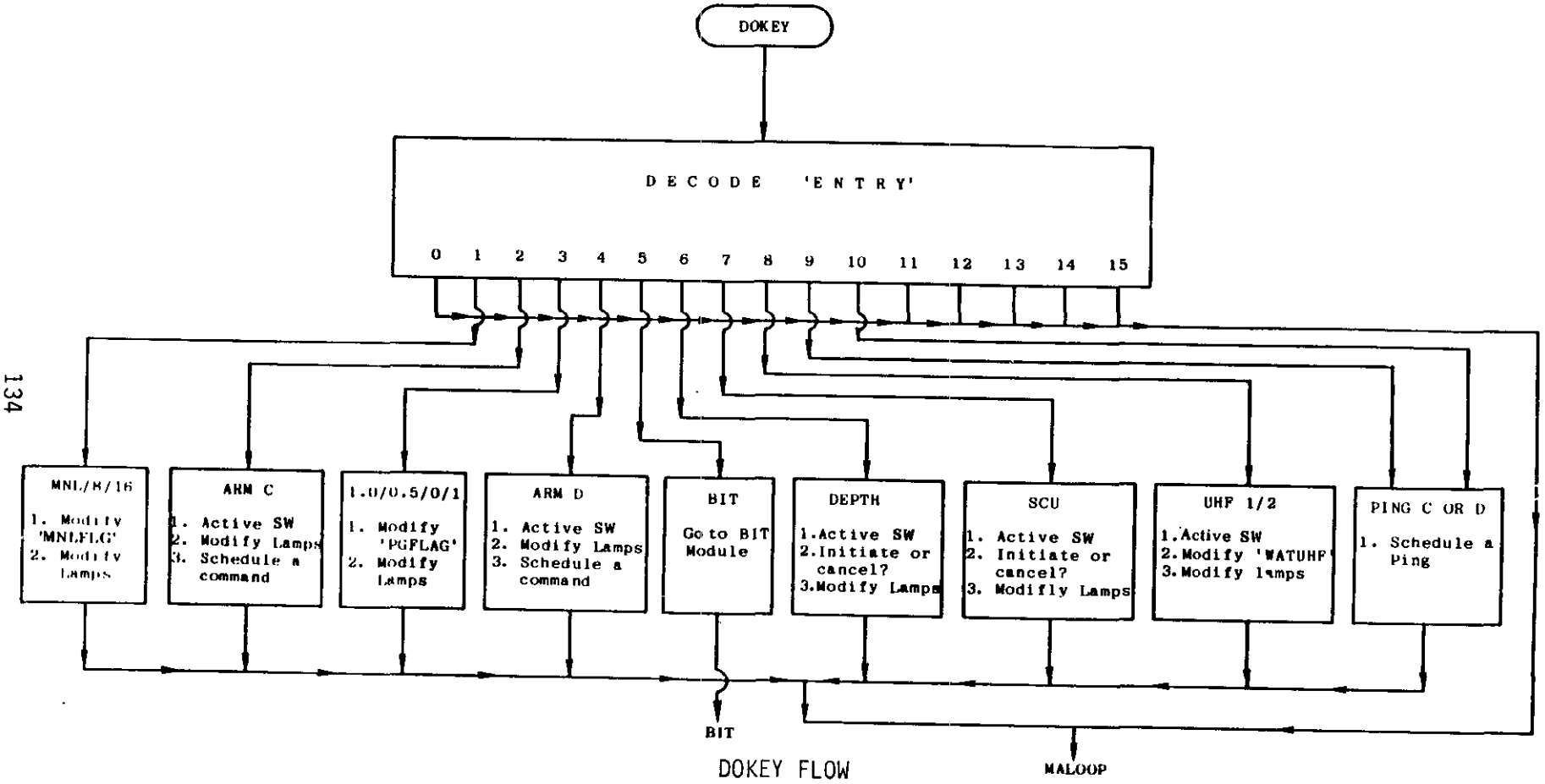
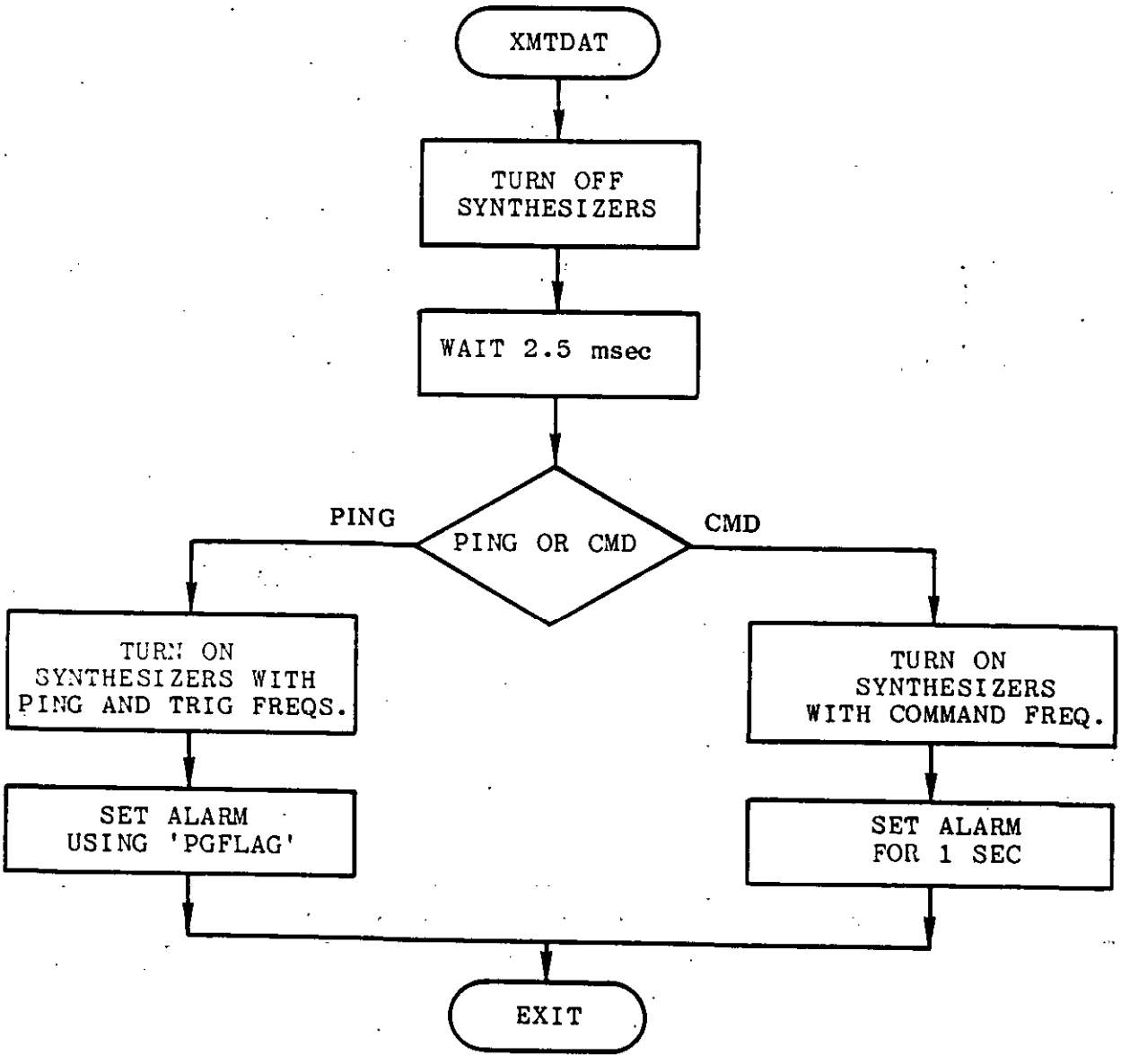


FIGURE 37 CPU Flow Charts.(continued)

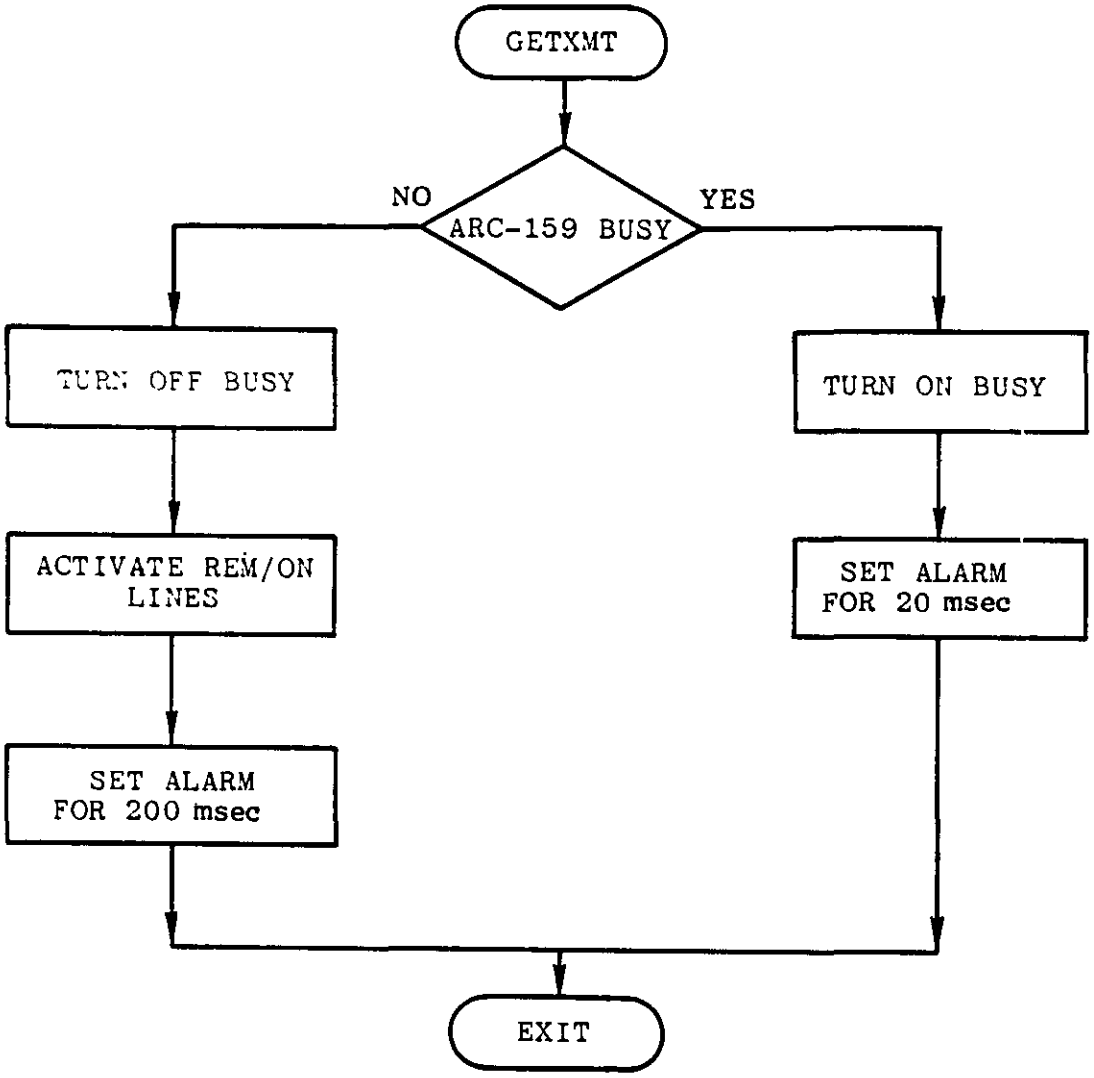
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INTERRUPT - TRANSMIT PING/COMMAND TONE FLOW

FIGURE 37 CPU Flow Charts. (continued)

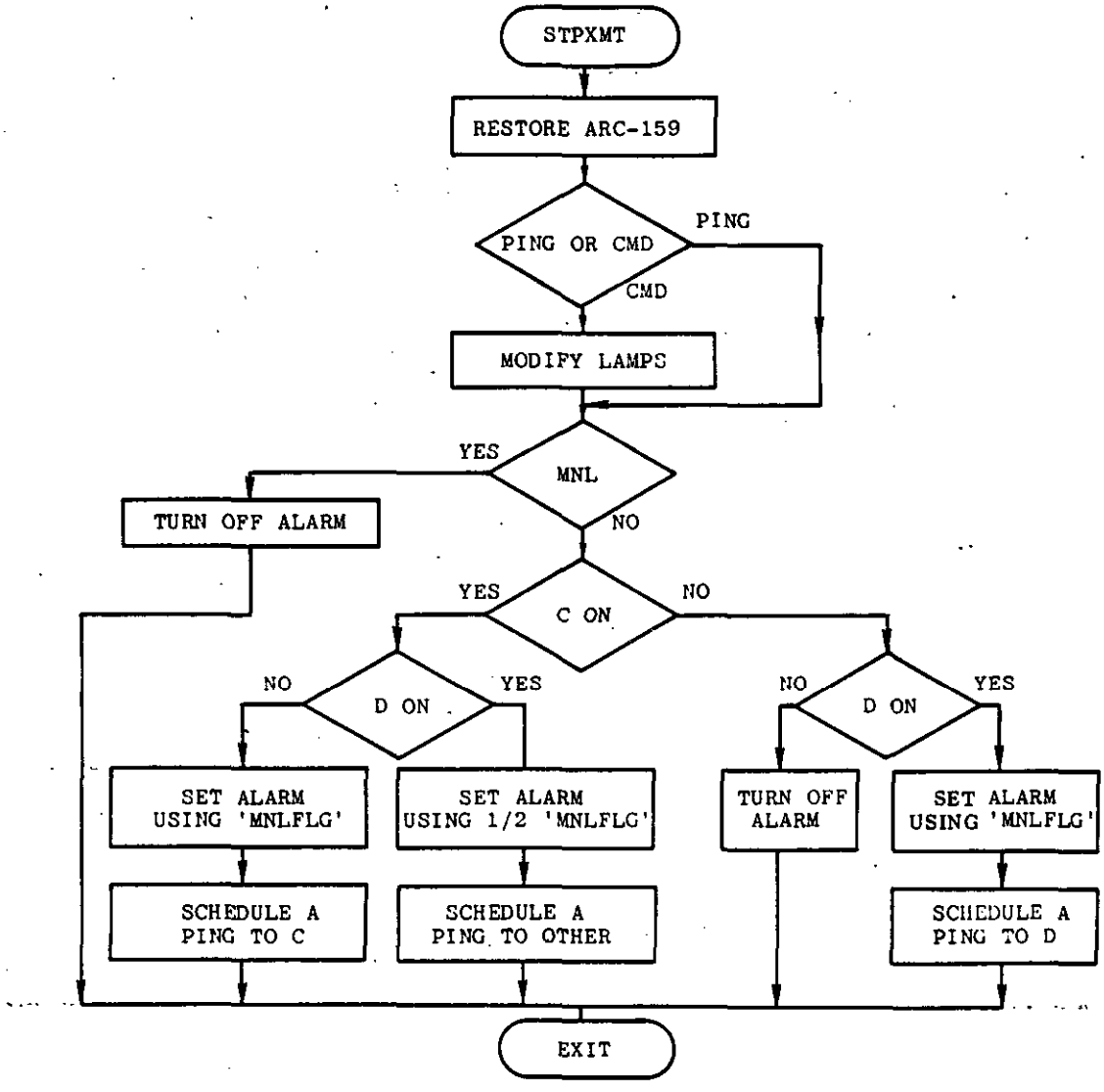
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INTERRUPT - ACTIVATE REM/ON LINES OF ARC-159 FLOW

FIGURE 37 CPU Flow Charts. (continued)

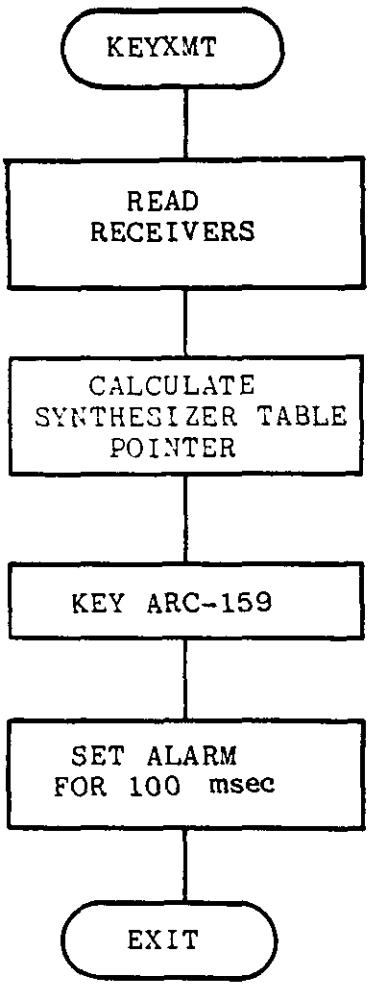
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INTERRUPT - COMPLETE TRANSMISSION FLOW

FIGURE 37 CPU Flow Charts. (continued)

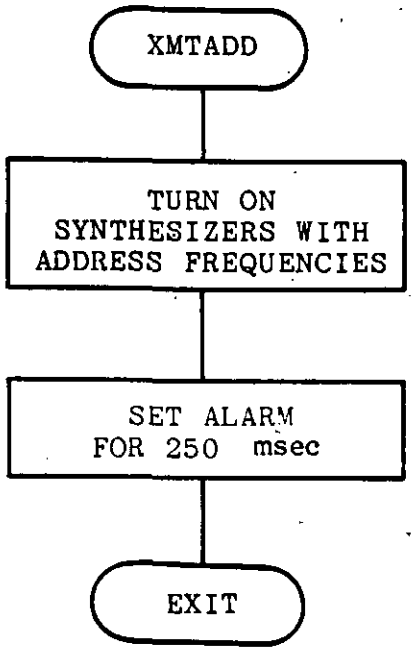
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INTERRUPT - KEY ARC-159 FLOW

FIGURE 37 CPU Flow Charts. (continued)

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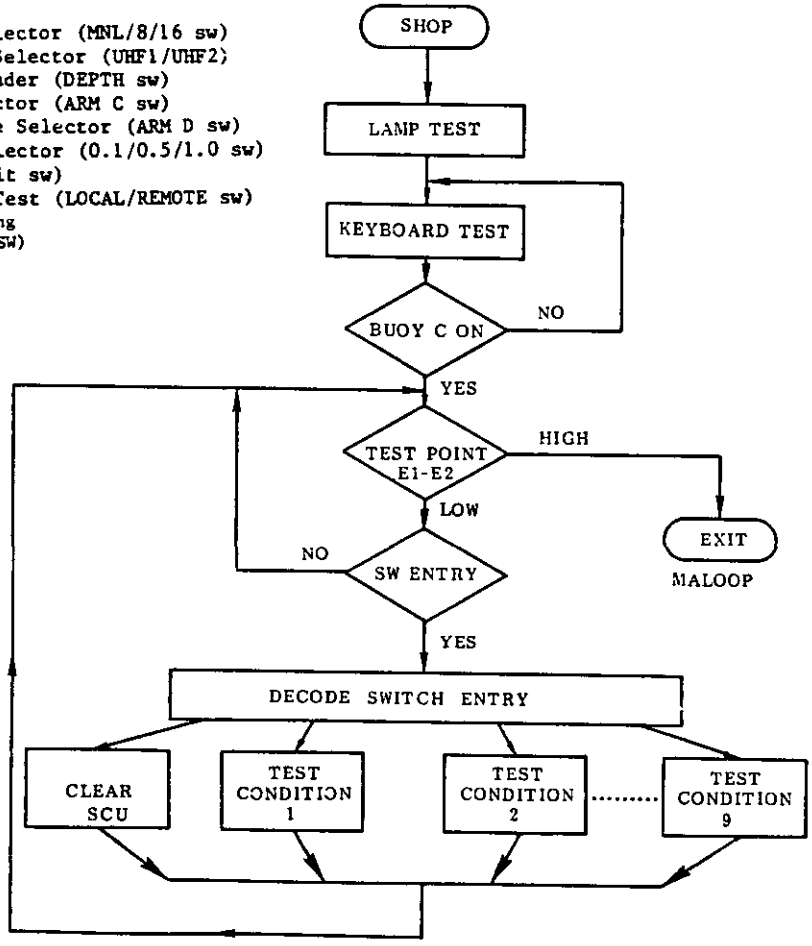
INTERRUPT - TRANSMIT ADDRESS TONE FLOW

FIGURE 37 CPU Flow Charts. (continued)

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TEST #

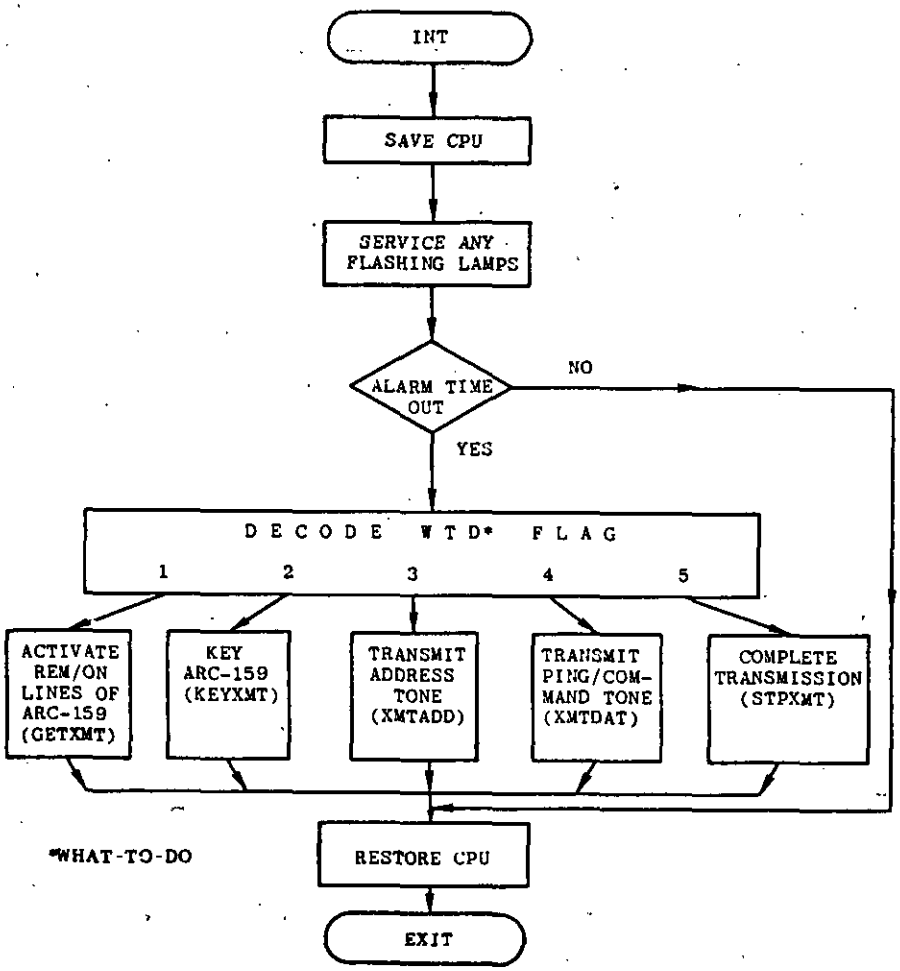
1. Frequency Selector (MNL/8/16 sw)
2. Synthesizer Selector (UHF1/UHF2)
3. Frequency Loader (DEPTH sw)
4. ARC-159 Selector (ARM C sw)
5. ARC-159 State Selector (ARM D sw)
6. Tone Type Selector (0.1/0.5/1.0 sw)
7. CPU Tests (Bit sw)
8. Remote Port Test (LOCAL/REMOTE sw)
9. Address and Ping
   (Buoy C and D SW)



INTERMEDIATE AND DEPOT LEVEL BIT FLOW

FIGURE 37 CPU Flow Charts. (continued)

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REAL TIME INTERRUPT DRIVEN TONE GENERATOR (50 Hz) FLOW

FIGURE 37 CPU Flow Charts. (continued)

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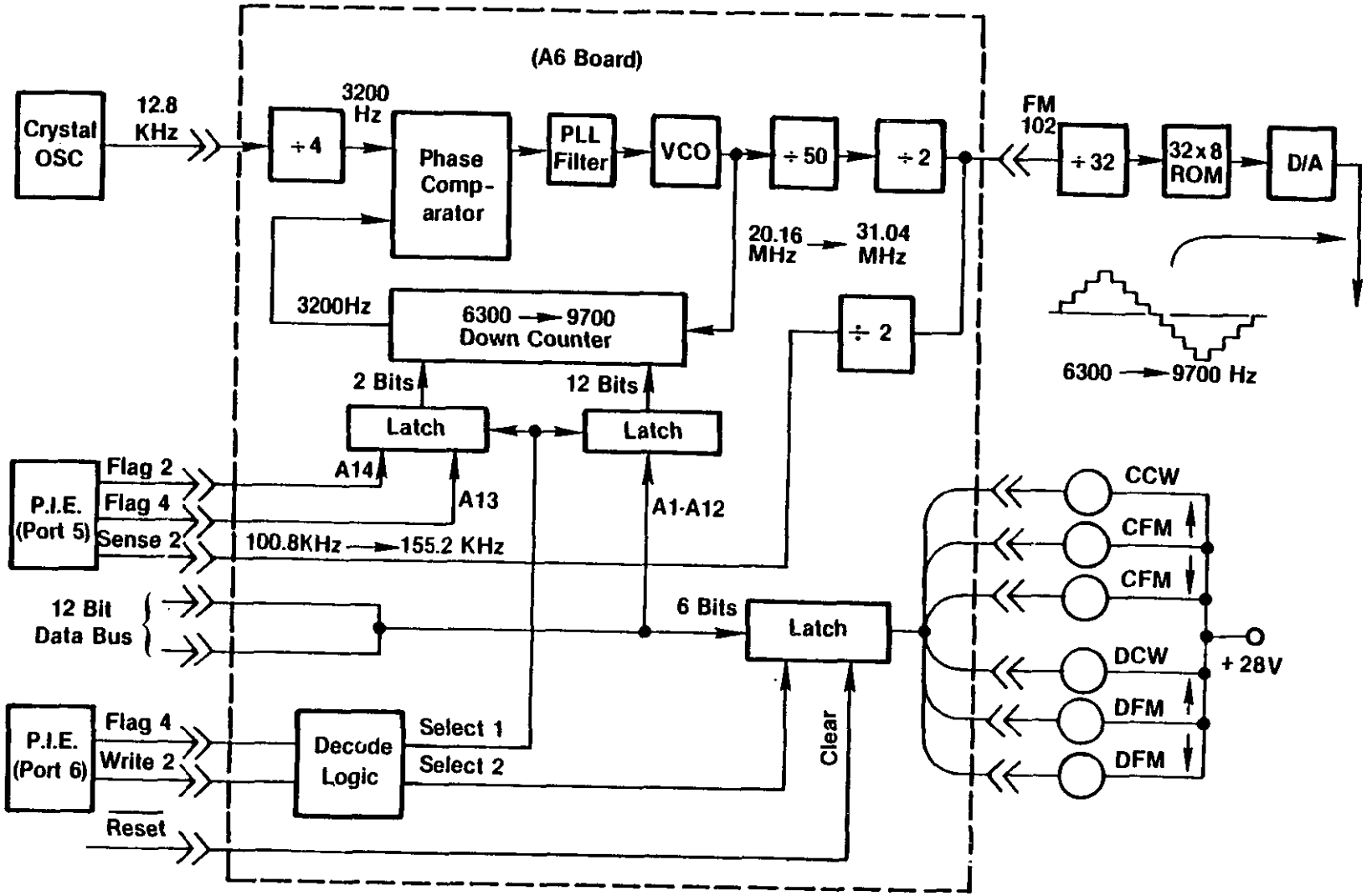


FIGURE 38 FM –SLIDE SUBASSEMBLY ELECTRICAL BLOCK DIAGRAM

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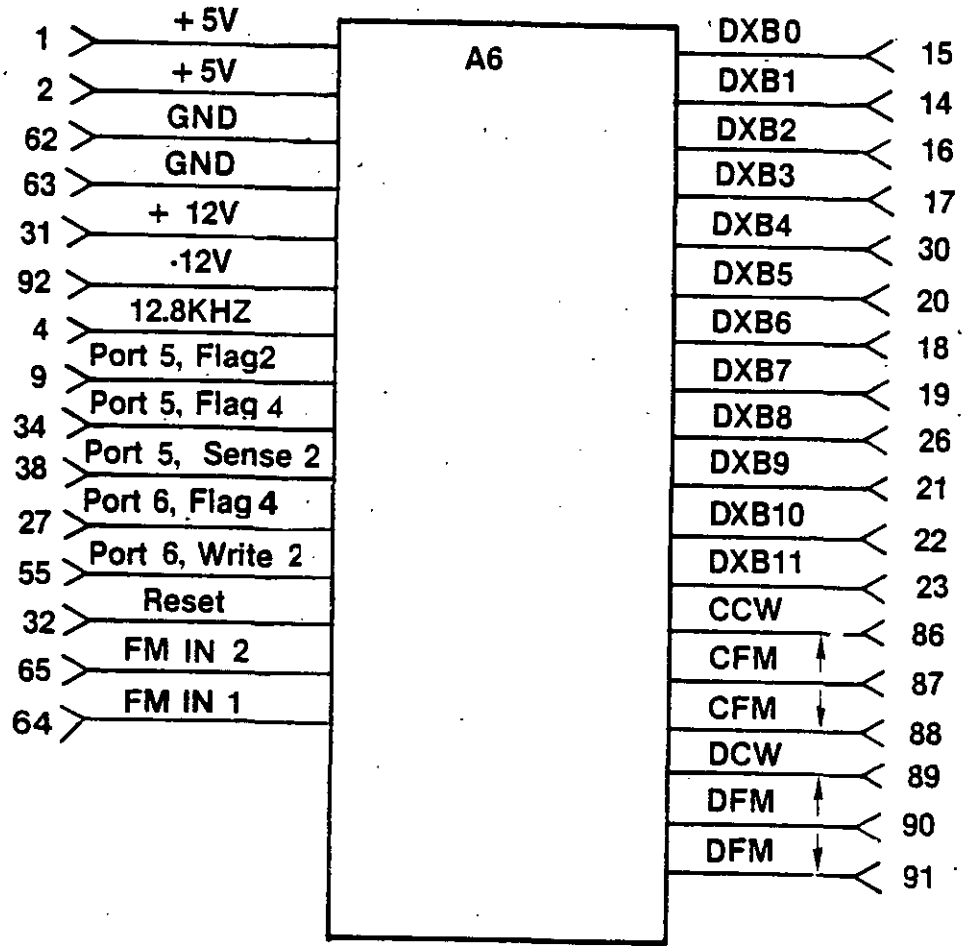


FIGURE 39 A6 BOARD INTERFACE PIN ASSIGNMENTS

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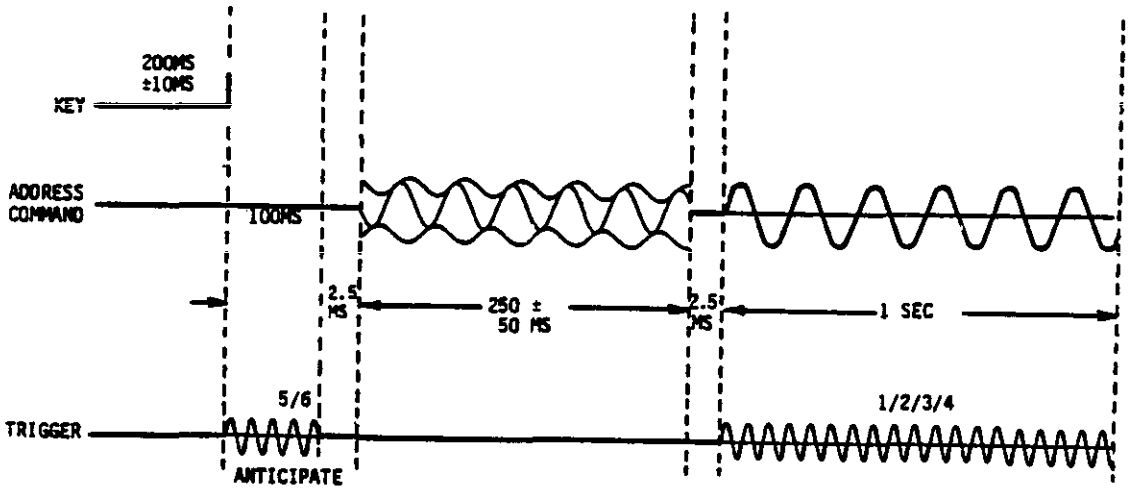


Figure 40 FM Slide Timing

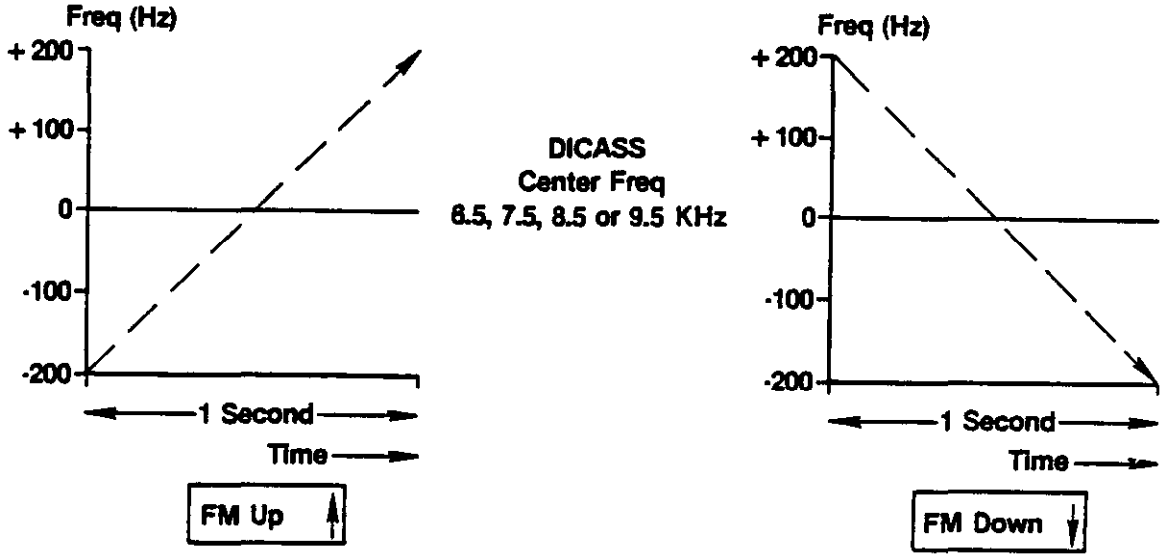


FIGURE 41 FM SWEEP FUNCTIONS

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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Check the source to verify that this is the current version before use.